



Securing Vehicle-to-Everything (V2X) Communication: Challenges, Mechanisms, and Future Directions

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Abstract

Vehicle-to-everything (V2X) communication enables vehicles to exchange information with other vehicles and infrastructure, such as traffic lights and road sensors. However, the communication channels used in V2X are susceptible to security threats, such as eavesdropping, message tampering, and denial-of-service (DoS) attacks. Therefore, V2X security is critical to ensure the confidentiality, integrity, and availability of communication in the transportation system. This paper provides an overview of the key security challenges in V2X communication and its security mechanisms. In addition, the various security testing methods are also addressed along with different security techniques adopted to protect V2X services. The performance metrics for evaluating the bandwidth, latency, dependability, and security of V2X communication are also investigated. This study leverages the tampering, repudiation, information disclosure, DoS, and elevation of privilege (STRIDE) threat model to identify backdoors and other loopholes in V2X communication. Further, the importance of standardization, interoperability, and scalability of security solutions across various V2X communication systems is also discussed, along with potential future developments and security challenges in V2X. This paper thus provides a comprehensive reference for researchers and academics in the fields of computer science, engineering, and transportation who seek to gain a thorough grasp of V2X systems and the security implications associated with them.

Keywords V2X Security · Vehicle-to-everything · STRIDE · V2X threat modelling

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1 Introduction

The concept of V2X communication has been around since the early 2000s, when researchers first began exploring methods to connect vehicles to the surrounding infrastructure. The initial emphasis was on vehicle-to-vehicle (V2V) communication, which was intended to enable vehicles to share information about their speed, location, and other vital data. In the years that followed, there was a growing interest in the potential advantages of V2X technology, such as improved road safety, increased traffic efficiency, and improved driving experiences. This resulted in the creation of vehicle-to-infrastructure (V2I) communication, which enabled vehicles to communicate with traffic signals, road signs, and other connected devices [1, 2]. The first successful deployment of V2X technology took place in Ann Arbor, Michigan, in 2012, where 3,000 vehicles were equipped with V2V communication devices. This pilot initiative demonstrated the potential of V2X technology and accelerated its development and implementation [3, 4].

The V2X technology has continued to develop since then, with the introduction of dedicated short-range communication (DSRC) and cellular communication for information transmission [5]. This technology for autonomous vehicles has also shown growing interest, as it can provide real-time information about the road network and surrounding environment. It also helps improve significantly improving road safety, reduce traffic congestion, and improve driving experience [6].

Figure 1 shows how V2X technology allows communication between vehicles and other entities through wireless network. V2X networks enable the exchange of information such as location, speed, and direction, which can be used to improve driver and vehicle situational awareness. On-board unit (OBU) is a vehicle-mounted device that enables V2X communication by processing, sending, and receiving messages between vehicles, infrastructure, pedestrians, and networks. OBUs are responsible for executing V2X protocols, applying security mechanisms such as encryption and digital signatures, and managing certificates. Highlighting OBUs emphasizes the vehicle-side hardware component critical for secure and reliable V2X operations. Complementing this, hardware security module (HSM) is a dedicated tamper-resistant hardware device integrated within the OBU or vehicle systems to securely store cryptographic keys and perform sensitive operations, such as signing

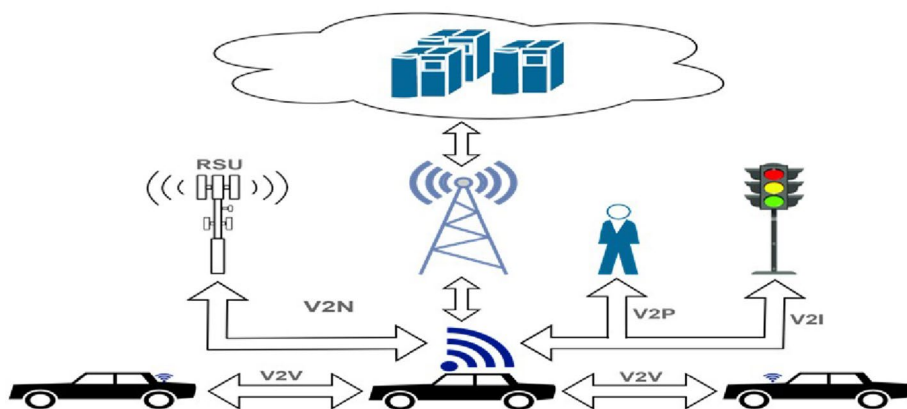


Fig. 1 General V2X structure

messages or generating certificates. The benefits of V2X technology are numerous [7]. For example, V2X helps improve road safety by alerting vehicles to potential hazards such as accidents or road closures. It can also reduce traffic congestion by providing real-time traffic data and optimizing traffic flow. Furthermore, V2X can enhance the driving experience by providing information on weather, parking availability, and other factors that can influence the driver's journey [8]. As such, it is a crucial development area for the automotive vehicle industry and its prevalence is likely to increase in the future.

Problem Statement: The extensive use of V2X communication introduces several security vulnerabilities and threats. Its wireless networks and protocols are vulnerable to many forms of security threats. The scope and impact of these risks are not discovered, hindering the development of effective security solutions [9]. Furthermore, the distinct properties of V2X systems, such as high mobility, low latency requirements, and resource-constrained devices, offer substantial challenges to existing security procedures. The lack of standardized security frameworks across V2X deployments, including technologies, protocols, quality of services, and testing methods, hinder inconsistency in security practices.

The objectives of the V2X security survey are to consolidate and synthesize existing knowledge and research in V2X security, identify gaps and research opportunities, distribute knowledge to researchers, and provide suggestions and guidance for future research in V2X security. By fulfilling these goals, the survey hopes to promote V2X security by improving awareness of current state-of-the-technology identifying vulnerabilities, and increasing knowledge diffusion for V2X communication.

Scope of this Survey: This study provides a detailed investigation of the V2X communication domain. It offers a comprehensive understanding of V2X, encompassing its foundational ideas and the most recent breakthroughs in technology. The article initially acquaints readers with the concept of V2X, providing an overview of its importance, practical uses, and its relationship to intelligent transportation systems. Furthermore, this discourse examines the fundamental standards and technologies that serve as the foundation for V2X communication, elucidating the protocols and infrastructure implicated in this domain. In addition, examines the distinct issues encountered by V2X systems and explores the security techniques implemented to protect these systems. The reliability and effectiveness of V2X communication depend greatly on the quality of service (QoS) factors, which are therefore thoroughly examined. The study highlights the importance of these characteristics in ensuring the smooth functioning of V2X systems. This resource offers valuable perspectives on the practice of possible testing, vulnerability assessments, threat modeling, and related methodologies, including an examination of their respective benefits and limitations.

Comparison with state-of-the-art Surveys: Table 1 contains the contributions and limitations of existing surveys in various areas, including vehicular networks, VANET, and V2X. It highlights that the limitations of previous research have motivated us to create a comprehensive survey with detailed information on V2X security. Table 2 enumerates the different methodologies covered in this document and compares them with existing survey documents. It is important to note that existing survey papers lack security threat modeling (STRIDE threat model), security testing methods based on the layered approach, and essential QoS parameters for V2X security. The authors in [15] discussed analyses of cryptographic techniques and the establishment of trust. They presented a comparative analysis of existing solutions in terms of their identity and location. The key contribution of the work [21] provided an exhaustive coverage of security and privacy concerns within V2X

Table 1 Comparison of previous survey on V2X security

References	Type of Network	Major Contributions	Limitations
[10]	VANET	It helps VANET security by revealing cryptography, authentication, and intrusion detection systems.	The scope of this study may be limited in addressing the whole of developing threats and solutions in VANET security, given the ongoing evolution of this sector.
[11]	VANET	This study sheds light on VANET security issues and attacks. Highlighting vehicular communication threats increases researcher and practitioner awareness. Additionally, the study analyzes countermeasures and solutions that can reduce security threats.	The paper may not provide an exhaustive review of all security assaults and countermeasures in VANETs, omitting recent advancements and comprehensive coverage.
[12]	V2X	It offers helpful insights into assessing the security and functionality of V2X communication systems by addressing a broad range of testing methodologies and tools.	It might restrict its coverage to broader V2X security and communication concerns.
[13]	VANET	The paper classifies and explains the security services and requirements necessary for VANETs to ensure safe and secure vehicular communication.	It may not include the most recent VANET security developments.
[14]	Vehicular	They concentrate on an in-depth review of the pseudonymity mechanism used for anonymous authentication schemes. Additionally, they elaborate on location-based privacy protection mechanisms designed to protect the privacy of vehicles and location-based services.	It does not extensively address the wider obstacles related to interoperability and standards, which are equally vital for the effective implementation of such networks.
[15]	V2X	It provides a contemporary examination of the latest developments in the field of security and privacy related to V2X communications.	It does not explore extensively the intricate technical aspects of security solutions.
[16]	V2X	This study offers significant information on the incorporation of cellular networks for V2X communication, specifically focusing on key concerns such as latency, dependability, and scalability.	It may not offer a thorough analysis of additional V2X communication technologies or hybrid approaches.
[17]	Vehicular networks	The author employs a layered approach to identify security and privacy issues and also presents a taxonomy for the identification of threats.	It is primarily concerned with summarizing and classifying present problems and potential solutions, which may be regarded as a more passive approach.
[18]	Vehicular networks	it encompasses a detailed analysis of various attack types and an overview of defense mechanisms.	Limits the coverage of broader aspects of autonomous vehicle technology and emerging trends in cybersecurity.
[19]	VANET	It offers helpful information on the current state of the field by providing a structured and comprehensive overview of existing research on VANET security.	It does not present new empirical research or propose novel solutions to VANET security issues.

Table 1 (continued)

References	Type of Network	Major Contributions	Limitations
[20]	V2X	It provides a comprehensive examination of the current security measures in place, helping readers to better understand the cybersecurity domain of V2X.	The emphasis on cybersecurity mechanisms may not encompass other significant aspects of V2X communication, including protocol standards, network infrastructures, and applications.
[21]	V2X	It covers a wide range of issues and is thorough, addressing both technical and privacy-related issues.	The domain of V2X communication is undergoing significant advancements, and it is possible that the survey does not fully capture the latest breakthroughs.

systems. It analyzes the diverse vulnerabilities, threats, and attack vectors encountered by V2X networks.

The authors in [20] classified security systems based on their defensive techniques, distinguishing between proactive and reactive approaches. This study examines the potential offered by developing security techniques that take advantage of artificial intelligence tools to achieve security objectives. Work [16] critically analyzes the existing body of research and the challenges related to long-term evolution (LTE) and 5G technologies to enable efficient V2X connections. Subsequently, the authors present a comprehensive examination of the LTE V2X framework, together with an analysis of the operational scenarios currently being deliberated. Following this, the author examines the challenges that arise in current LTE systems.

All of these previous researches fall short of offering a systematic method to identify possible threats and security vulnerabilities in communication networks. Therefore, we provide a STRIDE threat model. This entails providing a comprehensive analysis of the practical feasibility and implementation details for each kind of attack within the domain of V2X communication. Furthermore, we broaden our analysis of the V2X attack landscape to incorporate all aspects of cybersecurity issues. The focus of our review is primarily on assessing the compatibility of current mechanisms with fundamental security and privacy requirements [23]. Additionally, we propose a layered approach to security testing methods that can effectively identify security-related issues and threats in V2X communication. Additionally, our work emphasizes the significant role that modern technologies play in enabling self-regulating security features and raising the level of trust.

The following are the major contributions of our work:

- This paper provides a comprehensive taxonomy of V2X communication (Fig. 2), highlighting the main security issues, threats, vulnerabilities, and mechanisms that help improve the security of V2X systems.
- It conducts an in-depth examination of the V2X security landscape, categorizing and identifying potential threats using the STRIDE threat model.
- It emphasizes the importance of quality of service (QoS) parameters for ensuring reliable and secure V2X communication. Furthermore, the importance of considering QoS metrics to improve the performance and efficiency of security systems is outlined.
- A thorough investigation is conducted into the testing methodologies (layer approach)

Table 2 Comparison of previous survey papers with an emphasis on the security topics they cover and the progress made in those areas

Reference number	Standards and technologies	Security issues and threats	General attacks classification	Attacks classification based on the STRIDE threat model	Security mechanisms	QoS parameters based on security concerns	Testing methods and classification based on the layered approach
[15]	×	✓	✓	×	✓	×	×
[21]	✓	✓	✓	×	×	×	×
[12]	×	✓	×	×	×	×	✓/×
[16]	✓	✓	×	×	×	×	×
[20]	✓	✓	✓	×	×	×	×
[10]	×	×	✓	×	✓	×	×
[11]	×	×	✓	×	✓	×	×
[13]	✓	✓	✓	×	×	×	×
[19]	×	✓	✓	×	×	×	×
[18]	×	✓	✓	×	✓	×	×
[22]	×	✓	✓	×	×	×	×
[14]	✓	✓	✓	×	×	×	×
[17]	×	✓	✓	×	×	×	×
Our paper	✓	✓	✓	✓	✓	✓	✓/✓

✓ - Topics Covered

✓/× - Topics Covered that do not align with our suggested model.

× - Topics not Covered

✓/✓ - Topics Covered in accordance with our recommended model.

used to assess V2X security. It also emphasizes contemporary technologies in V2X communication and the critical role they play in improving security mechanisms and identifying and mitigating threats in V2X services.

This paper also highlights and addresses the open issues and future security challenges in V2X communication, providing valuable information to guide future research directions.

Figure 3 show the roadmap of this survey. The paper is summarized as follows. Section 2 provides an overview of how V2X standards and technologies are crucial in the development of intelligent vehicle transportation systems to improve road safety and improve driving experiences. Section 2.2 highlights various security concerns, potential security attacks, and challenges associated with V2X communication technologies. Section 4 briefly describe and analyses the protective measures available for V2X security, including a comparison of their pros and cons. Section 7.2.1 briefly describes the testing methods for V2X communication. The QoS parameters play an important role in ensuring the reliability and security of V2X security, which is summarized in Sect. 6. Section 7 discusses how modern technologies have positive impacts on V2X communication and security. The open issues and research directions in V2X are highlighted in Sect. 8 and the study inferences are presented in Sect. 9. The conclusions of our research are presented in Sect. 10.

1.1 Types of V2X Communication

This section demonstrates various V2X communication models [24, 25].

- *Vehicle-to-vehicle (V2V) communication*: It refers to direct communication between two or more vehicles that allows one to share information about speed, position, and road conditions. This information can be used to improve traffic flow and road safety [26].
- *Vehicle-to-infrastructure (V2I) communication*: This refers to communication between a vehicle and the road infrastructure, such as traffic signals, road indicators, and traffic management systems. It allows the vehicle to receive real-time information on the road ahead, such as traffic conditions and road closures, which can improve the driving experience [27].
- *Vehicle-to-pedestrian (V2P) communication*: This refers to the communication between a vehicle and adjacent pedestrians using Wi-Fi or Bluetooth technology [28].
- *Vehicle-to-network (V2N) communication*: This involves communication between a vehicle and the wider network, which can include the Internet and other connected devices. This allows the vehicle to access information and services beyond the immediate vicinity of the vehicle, such as traffic updates and navigation services in real time [29].
- *Vehicle-to-cloud (V2C) communication*: This involves communication between a vehicle and cloud-based application services. This enables the vehicle to access and process immense amount of data, thereby enhancing functionality such as predictive maintenance and advanced navigation [30].
- *Vehicle-to-grid (V2G) communication*: This refers to the bidirectional communication and energy exchange between electric vehicles (EVs) and the power grid. Through V2G technology, vehicles can supply stored energy back to the grid during peak demand or draw energy during low-demand periods. This interaction supports grid stability, renewable energy integration, and efficient energy utilization within smart city infrastructures.

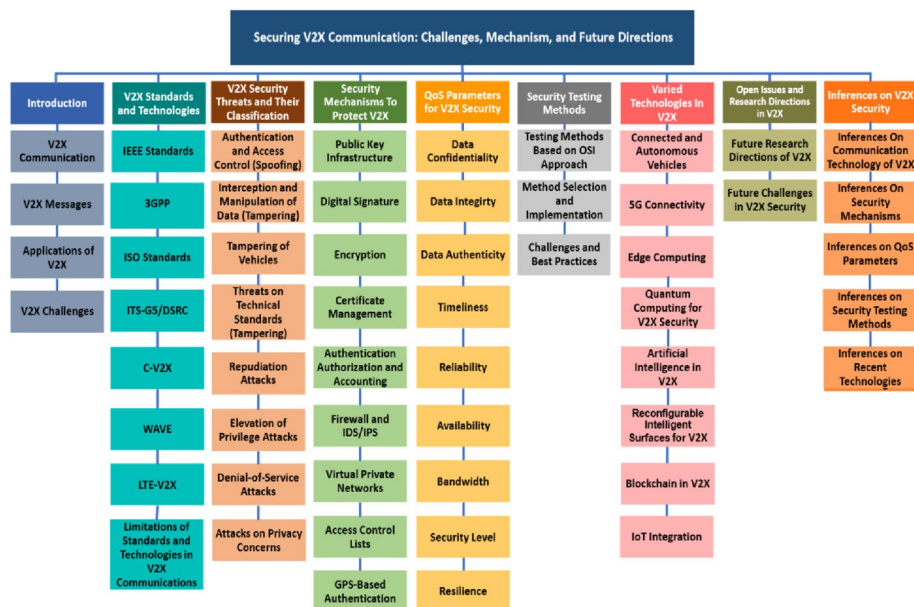


Fig. 2 Taxonomy of this review

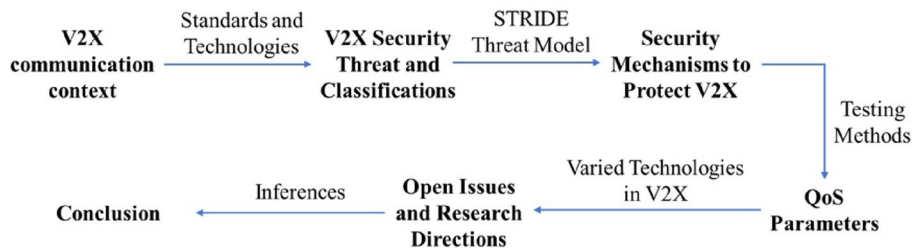


Fig. 3 Roadmap of this survey

V2G communication relies on secure protocols to ensure data integrity, authentication, and safe power transactions between EVs and grid operators [31].

1.2 V2X Messages

The V2X communication involves the exchange of various types of messages between vehicles and other entities [32]. The categories of messages in V2X communication are shown in Fig. 4.

- *Safety messages*: These are crucial communications about road safety that are typically exchanged in real time between vehicles. Warnings about imminent collisions, road hazards, and the presence of emergency vehicles are some examples of safety messages [33].

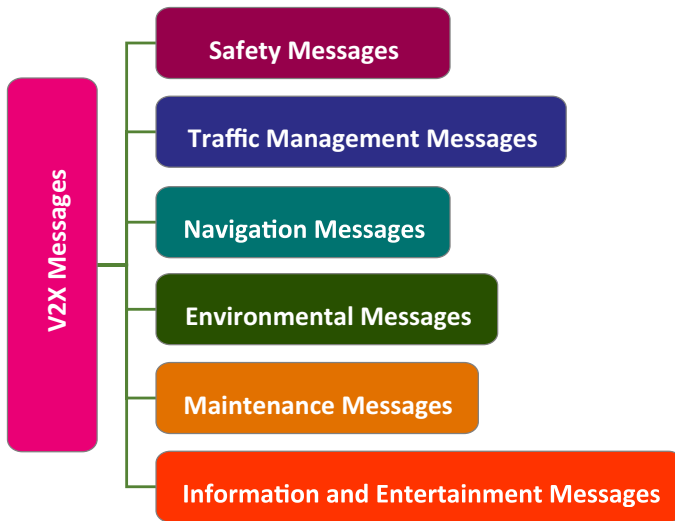


Fig. 4 Types of V2X messages

- *Traffic management messages*: These messages are used to increase traffic flow and reduce congestion. Information about traffic conditions, road closures, and traffic signals are examples of traffic management messages [34].
- *Navigation messages*: These messages provide vehicles with real-time navigation information, such as traffic updates, suggested alternate routes, and road conditions [35].
- *Environmental messages*: These messages monitor and manage the environment including information about weather conditions, air quality, and road surface conditions [36].
- *Maintenance messages*: These messages are extremely useful for managing vehicle maintenance and repair, as they contain information on vehicle performance, fuel economy, and wear and tear [37].
- *Information and entertainment messages*: These messages aim to provide drivers or passengers with valuable information, such as weather updates, news, and traffic reports. They can also provide entertainment features such as streaming audio and video [38].

These message categories are not mutually exclusive; a single message can belong to multiple categories. The varieties of messages used in V2X communication will depend on the specific requirements of the application and the underlying technology.

1.3 Applications of V2X

V2X communication has several uses in a variety of sectors including transportation, logistics, and automotive. The following are some of the most well-known V2X applications:

- *Intelligent transportation systems (ITS)*: ITS, a collection of cutting-edge technologies that aims to improve the sustainability, efficiency, and safety of the transportation system, can be supported by V2X. To improve traffic management and ease congestion,

V2X offers real-time information on road conditions, traffic flow, and accidents [36, 39].

- *Advanced driver assistance systems (ADAS)*: Advanced driver assistance systems, such as adaptive cruise control, lane departure warning, and automatic emergency braking, are supported by V2X. V2X can provide real-time information about the road ahead, such as other vehicles, road conditions, and traffic signals, which can be used to improve the performance of these systems [40, 41].
- *Autonomous vehicles*: V2X is essential for the advancement of autonomous vehicles. since it provides real-time data on traffic, other vehicles, and road conditions. This information can be used to help make autonomous vehicle decision and improve their performance and safety [42, 43].
- *Road safety*: V2X improves traffic safety by providing real-time data on road conditions, other vehicles, and potential dangers. These data can be used to warn drivers of potential hazards and prevent accidents [44, 45].
- *Traffic management*: Traffic can be managed using V2X, as it provides vehicles with real-time data on congestion, road closures, and alternative routes [34, 46].
- *Energy efficiency*: V2X improves the vehicle energy efficiency by providing data on traffic flow and road conditions, allowing drivers to optimize their driving style and reduce fuel consumption [47].
- *Parking management*: V2X manages parking by providing real-time information on available parking spaces and allowing vehicles to make advance reservations for parking [48, 49].

As technology continues to advance, it is likely that new applications will emerge that can further improve the safety, efficiency, and sustainability of the transportation system.

1.4 V2X Challenges

The V2X communication technology is still in its early stages of development and must overcome a number of obstacles to be widely adopted and successful. Figure 5 shows several primary challenges of V2X.

1.4.1 Interoperability

It refers to the ability of various V2X systems, regardless of the manufacturer or technology, to communicate with each other [50]. The V2X technology must support communication between vehicles from various manufacturers, as well as between vehicles and other entities, such as infrastructure, pedestrians, and other road users. This requires the development of universally applicable standards and protocols for all V2X systems. However, the process of developing these standards and protocols is complex and time-consuming. Despite the development of several technologies and methodologies, there is currently no consensus on the optimal V2X communication approach. Different V2X systems that are incompatible with each other can result in a fragmented market. Organizations like the society of automotive engineers (SAE) [51] and the institute of electrical and electronics engineers (IEEE) are developing V2X communication standards to address this problem [52]. These standards will ensure that various V2X systems, regardless of the manufacturer or technology, can communicate with each other [53].

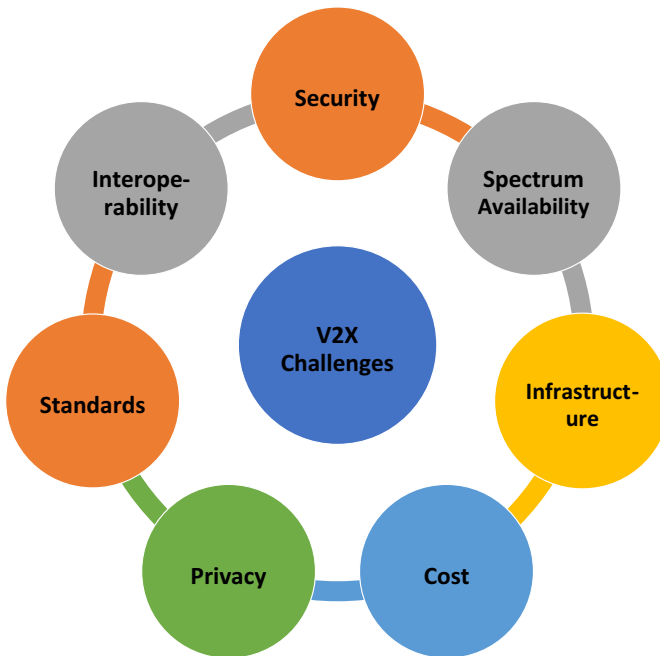


Fig. 5 General V2X challenges

1.4.2 Security

For the development and deployment of V2X technology, ensuring the security of V2X communication is a crucial challenge. V2X communication involves the exchange of confidential data between vehicles and other entities, including infrastructure, pedestrians, and other road users. These data may include the location and speed of the vehicle, as well as other information that could be used to compromise the safety and privacy of individuals. V2X systems must be designed to prevent unauthorized access and manipulation to address these security concerns. This requires the application of security technologies, such as encryption and digital signatures, to safeguard the confidentiality and integrity of data transmitted between vehicles and other entities [54]. In addition, V2X systems must be designed to be extremely resistant to attacks, such as DoS attacks [55], which can impede system availability. This requires the implementation of comprehensive security protocols and network architecture, as well as continuous monitoring and threat analysis to detect and respond to potential security threats.

The National Institute of Standards and Technology (NIST) [56] and the International Organization of Standardization (ISO) [57] have developed standards for secure V2X communication, providing guidance on the use of security technologies and best practices to ensure the security of V2X systems [58, 59].

1.4.3 Sufficient Spectrum Availability

To support the high-speed and reliable exchange of data between vehicles and other entities, V2X requires a substantial amount of bandwidth. However, the available spectrum is limited and in high demand, making it difficult to allocate sufficient bandwidth to support the anticipated increase in V2X devices. This can result in congestion and interference, which can reduce the quality and dependability of V2X communications [60]. To address this issue, spectrum allocation must be meticulously managed to ensure that it is used effectively and efficiently. This may involve the use of dynamic spectrum management techniques, such as spectrum sharing [61] and frequency hopping [62], which allow multiple devices to use the same spectrum band without interference. New spectrum frequencies can also be used, such as the use of unlicensed spectrum- 5 GHz band, or licensed spectrum- DSRC spectrum [63].

1.4.4 Infrastructure

The deployment of the V2X infrastructure requires substantial investment and planning, as well as the collaboration of numerous stakeholders, such as government agencies, private companies, and other organizations. This can be a complicated and time-consuming process, especially in regions with existing infrastructure such as rural or remote areas [64]. To address this issue, it is essential to prioritize the deployment of V2X infrastructure in locations where it is most needed and will have the greatest impact. This involves the deployment of DSRC networks, which can support highspeed and low-latency V2X communication, and cellular networks, which can provide wide-area coverage for V2X communication. Furthermore, it is essential to develop and implement effective planning and management processes for the V2X infrastructure in order to ensure its deployment in a timely and cost-effective manner. This involves the formation of public-private partnerships and the implementation of new technologies and approaches, such as edge computing and network slicing [65], to optimize the use of existing infrastructure and reduce the cost and complexity of V2X deployment.

1.4.5 Cost

V2X systems require significant investments in infrastructure, equipment, and research and development, which can be a significant barrier to the pervasive adoption of this technology. Specifically, the deployment of V2X infrastructure, such as DSRC networks [66], can be expensive due to the widespread deployment of communication devices and their maintenance. It also requires costly hardware and software components, as well as extensive testing and certification procedures [40, 67]. Therefore, it is essential to find solutions to reduce the cost of implementing the V2X technology and make it more accessible to a greater number of users.

1.4.6 Privacy

V2X communication involves the exchange of sensitive data, such as vehicle location, speed, and other data, which can raise concerns about privacy if mishandled or intercepted

by unauthorized parties [68]. To address this challenge, it is essential to implement robust privacy protection mechanisms capable of ensuring the security of V2X data and preventing unauthorized access. This involves the usage of encryption [69], authentication, and access control technologies [70], in addition to the implementation of privacy policies that regulate the collection, use, and storage of V2X data. It is also essential to consider the privacy implications of V2X technology throughout its development and deployment, from the design of communication protocols and systems to the selection of privacy-enhancing technologies [58, 59]. Subsequently, it is necessary to educate users about the privacy implications of V2X technology and provide them with the resources, as well as the information they need to make informed decisions about the use of this technology. This includes the creation of privacy-aware user interfaces, the provision of privacy-related information and controls, and the implementation of user education and training programs.

1.4.7 Standards

To ensure that the data are transmitted and received reliably and securely, it is essential to have a distinct and well-defined set of V2X communication standards [71]. However, the development of standards for V2X communication has been slow, due in part to the complexity and rapid evolution of this technology, as well as the need for global standardization to facilitate the deployment of V2X systems in various regions and countries. There are multiple competing V2X communication standards, each with its advantages and disadvantages, making it difficult to select the optimal standard for a given application or deployment. This leads to confusion and fragmentation in the market, slowing the adoption of V2X technology. The creation of global standards by organizations such as the international organization for standardization (ISO) or IEEE, as well as the creation of regional and local standards that can support the deployment of V2X technology in specific regions and countries are required [53]. Having introduced the various types of V2X communication and their applications, the next section examines the standards and underlying technologies that enable secure and reliable vehicular connectivity. Understanding these standards is essential before evaluating the security challenges inherent in V2X systems.

2 V2X Standards and Technologies

As V2X communication evolves, the requirement for standardized protocols and solutions becomes increasingly important. This section delves into a significant aspect of V2X standards and technologies. Understanding the established standards and new technologies in the V2X ecosystem is critical to ensuring seamless and secure communication [72]. The various standards and V2X technologies are shown in Fig. 6.

2.1 IEEE Standards

IEEE has developed several standards to facilitate the deployment of V2X communication systems. IEEE standards help to ensure the dependability, security, and interoperability of V2X systems by providing explicit specifications and guidelines [73]. The most important IEEE standards for V2X technology are:

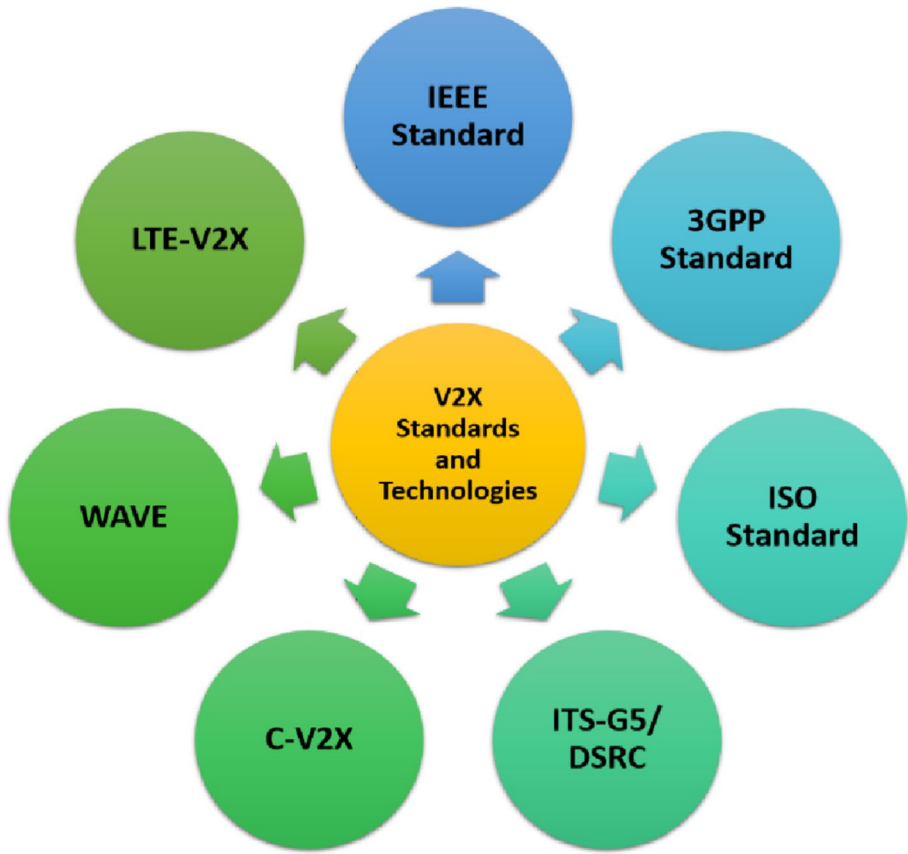


Fig. 6 V2X standards and technologies

- *IEEE 802.11p*: This standard defines the physical and media access control (MAC) layers of wireless communication used for V2V communication. It operates in the 5.9 GHz band and provides a high data rate for short-range communication [74].
- *IEEE 1609*: This standard defines the upper layers of the V2X communication stack and covers areas such as security, network management, and application-level protocols [75].
- *IEEE 1733*: This standard defines the architecture for V2X communication and provides a framework for developing V2X applications [76].
- *IEEE 1905.1*: This standard defines the architecture to integrate home and access networks with broadband networks [77].

2.2 3GPP

The 3rd Generation Partnership Project (3GPP) is a standardization organization that creates and maintains global mobile communications standards [78].

The organization was founded in 1998 with the mission of developing and promoting third-generation (3G) and beyond mobile telecommunication systems. 3GPP is responsible for developing and maintaining standards for various aspects of mobile telecommunications, such as radio access technologies (such as LTE and 5G), system architecture, security, and services [79]. The 3GPP standards are widely adopted by mobile network operators, device manufacturers, and other stakeholders in the mobile telecommunications industry [80]. The main focus on 3GPP's is the development of standards for cellular-based V2X (C-V2X) communication [81], which allows direct communication between vehicles and other entities of the transportation system. The 3GPP standards for C-V2X communication address multiple aspects of the technology, including the physical and MAC layers, the architecture of the system, the security, and the services [82]. It also contributes to the interoperability, dependability, and security of mobile telecommunication systems by creating and maintaining internationally recognized standards.

2.3 ISO Standards

ISO standards play a crucial role in ensuring the safety, security, and reliability of V2X communication systems, focusing primarily on engineering processes and system-level assurance rather than direct communication protocols. ISO has developed a variety of standards to aid in the adoption of V2X communication systems and their interoperability. Some of the key ISO standards related to V2X communication are:

- *ISO 26,262*: This standard defines the functional safety requirements for road vehicles and provides a framework for ensuring the safe operation of V2X communication systems [83].
- *ISO 21,434*: This standard provides guidelines for the cybersecurity of road vehicles and covers various aspects of V2X communication security, such as threat modeling, risk management, and security measures [84].
- *ISO 17,987*: Unlike the previous two standards, ISO 17,987 is focused on in-vehicle communication networks, providing specifications for low-speed communications within a vehicle [85]. While it contributes indirectly to V2X security by ensuring robust internal network communications, it is not designed for external V2X interactions such as V2V, V2I, or V2N.

2.4 ITS-G5/DSRC

ITS-G5/DSRC is a wireless communication technology intended to facilitate communication between vehicles and other road users [86]. It utilizes a dedicated frequency band and Wi-Fi-based communication to provide low-latency, high-reliability communication over a range of approximately 1000 m. ITS-G5/DSRC has built-in security features, making it a secure option for V2V communication [74].

One of its advantages is a dedicated spectrum for vehicular communication, which reduces the risk of interference from other devices. Another benefit is its low latency, which makes it suitable for real-time applications such as collision avoidance and traffic manage-

ment. Moreover, it is widely used in the US, leading to a large installed base of devices and infrastructure. These substantial advantages make it a promising solution for the connected vehicle market [87, 88] and [89]. However, it comes with disadvantages. One limitation is its limited range, which may not be sufficient for large-scale deployments or rural areas. Another drawback is the limited data rate, which may restrict the types of applications that can be supported. Additionally, it has limited global deployment, meaning that compatibility issues may arise when traveling between countries.

2.5 C-V2X

C-V2X is used predominantly for V2V and V2I communication, with applications that include collision avoidance, traffic management, and road safety [86]. It is supported by organizations like 3GPP, a consortium of telecommunications standards organizations. Compared to ITS-G5/DSRC [86], C-V2X has higher latency and depends on the security offered by the current cellular network architecture [74].

The technology chosen will depend on the specific demands and requirements of a given scenario [90]. One of its main advantages is its wide global deployment, making it a suitable choice for large-scale and multi-national deployments. Another benefit is its high data rate, which enables the support of data-intensive applications such as video streaming and high-definition mapping. Additionally, it offers integration with the cellular network, allowing seamless communication with other devices, including smartphones and infrastructure. However, it also has some disadvantages. One drawback is the potential for interference from other devices that use the same frequency bands. Another concern is that latency may be higher than dedicated vehicular technologies, such as ITS-G5/DSRC and Wireless Access for Vehicular Environments (WAVE), due to the longer signal travel distance. Additionally, it comes with higher costs compared to dedicated vehicle technologies, primarily due to the need for additional infrastructure and equipment.

2.6 WAVE

WAVE, the protocol for V2X communication, facilitates communication between vehicles and infrastructure such as roadside units (RSU) and traffic lights [91]. The protocol offers low-latency, high-reliability communication, and protection against unauthorized access [92]. It is also used for incident notification, traffic information dissemination, and emergency alerts. It is utilized by organizations such as the IEEE and the European Telecommunications Standards Institute (ETSI) [41].

The advantage of this technology is that it facilitates both broadcast and unicast communication. Vehicles have the ability to broadcast safety messages to all other nearby vehicles, increasing everyone's awareness of road conditions and potential hazards. Additionally, it can send signals to specific geographic locations, making it suitable for traffic control or personalized warnings. The geocasting feature enhances the effectiveness of V2X communication in particular scenarios. However, a notable disadvantage is the specialized frequency band used for WAVE, which may encounter issues related to spectrum allocation and regulatory constraints, especially as the demand for wireless communication continues to rise. While the WAVE protocol performs well for limited V2X communication, it may face scalability issues when dealing with a large number of vehicles and an extensive road network.

2.7 LTE-V2X

It enables communication between automobiles and other road users by using existing long-term evolution (LTE) cellular networks [79]. LTE-V2X expands LTE's capabilities to provide low-latency, high-reliability communication for the market of connected vehicles. Vehicles and RSUs serve as communication nodes, enabling direct communication between vehicles, as well as between vehicles and infrastructure [93]. LTE-V2X is a flexible solution for the V2X market [94]. It is supported by 3GPP and is viewed as a technology that complements ITS-G5/DSRC and C-V2X. It has a higher latency than ITS-G5/DSRC and offers the advantage of leveraging existing LTE network infrastructure [6].

2.8 Limitations of Standards and Technologies in V2X Communications

The limitations of security standards in V2X communication include:

- *Complexity*: The V2X security standards are complex and require a high level of technical knowledge to implement effectively.
- *Interoperability*: Different V2X technologies, such as ITS-G5/DSRC, WAVE, and C-V2X, use different security protocols and standards, which can lead to compatibility issues between devices and networks.
- *Scalability*: The V2X security standards must be scalable to accommodate the growth in the number of connected vehicles, which is a challenge given the limited resources and processing power available in vehicles.
- *Vulnerability to attacks*: Despite the use of strong encryption algorithms and authentication mechanisms, V2X security standards are still vulnerable to various types of attack, such as eavesdropping, man-in-the-middle, and denial-of-service attacks.
- *Cost*: Implementing strong V2X security standards can be expensive, both in terms of hardware/software upgrades and in terms of ongoing maintenance and management costs.
- *Balance of security and performance*: Finding a balance between high levels of security and optimal performance is a challenge in V2X security standards, as strong security measures can have a negative impact on performance.

Cross-standard interoperability introduces challenges. Vehicles equipped with different standards may face authentication mismatches, certificate incompatibilities, or inconsistent message integrity verification. For example, a DSRC-equipped vehicle communicating with a 5G-V2X-enabled roadside unit may require protocol translation or hybrid security frameworks to ensure mutual trust. Furthermore, security policies optimized for one standard may increase latency or reduce reliability when applied in a mixed network. Therefore, ensuring the security of V2X communication requires collaboration between stakeholders, including vehicle manufacturers, technology providers, and government agencies. It is clear from our investigation of the existing V2X standards and the deployment of cutting-edge communication technologies that these components form the basis of effective V2X networks. However, protecting V2X systems against threats requires a thorough analysis of all possible issues and challenges that can compromise the reliability and integrity of communication. We explore the most important aspect of security threats in V2X and its classifications

in the section that follows. To develop effective countermeasures and protect the future of V2X communication, we want to fully understand the security landscape by identifying and analyzing possible threats and vulnerabilities within V2X environments. Section 2.2 categorizes the main security threats and attack vectors that exploit weaknesses in V2X communication and standards.

3 V2X Security Threats and its Classification

V2X networks face distinct security challenges that differ fundamentally from traditional enterprise systems due to their dynamic, distributed, and safety-critical nature. Unlike static enterprise environments, vehicles continuously join and leave the network, complicating authentication and trust management. Real-time, latency-sensitive applications such as collision warnings and emergency vehicle alerts impose strict constraints on security mechanisms, whereas enterprise systems can tolerate greater computational overhead. Moreover, V2X components are physically exposed, making them susceptible to tampering, GPS spoofing, and relay attacks, while enterprise infrastructures are usually housed in controlled environments. The coexistence of multiple communication standards introduces interoperability vulnerabilities, and security breaches can directly threaten human life rather than merely causing data or financial loss. Limited computational resources in onboard units further restrict the use of complex cryptographic operations, necessitating lightweight and efficient solutions. Additionally, emerging threats such as adversarial attacks on AI-enabled traffic systems and privacy risks in location sharing further distinguish V2X from conventional networks. These factors demonstrate why traditional enterprise security models cannot be directly applied to V2X, underscoring the need for distributed, latency-aware, and resilient security mechanisms. Therefore, a complete framework for identifying and analyzing potential security threats in V2X communication is essential. Therefore, a complete framework for the identification and analysis of various possible security threats in V2X communication is required. The authors in [95] conducted a survey on various threat modeling approaches and their processes. They evaluated the existing body of research, offering an overview of methodology, approaches, and best practices related to threat modeling [96]. conducted a comprehensive examination of the security credential management system (SCMS) used in vehicular communications, with a specific focus on identifying potential threats. The authors also identified and analyzed multiple threat scenarios and security vulnerabilities.

The authors in [97] develop a layered methodology for threat modeling that is specifically designed for systems operating on the 5G network. By using this layered strategy, the authors provide a framework to help create robust security measures to protect 5G-based systems. The work in [98] examined the fundamental elements of security and privacy in intelligent autonomous vehicles. It examined the challenges and problems of ensuring and protecting the privacy of autonomous vehicles, highlighting the growing significance of these aspects within the automotive sector. It emphasizes the need to implement strong security measures to protect vehicle systems and ensure the privacy of users. We provide a comprehensive methodology based on the STRIDE threat model to analyze and address security vulnerabilities in V2X communication [99]. The threat modeling process is of significant importance within the field of cybersecurity, as it serves the purpose of detecting

and addressing potential threats and vulnerabilities that may exist within various systems and applications. Spoofing [100], tampering [101], repudiation [102], information disclosure [103], denial of service [104], and elevation of privilege [105] are the six dimensions used to categorize threats in V2X. Each category serves as a representation of a particular type of attack and guides the discovery and evaluation of pertinent security issues.

The STRIDE model [105, 106] is well suited for V2X security challenges because:

- *Comprehensive coverage:* The STRIDE model covers a wide variety of potential threats, including attacks on authenticity, data integrity, privacy, availability, and system trustworthiness. It guarantees a thorough examination of the security environment in V2X communication.
- *Systematic approach:* The STRIDE model provides a methodical way to classify and examine threats. Researchers and practitioners in the field of security can analyze the distinct vulnerabilities related to V2X communication by looking at each category, which helps to determine the priority of security precautions and countermeasures.
- *Relevance to the V2X context:* The taxonomy of V2X security challenges presented in this survey is derived from a comprehensive review of the literature, encompassing both documented threats from prior studies and insights from threat modeling approaches applied to V2X systems. The unique characteristics and needs of V2X communication are nicely aligned with the STRIDE concept. The authors discuss the particular difficulties with vehicle centric communication, authentication, privacy issues, data manipulation, and the potential effects on safety and trust.
- *Established framework:* The STRIDE model is a well-known and commonly used threat modeling methodology. Its application in the study of V2X security offers a framework and common language for interaction and cooperation among academics, professionals, and decision makers.

Let us consider the following example in the V2X communication scenario, vehicle (A), vehicle (B) and RSU (C) as a case study for this threat modeling scenario. Node A sends a message to node B, which relays it to node C. The message is eventually delivered to node C. Figure 7 shows possible potential attacks on this communication that are categorized using the STRIDE threat model.

3.1 Authentication and Access Control (spoofing)

It is critical to recognize potential threats such as impersonation and unauthorized access to protect V2X communication from these attacks. The security, dependability, and integrity of V2X communication can be compromised by spoofing attacks. Reliable identity verification processes [107], secure key management [108], and robust authentication mechanisms [70] should be thoroughly discussed and effectively implemented to prevent spoofing attacks. These measures help mitigate the risks posed by spoofing and maintain a trustworthy and secure communication environment. A trust anchor refers to a reliable and trusted entity such as a certificate authority that serves as the root of trust within a security framework. It ensures the authenticity and integrity of messages exchanged in the V2X network.

The following attacks that are classified under 'authentication and access control' in V2X communication systems are listed below:

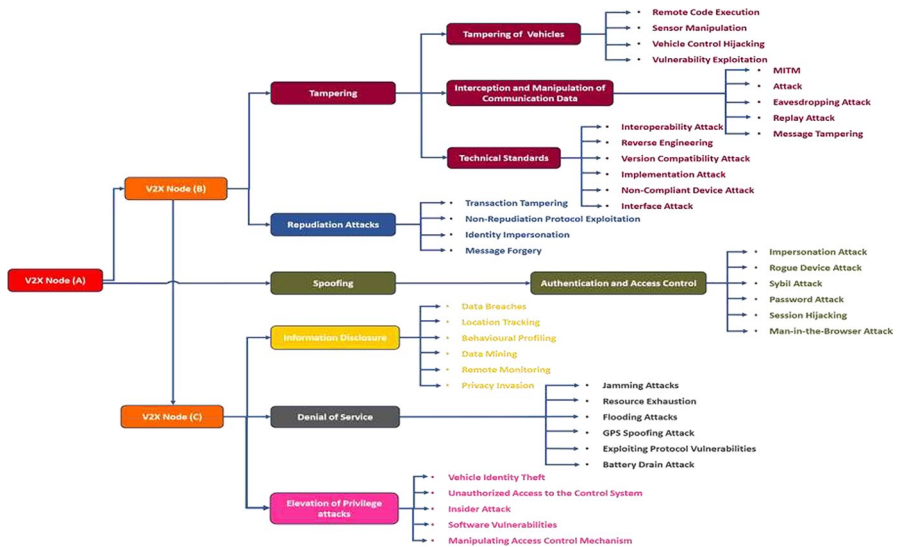


Fig. 7 Taxonomy of V2X security challenges based on the STRIDE threat model

- *Impersonation attacks*: An attacker impersonates a legitimate device and gains unauthorized access to the V2X network [109].
- *Password attacks*: An attacker uses techniques such as cracking or guessing to gain access to a device or network [110].
- *Sybil attacks*: The attacker creates numerous fictitious vehicle identities, often known as Sybil nodes. These fraudulent vehicles pose as authorized users of the V2X network and are capable of participating in a number of dishonest actions that jeopardize the integrity and security of the network [111].
- *Rogue device attacks*: An attacker can use a rogue device to gain unauthorized access to the V2X network [112].
- *Man-in-the-browser attack*: An attacker infects a browser with malware that allows them to intercept and alter communication between a device and the V2X network [113].
- *Session hijacking*: An attacker intercepts a session between a device and the V2X network and uses it to gain unauthorized access [114].

3.2 Interception and Manipulation of Communication Data (tampering)

Tampering threats can have an impact on the precision, dependability, and security of V2X messages. Countermeasures such as cryptographic protocols [69], digital signatures [68], and integrity [115] checks aid in detecting and preventing tampering attempts. These safeguards limit the risks of data manipulation while also providing a secure and reliable V2X communication environment.

- *Man-in-the-middle attack (MITM)*: An attacker intercepts and alters the communication between two devices without the knowledge of either party [116].

Table 3 Review of security attacks on V2X nodes

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[147]	To evaluate the performance of LTE and IEEE 802.11 p for vehicular networking	Performance evaluation	Compared the performance of LTE and IEEE 802.11 p for vehicular networking.	Performance evaluation of LTE and IEEE 802.11 p.	Provides information on the performance of both technologies	Limited to performance evaluation, may not cover other aspects.
[148]	To evaluate the impact of attacker strategies on man-in-the-middle attacks in vehicular ad-hoc networks	Simulation and analysis	Evaluated different attacker strategies and their impact on the network.	Evaluation of the impact of attackers' strategies	Provides insights into the impact of attacker strategies on vehicular networks	Limited to evaluating man-in-the-middle attacks and their impact on the network.
[109]	To investigate security breaches during handover in 5G-V2X communications	Simulation and analysis	Identified potential security breaches during handover in 5G-V2X networks.	Identification of security breaches during handover in 5G-V2X communications	Provides insights into the vulnerabilities of 5G-V2X networks during handover	Limited to investigating security breaches during handover in 5G-V2X networks.
[149]	To identify and discuss cybersecurity challenges in vehicular communications	Literature review and analysis	Identified and discussed various cybersecurity challenges in vehicular communications.	Comprehensive overview of cybersecurity challenges in vehicular communications	Provides insights into the vulnerabilities and issues in vehicular communications	Does not include empirical experiments or case studies to validate the identified challenges.
[150]	To analyze the techno-economic aspect of man-in-the-middle attacks	Economic analysis	Explored the economic impact of man-in-the-middle attacks.	Provides insight into the economic implications of man-in-the-middle attacks.	Highlights the economic considerations in addressing and mitigating such attacks.	May not cover technical details or specific attack scenarios.

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[151]	To investigate security measures against man-in-the-middle attacks in vehicular ad-hoc networks	Literature review and analysis	Explored the concept of man-in-the-middle attacks in vehicular Ad hoc networks.	Provides an overview of the security challenges and counter-measures for man-in-the-middle attacks in vehicular ad-hoc networks.	Helps to understand the importance of securing vehicular ad-hoc networks against such attacks.	May not provide a comprehensive analysis of all existing security measures.
[152]	To detect low-rate replay-based injection attacks on in-vehicle networks	Experimental setup and data collection	Identified vulnerabilities in in-vehicle networks to low-rate replay-based injection attacks.	Proposed a detection algorithm for low-rate replay-based injection attacks.	Effective in detecting low-rate replay-based injection attacks.	Evaluation may have been limited to specific scenarios or datasets.
[153]	Distributed platooning control of automated vehicles subject to replay attacks based on proportional integral observers	Proportional integral observers	Proposed a distributed platooning control scheme resilient to replay attacks	Addressed the issue of replay attacks in distributed platooning control	Enhanced the security of automated vehicle platooning	Limited to proportional integral observer-based approach
[154]	Machine learning based detection of replay attacks in VANET	Machine learning techniques	Proposed a machine learning-based approach for detecting replay attacks in VANET	Improved the detection of replay attacks in vehicular networks	Effective in identifying and mitigating replay attacks	Relies on the availability of sufficient training data
[155]	Pseudonym Tracking Certificateless Aggregate Signature Scheme for Preventing Replay Attacks in a Platoon of Vehicles	Certificateless aggregate signature scheme	Proposed a pseudonym tracking certificateless aggregate signature scheme for preventing replay attacks in vehicle platoons	Addressed the security challenges of replay attacks in platoon-based systems	Provided an efficient and secure mechanism for preventing replay attacks	Enhanced security and privacy in platoon-based vehicle systems

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[156]	Replay Attack and Defense of Electric Vehicle Charging on GB/T 27,930–2015 Communication Protocol	GB/T 27,930–2015 communication protocol analysis	Investigated replay attack vulnerabilities and proposed defense mechanisms for electric vehicle charging based on the GB/T 27,930–2015 communication protocol	Identified vulnerabilities and proposed countermeasures for replay attacks in EV charging systems	Improved the security of electric vehicle charging infrastructure	Addressed a specific communication protocol and may not cover all possible attack scenarios
[157]	Jamming and Eavesdropping Defense Scheme Based on Deep Reinforcement Learning in Autonomous Vehicle Networks	Deep reinforcement learning	Proposed a defense scheme against jamming and eavesdropping attacks using deep reinforcement learning in autonomous vehicle networks	Mitigated the impact of jamming and eavesdropping attacks on autonomous vehicle networks	Enhanced the security and reliability of autonomous vehicle communications	Requires sufficient computational resources and training data for deep reinforcement learning
[158]	Intermittent jamming for eavesdropping defense in WAVE-based vehicular networks	Intermittent jamming technique	Proposed an intermittent jamming scheme for defending against eavesdropping attacks in WAVE-based vehicular networks	Reduced the effectiveness of eavesdropping attacks and improved the security of vehicular networks	Provided a practical approach for enhancing the security of WAVE-based vehicular networks	Intermittent jamming may affect the quality of service and require careful parameter tuning
[159]	Secrecy performance for reconfigurable intelligent surfaces(RIS)-based integrated satellite vehicle networks with a unmanned aerial vehicle (UAV) relay and MRC eavesdropping	RIS-based satellite vehicle network with UAV relay and maximal ratio combining (MRC) eavesdropping analysis	Evaluated the secrecy performance of a RIS-based integrated satellite vehicle network with a UAV relay and MRC eavesdropping	Investigated the impact of RIS, UAV relay, and MRC eavesdropping on the secrecy performance of satellite vehicle networks	Improved the understanding of the secrecy performance in such networks	Provided insights for designing secure RIS-based integrated satellite vehicle networks

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[160]	GPS location spoofing attack detection for enhancing the security of autonomous vehicles	GPS location spoofing attack detection technique	Proposed a GPS location spoofing attack detection mechanism to enhance the security of autonomous vehicles	Detected GPS location spoofing attacks and improved the security of autonomous vehicles	Provided a practical approach for detecting GPS location spoofing attacks in autonomous vehicles	Enhanced the security and reliability of autonomous vehicle navigation
[161]	Selection of Areas for Effective Global navigation satellite system (GNSS) Spoofing Attacks to a Vehicle-mounted Multi Sensors Fusion (MSF) System based on Scenario Classification Models	Selection of effective areas for GNSS spoofing attacks based on scenario classification models	Proposed a method to select areas for effective GNSS spoofing attacks on a vehicle-mounted MSF system using scenario classification models	Identified areas with high vulnerability to GNSS spoofing attacks in a vehicle-mounted MSF system	Improved the understanding of GNSS spoofing attack scenarios and their impact on vehicle-mounted MSF systems	Provided insights for designing robust and secure GNSS systems for vehicles
[162]	Novel approach of spoofing attack in VANET location verification for non-line-of-sight (NLOS)	Novel approach for spoofing attack in VANET location verification for NLOS scenarios	Proposed a novel approach for spoofing attacks in VANET location verification under NLOS scenarios	Demonstrated the feasibility of spoofing attacks in VANET location verification under NLOS scenarios	Introduced a new perspective on spoofing attacks in VANETs	Increased awareness of the security vulnerabilities in VANET location verification
[163]	Localizing spoofing attacks on vehicular GPS using V2V communications	Localizing spoofing attacks on vehicular GPS using vehicle-to-vehicle communications	Proposed a method to localize spoofing attacks on vehicular GPS by leveraging vehicle-to-vehicle communications	Localized the source of GPS spoofing attacks in vehicular networks using vehicle-to-vehicle communications	Improved the ability to identify the source of GPS spoofing attacks in vehicular networks	Only enhanced the security and reliability of GPS-based navigation in vehicular networks

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[164]	Machine learning approach of hybrid KSVN algorithm to detect DDoS attack in VANET	Hybrid KSVN algorithm with machine learning for DDoS attack detection in VANET	Proposed a machine learning-based hybrid KSVN algorithm for detecting DDoS attacks in VANET	Detected and mitigated DDoS attacks in VANET using the hybrid KSVN algorithm	Introduced a new approach for detecting DDoS attacks in VANET using machine learning	Enhanced the resilience and security of VANETs against DDoS attacks
[165]	Resilient distributed event-triggered platooning control of connected vehicles under denial-of-service attacks	Resilient distributed event-triggered platooning control of connected vehicles under DoS attacks	Proposed a resilient distributed event-triggered platooning control mechanism for connected vehicles under DoS attacks	Achieved resilient platooning control in connected vehicles under DoS attacks	Improved the resilience and reliability of platooning control in connected vehicles under DoS attacks	Only improved connected vehicles platooning systems security
[166]	Privacy-preserving cloud establishment and data dissemination scheme for vehicular cloud	Privacy-preserving cloud establishment and data dissemination scheme for vehicular cloud	Proposed a privacy-preserving cloud establishment and data dissemination scheme for vehicular cloud	Ensured privacy preservation in cloud-based services for vehicular networks	Provided a practical approach for preserving privacy in vehicular cloud environments	The security and privacy vehicular cloud services is only enhanced
[167]	A scalable blockchain based trust management in VANET routing protocol	Scalable blockchain-based trust management in VANET routing protocol	Proposed a scalable blockchain-based trust management mechanism for VANET routing protocols	Improved the trust management in VANET routing protocols using blockchain technology	Enhanced the security and reliability of VANET communication through trust management	Provided a scalable solution for trust management in VANET routing protocols

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[168]	Jamming attack in vehicular networks: adaptively probabilistic channel surfing scheme	Adaptively probabilistic channel surfing scheme for mitigating jamming attacks in vehicular networks	Proposed an adaptively probabilistic channel surfing scheme to mitigate jamming attacks in vehicular networks	Mitigated the impact of jamming attacks on vehicular networks through adaptively probabilistic channel surfing	Provided a practical approach for improving the resilience of vehicular networks against jamming attacks	Concentrated the security and reliability of vehicular communication in the presence of jamming attacks
[169]	Jamming attack model and detection method for beacons under multichannel operation in vehicular networks	Jamming attack model and detection method for beacons in multichannel vehicular networks	Proposed a jamming attack model and detection method for beacons in multichannel vehicular networks	Modeled jamming attacks and developed a detection method for beacons in multichannel vehicular networks	Improved the understanding of jamming attacks and their detection in multichannel vehicular networks	Enhanced the security and reliability of beacon communication in vehicular networks
[170]	Jamming attack detection in a pair of radio frequency (RF) communicating vehicles using unsupervised machine learning	Jamming attack detection using unsupervised machine learning in RF communicating vehicles	Proposed a jamming attack detection mechanism using unsupervised machine learning in RF communicating vehicles	Detected jamming attacks in RF communicating vehicles using unsupervised machine learning	Provided a practical approach for detecting jamming attacks in RF communication between vehicles	only focused the security and reliability of RF communication in vehicular networks

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[103]	Security and privacy in V2X communications: How can collaborative learning improve cybersecurity?	Collaborative learning to improve security and privacy in V2X communications	Proposed a collaborative learning-based approach to enhance security and privacy in V2X communications	Improved the security and privacy of V2X communications through collaborative learning	Provided a novel approach for enhancing cybersecurity in V2X communications	Enhanced the resilience and reliability of V2X communication systems
[171]	5G V2X: Gearing up for security and privacy	Security and privacy considerations for 5G V2X services	Discussed security and privacy considerations for 5G V2X services	Identified security and privacy challenges and proposed solutions for 5G V2X services	Provided insights into the security and privacy aspects of 5G V2X services	Enhanced the understanding of security and privacy in the context of 5G V2X services
[172]	A privacy-preserving and verifiable federated learning framework for V2X environments using blockchain and Zero-Knowledge Succinct Non-Interactive Argument of Knowledge(zkSNARK)	Privacy-preserving and verifiable federated learning framework for V2X environments using blockchain and zkSNARKs	Proposed a privacy-preserving and verifiable federated learning framework for V2X environments using blockchain and zkSNARKs	Ensured privacy preservation and verifiability in federated learning for V2X environments using blockchain and zkSNARKs	Provided a practical approach for preserving privacy and enhancing trust in V2X federated learning	the privacy and trustworthiness of V2X federated learning system is only enhanced systems
[173]	Matrix-based dynamic authentication with conditional privacy preservation for vehicular network security	Proposed a matrix-based dynamic authentication scheme with conditional privacy-preservation for vehicular network security	Enhanced the security and privacy of vehicular network communication	Developed a matrix-based dynamic authentication scheme with conditional privacy-preservation for vehicular network security	Improved security and privacy in vehicular network communication	Focused on a specific authentication scheme and may not cover all aspects of vehicular network security

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[174]	Survey on existing authentication issues for cellular-assisted V2X communication	Conducted a survey on existing authentication issues in cellular-assisted V2X communication	Identified the authentication issues in cellular-assisted V2X communication	Provided an overview of existing authentication issues in cellular-assisted V2X communication	Comprehensive review of authentication issues in cellular-assisted V2X communication	Focused on authentication issues and may not cover all aspects of V2X communication
[175]	A V2X Access Authorization Mechanism based on Decentralized ID (DID) and Verifiable Credentials (VC)	Proposed a V2X access authorization mechanism based on DID and VC	Enhanced the security and efficiency of access authorization in V2X communications	Developed a V2X access authorization mechanism based on DID and VC	Improved security and efficiency in V2X access authorization	Focused on access authorization mechanism and may require further evaluation in real-world scenarios
[176]	Post-Quantum PHY-Layer Authentication for Secure Initial Access in V2X Communications	Proposed a post-quantum PHY-layer authentication scheme for secure initial access in V2X communications	Improved the security of initial access in V2X communications	Developed a post-quantum PHY-layer authentication scheme for secure initial access in V2X communications	Enhanced security in V2X communications	Only focused on PHY-layer authentication and may not cover all aspects of V2X security
[70]	C-V2X equipment identification management and authentication mechanism	Proposed a C-V2X equipment identification management and authentication mechanism	Enhanced the equipment identification and authentication in C-V2X communications	Developed a C-V2X equipment identification management and authentication mechanism	Improved equipment identification and authentication in C-V2X communications	Focused on C-V2X equipment and may not cover all aspects of V2X communication security

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[177]	AVoD: Advanced Verify-on-Demand for efficient authentication against DoS attacks in V2X communication	Proposed an advanced Verify-on-Demand (AVoD) scheme for efficient authentication against DoS attacks in V2X communication	Improved the efficiency and security of authentication against DoS attacks in V2X communication	Developed an AVoD scheme for efficient authentication against DoS attacks in V2X communication	Efficient authentication against DoS attacks in V2X communication	Concentrated on DoS attacks and may not cover all aspects of V2X communication security
[178]	Linux-stack based V2X framework that need to hack connected vehicles	Investigated vulnerabilities in a Linux-stack based V2X framework	Identified vulnerabilities in a Linux-stack based V2X framework	Highlighted the vulnerabilities in a Linux-stack based V2X framework	Identified potential security risks in a Linux-stack based V2X framework	vulnerabilities in a specific V2X framework and may not cover all aspects of V2X communication security
[179]	Improving ITS-G5 Cybersecurity Features Starting from Hacking IEEE 802.11p V2X Communications	Explored vulnerabilities and proposed improvements in ITS-G5 cybersecurity features	Identified vulnerabilities in ITS-G5 cybersecurity features and proposed improvements	Proposed improvements in ITS-G5 cybersecurity features starting from the vulnerabilities found	Identified potential security improvements in ITS-G5 cybersecurity features	Only focused on ITS-G5 and may not cover all aspects of V2X communication security
[100]	Integrated sensing and communication for joint GPS spoofing and jamming detection in vehicular V2X networks	Developed an integrated sensing and communication for joint GPS spoofing and jamming detection in vehicular V2X networks	Enhanced the detection of GPS spoofing and jamming attacks in vehicular V2X networks	Developed an integrated sensing and communication system for joint GPS spoofing and jamming detection in vehicular V2X networks	Improved detection of GPS spoofing and jamming attacks in vehicular V2X networks	Concentrated on GPS spoofing and jamming attacks and may not cover all aspects of V2X communication security

Table 3 (continued)

Reference number	Aim and scope of the study	Methodology/ Technique used	Result or Outcomes	Contribution	Benefits	Limitations
[146]	A Pseudorange-based GPS Spoofing Detection Using Hyperbola Equations	Proposed a pseudorange-based GPS spoofing detection method using hyperbola equations	Improved the detection of GPS spoofing attacks using pseudorange measurements and hyperbola equations	Developed a pseudorange-based GPS spoofing detection method using hyperbola equations	Improved detection of GPS spoofing attacks using pseudorange measurements	Focused on GPS spoofing detection using a specific method and may not cover all aspects of V2X communication security
[180]	Multi-radio V2X communications interoperability through a multi-access edge computing (MEC)	Investigated multi-radio V2X communications interoperability through MEC	Explored the potential of MEC for enhancing multi-radio V2X communications interoperability	Highlighted the potential of MEC for multi-radio V2X communications interoperability	Identified potential benefits of MEC for V2X communications	multi-radio communications and MEC may not cover all aspects of V2X communication security

- *Replay attack*: An attacker intercepts a valid message, records it, and then retransmits it at a later time to impersonate the original sender [117].
- *Eavesdropping*: An attacker listens to communication between two devices without altering or blocking the communication [118].
- *Message tampering*: An attacker alters the content of a message transmitted between two devices [119].

These attacks pose a significant risk to the security of V2X communication systems and can result in compromise of sensitive data, interruption of communication, and loss of confidence in the system.

3.3 Tampering of Vehicles

To maintain vehicle safety, potential threats from software, firmware, or system vulnerabilities must be identified. This includes thinking about attacks that could compromise vehicle control, change features, or steal data. Vehicle hacking has serious implications for security, privacy, and the V2X ecosystem. To mitigate these potential risks, it is critical to incorporate security solutions such as safe coding, over-the-air upgrades, intrusion detection systems, and secure firmware updates. These precautions reduce the risk of hacking and provide a safe environment for automobiles and the V2X ecosystem.

- *Remote code execution*: An attacker executes code remotely on a vehicular system, potentially compromising its safety and security [120].
- *Vehicle control hijacking*: An attacker can gain control of vehicle systems and manipulate its movements, which can lead to safety hazards [121].
- *Sensory manipulation*: An attacker can manipulate the sensory inputs of a vehicle, which can lead to incorrect readings and decisions [122].
- *Vulnerability exploitation*: An attacker can exploit vulnerabilities in a vehicle system to gain unauthorized access or cause harm [123].

For the safe and reliable operation of these systems, preventing these types of attack and ensuring the security of vehicles connected to the V2X network are essential. This can be accomplished by using secure protocols, software and firmware updates, and regular monitoring and auditing of vehicle systems.

3.4 Threats To Technical Standards (tampering)

Attackers can take advantage of interoperability problems or vulnerabilities introduced by faulty implementations. Weak technical standards create a negative impact on communication efficiency and safety.

- *Inter-operability attacks*: An attacker can exploit incompatibility issues in the technical standards of V2X potentially leading to communication failures [124].
- *Implementation attacks*: An attacker uses weaknesses in the implementation of V2X technical security standards, leading vulnerabilities in security and communication failures [125].
- *Reverse engineering*: An attacker can reverse engineer the V2X technical standards, potentially exposing sensitive information or discovering vulnerabilities [126].
- *Noncompliant device attacks*: An attacker can use non-compliant devices to gain unauthorized access to the V2X network, possibly resulting in security breaches [127].
- *Interference attacks*: An attacker can interfere with V2X communication by transmitting signals that disrupt the intended communication [128].
- *Version compatibility attacks*: An attacker can exploit the differences between various versions of the V2X technical standards, potentially leading to interoperability issues and communication failures [129].

To mitigate these threats, secure protocols must be implemented, standards must be followed, security testing must be performed, and regular upgrades must be performed. These techniques mitigate the hazards associated with technical standards, thus improving communication dependability and safety in the V2X ecosystem. Recent studies emphasize the use of automated protocol verification tools and cross-standard testing environments to detect vulnerabilities early.

3.5 Repudiation Attacks

When an entity denies its involvement or action in a communication or transaction, possible repudiation assaults may take place. Within the V2X environment, these attacks generate responsibility and trust issues. Some examples of repudiation attacks in V2X communication are:

- *Message forgery*: Messages transmitted by a legitimate entity can be forged or fabricated by an attacker. As a result, there may be disagreements or difficulties in proving the integrity of the communications if the attacker claims that they were not involved in the message exchange [130].
- *Identity impersonation*: By impersonating a trustworthy entity, the attacker can deny being involved in the interaction, raising doubts about the authenticity and source of the communications [131, 132].
- *Non-repudiation protocol exploitation*: Non-repudiation protocols are intended to stop participants from claiming that they were not involved in a conversation. Attackers could use flaws or vulnerabilities in these protocols to fraudulently deny their participation, which might lead to legal or contractual issues [133, 134] and [135].
- *Transaction tampering*: Attackers may manipulate transaction data or change the conditions of a transaction that involves financial or contractual transactions. Later, they deny their involvement or claim that the transaction was not allowed, resulting in conflicts or difficulties that require the proof of the legitimacy and integrity of the transaction [136].

Digital signatures, PKI, and blockchain-based tamper-proof logs are effective in ensuring non-repudiation. Recent V2X research suggests combining lightweight cryptographic protocols with distributed logging to maintain accountability without impacting latency-sensitive communications.

3.6 Elevation of Privilege Attacks

It occurs when an unauthorized individual or vehicle obtains access to resources with more privileges or degrees of control than they are authorized. These attacks take advantage of system vulnerabilities to increase the privileges that allow them to control, manipulate or interfere with V2X communication without authorization. Possible Elevation of Privilege attacks include:

- *Vehicle identity theft*: An effort by an unauthorized entity or malicious actor to steal or pretend the identity of a valid vehicle within the V2X network is called a vehicle identity theft attack. The attacker can obtain higher access and privilege levels by posing as a trustworthy vehicle, giving them the ability to trick other vehicles or infrastructure elements, acquire access to restricted resources, or carry out malicious activities [137, 138].
- *Unauthorized access to control systems*: Attackers use vulnerabilities in electronic control units (ECUs) found in vehicle control systems to obtain unauthorized access. By

breaking into these systems, an attacker takes over crucial operations, changes sensor data, or injects malicious commands, posing a risk to user safety or approving unpermitted behavior [139].

- *Insider attacks*: Insider attacks occur when legitimately authorized people or organizations misuse their privileges to gain access beyond what is intended. This can happen when people misuse their access privileges or conspire with outside adversaries to seize control or manipulate V2X communication for their own benefit or malicious purpose [140, 141] and [142].
- *Exploiting firmware or software vulnerabilities*: To achieve higher privileges, attackers take advantage of vulnerabilities in the V2X communication stack, software, or firmware of the target vehicle. The attacker thus increases their privileges, compromises the integrity of the system, or carries out unauthorized actions [143, 144].
- *Manipulation of access control mechanisms*: Attackers may try to trick or get around access control systems, including secure boot procedures and user authentication systems. They are able to obtain enhanced privileges, pose as reliable sources of information or resources, or impersonate trustworthy entities by getting beyond these protections [100, 145] and [146].

Multi-layered approaches including role-based access control and frequent security audits can prevent privilege escalation. The AI-based behavioral monitoring to detect unusual privilege elevation attempts in real time.

3.7 DoS Attack

DoS attacks in V2X networks disrupt communication, degrade QoS, and may endanger human safety by preventing timely delivery of critical messages. Several possible attacks are feasible in V2X communication, which falls under the Denial-of-service category of the STRIDE threat model.

- *Jamming attacks*: The wireless communication channels that are used for V2X communication are disrupted during a jamming attack. The attacker prevents valid V2X communications from being received by sending out a powerful interference signal. V2X communication experiences a temporary or permanent denial of service as a result of jamming attacks [181, 182].
- *Resource exhaustion*: The attacker consumes available resources, including memory, computing power, and network bandwidth, thus overloading the system with an excessive number of requests, which in turn makes the system unresponsive. Legitimate V2X participants experience a denial of service due to resource exhaustion [183].
- *Flooding attacks*: Attackers using flooding attacks transmit a large amount of malicious or pointless messages to the V2X communication network. The network infrastructure is overwhelmed by this onslaught of communications, which uses up processing power and bandwidth. As a result, genuine communications are lost or delayed, disrupting V2X connectivity and resulting in denial of service [184, 185].
- *GPS spoofing*: Global positional system (GPS) signals are frequently used in V2X communication to provide precise positional data. In a GPS spoofing attack, an attacker

broadcasts bogus GPS signals to trick infrastructure or vehicle components into thinking they are somewhere else. This results in inaccurate calculations, errors in coordination, and interruptions in V2X communication.

- *Exploiting protocol vulnerabilities:* Attackers can take advantage of vulnerabilities or weaknesses in V2X communication protocols. The attacker can impede the smooth operation of the V2X communication system by sending specially crafted messages or taking advantage of protocol weaknesses. These actions can result in protocol failures, protocol-level errors, or complete protocol failures [186].
- *Battery drain attacks:* Attacks that target the power source of V2X devices, such as vehicles or infrastructure elements, are known as battery drain attacks. This attack shortens the battery life of a device, limiting its capacity to function and participate in V2X communication. Affected devices experience a denial of service as a result, or they may lose the capacity to participate in the V2X ecosystem [187, 188].

DoS attacks can be mitigated using a multifaceted strategy that includes traffic analysis and anomaly detection, strong access control mechanisms, network segmentation, encryption, intrusion detection and prevention systems, and redundancy in the communication infrastructure [189].

3.8 Attacks on Privacy Concerns (information disclosure)

Information disclosure refers to the unintentional exposure of private information exchanged between vehicles and components of the V2X network infrastructure. To preserve user privacy, pseudonymity, which involves the use of temporary or false identifiers instead of real identities, is often employed in V2X systems. Vehicles can periodically change their pseudonyms to prevent tracking by unauthorized entities. This vulnerability raises concerns about privacy and confidentiality, as it may allow private information to be revealed to unauthorized parties or adversarial actors. To mitigate the risks associated with information disclosure, privacy-enhancing techniques such as data anonymization, encryption, and secure communication protocols should be implemented. Some of the attacks that fall under 'privacy concerns' in V2X communication systems include:

- *Location tracking:* An attacker overhears the communication of V2X to track the location of vehicles and potentially identify the people associated with them [174].
- *Behavioral profiling:* An attacker can use V2X communication to gather information about an individual's driving habits and behavior, which can be used for malicious purposes [190].
- *Data Mining:* An attacker extracts personal and sensitive information from V2X communication data, such as credit card information or social security numbers [191].
- *Remote monitoring:* An attacker uses V2X communication to remotely monitor and control vehicles, potentially exposing sensitive information [192].
- *Privacy invasion:* In the context of V2X communication, privacy invasion refers to

situations in which the privacy and confidentiality of an individual's personal information and actions are at risk within the V2X network. When data are transferred between vehicles, infrastructure components, and other entities in V2X communication, privacy concerns may arise if the situation is not adequately controlled [193].

- *Data breaches*: An attacker can potentially breach the security of V2X communication systems and access sensitive information, leading to privacy violations [194].

3.9 Threat Mapping across V2X Communication Layers

Table 4 provides a concise and well-organized summary of security attacks, highlights important results, methodologies, and impact of the various security threats, and reviews the main findings of previous research. To provide a clearer understanding of the threat model, each attack type can be mapped to the corresponding V2X communication layer or component affected. V2X systems operate across multiple layers ranging from the physical layer (wireless channel) to the application layer (data exchange and services). Different threats exploit vulnerabilities specific to these layers. Table 4 presents a mapping of the major V2X attack types discussed in this paper to their affected layers and typical targets. This mapping highlights that V2X security threats are distributed across all communication layers. Physical layer attacks target the communication medium directly, while network and application layer threats often involve spoofing, data manipulation, or denial-of-service. A

Table 4 Mapping of V2X attack types to communication layers and components

Attack type	Primary layer affected	Targeted components	Typical impact
Spoofing (Authentication bypass)	Network/Application layer	OBU, RSU	False identity creation, misleading messages
Tampering/Message alteration	Data link/Network layer	Communication channel, RSU, Cloud server	Data integrity compromise
Vehicle tampering	Physical/Hardware layer	Electronic control unit (ECU), OBU	Unauthorized access, firmware modification
Repudiation attacks	Application layer	Cloud services, V2N components	Denial of actions, loss of accountability
Elevation of privilege	Network/Application layer	Access control systems	Unauthorized privilege escalation
DoS	Physical/Network layer	Wireless channel, RSU	Service disruption, increased latency
Privacy breach	Application layer	Cloud, Data aggregation platforms	Leakage of personal or vehicular data
Attacks on technical standards	All layers	Protocol stack (IEEE 1609, 3GPP)	Protocol exploitation, interoperability failure

multi-layered defense strategy that combines cryptographic protection, intrusion detection, and secure protocol design is therefore essential for achieving end-to-end V2X security. After identifying the key problems and challenges related to V2X security, it is essential to create robust safety measures for the communication environment. This section turns our attention to security methods to protect V2X. We try to protect V2X communication from potential threats by exploring various security strategies, authentication mechanisms, and encryption techniques. Section 4 presents cryptographic, protocol-based, and system-level security solutions that safeguard V2X communications.

4 Security Mechanisms to Protect V2X

There are several methods (Fig. 8) to provide security to V2X communication systems. As we explore the realm of V2X security, it is crucial to recognize that the following subtopics will thoroughly explore customized solutions. Each subtopic addresses a specific aspect of our customized security methodology, designed to mitigate vulnerabilities identified by analysis using the STRIDE model [195]. This study aims to investigate a range of security mechanisms tailored for V2X communication, including PKI, digital signatures, encryption, and certificate management. Authentication mechanisms like PKI, certificate management, digital signatures, and authentication authorization and accounting (AAA) ensure only legitimate entities communicate securely. Access control systems such as access control lists (ACLs) regulate permissions and prevent unauthorized privilege escalation. Encryption, virtual private network (VPN), and GPS-based authentication protect data confidentiality and location integrity [196]. At the network level, firewalls and IDS/IPS detect and block malicious activities. This categorization distinguishes PKIs overarching trust frame-

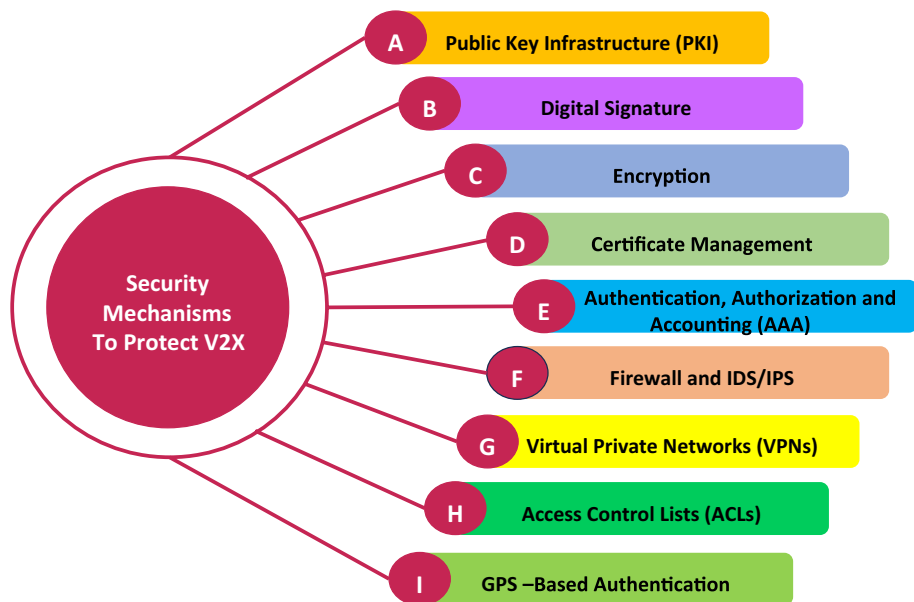


Fig. 8 Security mechanisms of V2X

work from operational components like certificate management and highlights how various mechanisms collectively enhance authentication, integrity, and resilience in V2X networks [197]. By examining these processes, we aim to better understand how they address the unique challenges of V2X communication. This investigation will demonstrate how these solutions can be integrated to establish a comprehensive defense strategy, effectively safeguarding the V2X ecosystem against a wide range of threats.

4.1 PKI

The PKI is a collection of protocols, procedures, and technologies used to manage the secure exchange of information over networks. It provides a secure means of communication between vehicles and other connected devices, such as roadside units and other vehicles, in the context of V2X security. Provides authentication and encryption services to ensure that data transmitted between devices is secure and tamper-proof in V2X communications [198]. The authentication process verifies the identity of each device to prevent unauthorized access, while encryption ensures confidentiality by encrypting data transmitted between devices [199]. According to ETSI, the V2X PKI consists of several key components:

- *Certificate authority (CA)*: The CA is a trusted third party in charge of issuing and revoking digital certificates for V2X network devices. Typically, the CA is a government agency or a private organization authorized to issue certificates according to established standards [200, 201].
- *Registration authority (RA)*: The RA has the responsibility to verify the authenticity of devices that request CA. Typically, the RA performs background checks on the owner or operator of the device to ensure that they are permitted to participate in the V2X network.
- *End entity (EE)*: The CA issues a digital certificate to the EE, which is a component of the V2X network. EE can be a vehicle, a component of the infrastructure, or any other device that participates in V2X communications.
- *Certificate revocation list (CRL)*: CRL is a list of certificates that have been revoked by the CA. EEs utilize the CRL to validate the validity of digital certificates.

Typically, the PKI structure for V2X communications is in accordance with the X.509 standard, which specifies the format for digital certificates. Digital certificates include information on the identity of the device, the public key associated with the device, and the duration of the validity of the certificate. It is used to authenticate the device's identity and establish secure communication channels between devices on the V2X network, as shown in Fig. 9. Devices use their private keys to sign and encrypt messages, and their public keys to validate the sender's identity and the integrity of the message [202, 203].

PKI in V2X communications provides multiple security advantages. It can be used to authenticate the sender and recipient of a message, thereby preventing spoofing and other attacks based on impersonation. Furthermore, PKI guarantees that a message has not been altered during transmission. By validating the sender's digital signature, the recipient can validate that the message has not been altered. Additionally, the PKI is used to encrypt communications, preventing unauthorized access to their contents. Additionally, it provides

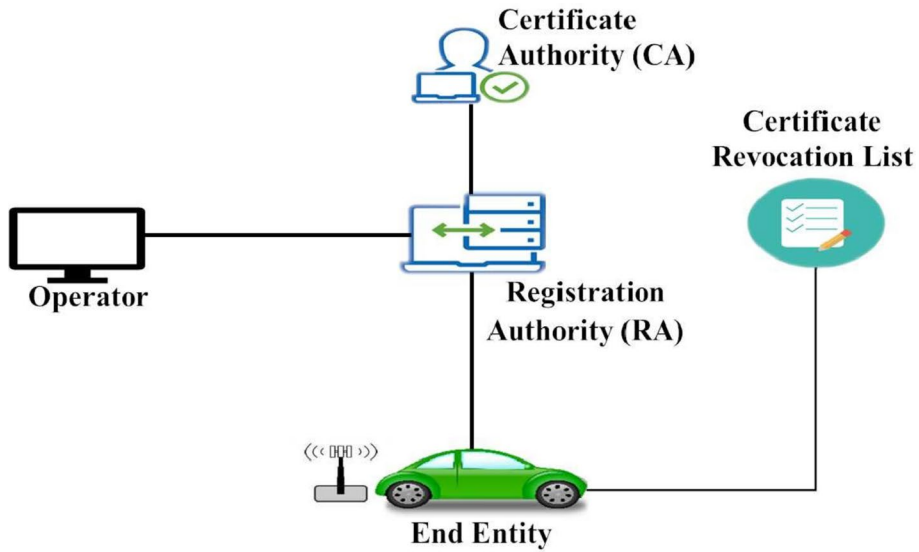


Fig. 9 Structure of PKI

evidence of the origin and receipt of the message, which can be invaluable for auditing and legal purposes. Furthermore, the PKI can be scaled to accommodate large-scale V2X communication systems, enabling secure communication for a variety of entities. V2X communication is based on standardized algorithms and protocols to ensure interoperability and compatibility between various systems and devices. PKI can also be used to support multiple applications in V2X communications, such as collision avoidance at intersections, cooperative adaptive cruise control, and emergency vehicle preemption.

At the same time, the implementation and management of PKI can be complicated, particularly in large-scale V2X communication systems, requiring specialized knowledge and dedicated resources. In addition, PKI introduces performance overhead because data encryption and decryption require substantial computing power, potentially influencing communication latency and bandwidth. The management of encryption keys can be particularly difficult, especially in large-scale V2X communication systems. PKI algorithms may also be susceptible to cryptanalysis and their security can be compromised by the development of cryptanalysis techniques or the discovery of algorithm flaws. In addition, PKI introduces a single point of failure by relying on a third-party CA to issue and manage digital certificates. When deciding,

whether to implement PKI in V2X communication systems, it is crucial to evaluate these pros and cons in light of the particular security and communication requirements of the system.

4.2 Digital Signature

A digital signature is a mathematical algorithm that verifies the authenticity and integrity of digital data. Encryption and hash functions are used to generate a digital signature. The sender generates a hash, which is a unique representation of the message data, and encrypts

it with his private key to generate a digital signature. Using the sender's public key, the recipient decrypts the signature and verifies the authenticity of the message by generating a hash of the received message. It then compared it to the decrypted signature (Fig. 10). It is commonly used for non-repudiation, which ensures that the sender of a message cannot afterward deny sending it [204, 205]. The most commonly used digital signature algorithms are [206, 207]:

- *Elliptic curve digital signature algorithm (ECDSA)*: ECDSA is a public-key cryptography algorithm that is based on the mathematics of elliptic curves. It is considered to be more secure than RSA for the same key size and is well suited for devices with limited resource [208, 209].
- *Digital signature algorithm (DSA)*: DSA is a public-key cryptography algorithm that is based on the mathematics of modular arithmetic. It is widely used for digital signature verification, particularly in secure data transmission protocols such as Secure Socket Layer (SSL) and Transport Layer Security (TLS) [210].
- *Hash-based message authentication code (HMAC)*: HMAC is a message authentication code that uses a cryptographic hash function in combination with a secret key [211, 212].

The type of algorithm used to generate digital signatures is determined by the security requirements of the V2X system and the available resources of the devices involved in the communication. Digital signatures provide numerous benefits for V2X communication. They provide a method for validating the integrity of the data, ensuring that it has not been altered or interfered with. The non-repudiation feature of digital signatures prevents senders from retracting V2X messages after they have been sent, ensuring accountability and facilitating dispute resolution. Moreover, digital signatures protect the data from illicit modifications or tampering by external parties. Furthermore, digital signatures provide protection

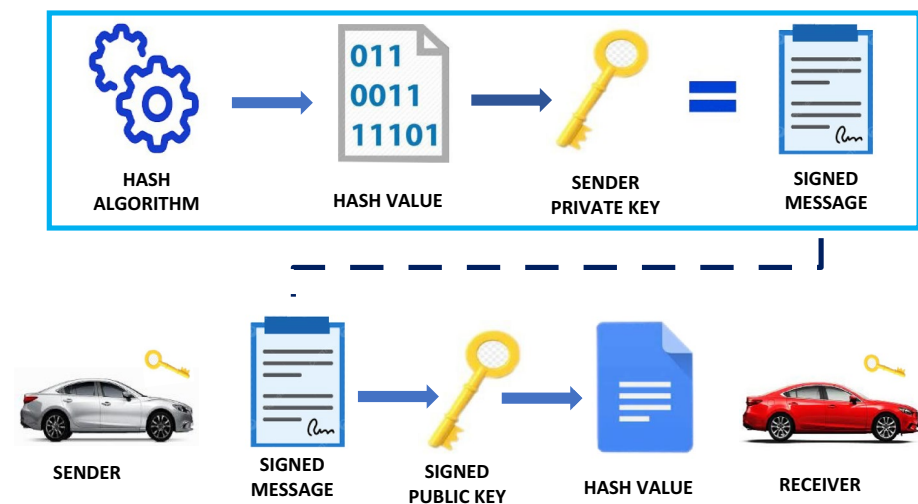


Fig. 10 Digital signature process

against replay attacks, in which assailants capture and replay V2X messages in an attempt to mislead participants. This is accomplished by limiting the validity periods of communications by including a timestamp in the signature, thereby rendering replay attacks ineffective.

While digital signatures offer significant advantages for V2X communication, there are also significant disadvantages to consider. First, digital signature operations require computational resources, which may influence the processing time and resource consumption of vehicles and other infrastructure components. In extensive V2X networks, managing public and private keys for digital signatures can be challenging. To mitigate key-related vulnerabilities, it is essential to meticulously plan key distribution, revocation, and renewal processes and implement stringent security measures. In addition, the use of digital signatures to verify the sender of V2X messages adds information to the messages themselves. Consequently, this can result in an increase in message size, requiring additional bandwidth, and potentially causing communication channel congestion. In time-sensitive V2X scenarios, such as collision avoidance, the time required for the verification of digital signature may also contribute to latency. In situations where safety is a top priority, it is imperative to reduce verification time.

4.3 Encryption

Encryption is used to protect against eavesdropping, tampering, and other security threats. In V2X communication, the data exchanged between vehicles and other entities is encrypted. Advanced encryption standard (AES), elliptic curve cryptography (ECC), and RSA are common encryption algorithms used for this purpose [213, 214]. The choice of encryption algorithm depends on factors such as the computing power of the devices involved, the intended level of security, and communication requirements, such as latency and bandwidth. Key management is a crucial component, as keys must be distributed and managed securely to prevent unauthorized access. PKI is typically used to issue and manage encryption keys to ensure secure key management. Here are some of the common encryption methods currently available for V2X:

- *AES*: AES is a symmetric key encryption algorithm that is widely used for secure data transmission. It supports key lengths of 128, 192, and 256 bits, providing a high level of security [214].
- *ECC*: ECC is an asymmetric key encryption algorithm that is well suited for resource-constrained devices used in V2X communication. It provides high security with relatively short key lengths compared to other encryption algorithms [215].
- *RSA*: RSA is a widely used public-key encryption algorithm that is based on the mathematical properties of large prime numbers. RSA is often used for digital signatures and key exchange in V2X communication [214].
- *Diffie-Hellman key exchange (DH)*: DH is a public-key encryption algorithm that is used for secure key exchange between two parties. DH allows two parties to agree on a shared secret key without exchanging the key in plain text [216].

Encryption techniques have several advantages that significantly improve the security, privacy, and integrity of shared data shared within the V2X network and provide substantial

benefits to V2X communication. Encryption facilitates the establishment of secure communication channels between vehicles and other infrastructure components by providing confidentiality, data privacy, and protection against tampering and replay attacks. The implementation of digital signatures combined with encryption strengthens the reliability of the V2X system by improving message authentication and verification. Encryption is implemented in V2X systems to help comply with data protection laws, protect sensitive data, and protect future-proof against emerging security threats. Although encryption techniques offer important security advantages for V2X communication, there are some disadvantages to take into account. These include increased message size, key management complexity, possible single points of failure, and computational cost. Furthermore, real-time applications may be affected by encryption due to increasing latency and interoperability issues between various V2X systems. To reduce security concerns, proper key management and maintenance of the infrastructure are essential. Despite these difficulties, encryption continues to be a crucial tool for maintaining data security and safe communication paths within the V2X network. The benefits of encryption can be fully realized with proper planning and execution, resulting in a secure and reliable environment for connected infrastructure and vehicles.

4.4 Computational Efficiency and Suitability in Low Latency Environments

Vehicular communication systems operate under stringent latency requirements, typically in the range of 10–100 ms, making the computational efficiency of cryptographic operations a crucial consideration. While asymmetric cryptography such as RSA and ECDSA provides strong security guarantees, their high computational cost can introduce delays in message authentication and verification. For instance, RSA-2048 consumes more processing time and energy compared to ECC, which achieves comparable security strength with smaller key sizes. Therefore, ECC-based approaches are widely preferred in V2X for their faster key generation and lower transmission overhead.

Symmetric encryption methods such as AES are suitable for securing bulk data transmissions due to their low computational delay, but they require efficient key management to avoid scalability issues. Lightweight cryptographic algorithms such as PRESENT, SPECK, and SIMON have also been explored to balance security with minimal latency in resource-constrained on-board units (OBU). To ensure both security and timeliness, hybrid schemes combining ECC for key exchange and AES for data encryption have shown promising results in recent studies [217]. These approaches achieve end-to-end confidentiality and authentication while maintaining latency within acceptable vehicular communication thresholds.

4.5 Certificate Management

A certificate is a digital document used to verify the identity of an entity, such as a vehicle or communication device, and to encrypt communication between entities [204] issued and verified by a CA [218]. It is used to authenticate and verify the identity of communicating entities in V2X security. Certificates include details such as the public key of the entity, its identity, and the validity duration of the certificate. The certificate management process in V2X includes the following steps:

- *Certificate request*: An entity, such as a vehicle or a communication device, generates a certificate request and sends it to the CA.
- *Certificate issuance*: The CA verifies the identity of the entity and issues a certificate if the identity is confirmed. The certificate is signed by the CA and contains the public key, identity, and other information of the entity.
- *Certificate distribution*: The certificate is securely distributed to the entity that requested it.
- *Certificate verification*: When two entities need to communicate, they verify each other's certificates to ensure that they are communicating with the correct entity. This step is important in preventing man-in-the-middle attacks.
- *Certificate renewal*: Certificates have a period of validity and when that period expires, the certificate must be renewed. The renewal procedure is comparable to the request procedure.

In V2X communication, certificate management has an extensive list of benefits, including improved security, data integrity, and non-repudiation. In addition to ensuring privacy preservation and establishing trust between entities, it scales well in massive networks. But handling certificates can be difficult and time-consuming, and the Certificate Authority might be a single point of failure. A thorough consideration of key management issues, initial setup efforts, and certificate revocation procedures must be given. Despite these issues, certificate management is still essential for building a secure and compliant V2X ecosystem that promotes secure communication and reliable interactions between connected vehicles and infrastructure components.

4.6 AAA

AAA is a security framework that provides three critical functions in V2X authentication, authorization, and accounting.

- *Authentication*: This function verifies the identity of communicating entities in V2X communication, ensuring that only authorized entities exchange data. Methods such as username/password, digital certificates, biometric authentication, or a combination of these [219] are used to do this.
- *Authorization*: This function specifies the access level and actions that an authenticated entity is allowed to perform. This is accomplished through the assignment of roles and permissions to entities according to their identity and intended use of the communication system [175, 219].
- *Accounting*: This function monitors and records the communication activities of V2X entities. This includes the nature of communication, its time and duration, and the amount of data transmitted. Accounting data can be used to invoice, audit and troubleshoot the communication system [70].

AAA provides numerous advantages for V2X communication, including improved security, granular access control, data integrity, and increased privacy protection. Prevents assailants and other malicious users from gaining access to the system by allowing only authorized users that allow access. The AAA authorization feature ensures the assignment of role-

based permissions to entities, enabling the enforcement of access policies. The accounting function provides a communication audit trail that helps in resource management and compliance with regulatory requirements. Despite its benefits, implementing AAA can be challenging, which can slow down or limit mobility-related tasks. Due to its dependence on a single AAA server, there is also the risk of a single point of failure.

4.7 Firewall and Intrusion detection/Intrusion Prevention System (IDS/IPS)

A firewall is a security device that monitors and controls network traffic based on predefined security protocols. It provides a barrier between internal and external networks in V2X security, preventing unauthorized access to the internal network and protecting it from external attacks [220]. IDS monitors network traffic and identifies suspicious activities that may indicate an attack. An intrusion prevention system (IPS) can take automatic measures to prevent the execution of malicious activities, such as blocking malicious traffic or isolating the compromised device. ID/IPS are utilized in V2X security to detect and prevent malicious attacks such as network intrusion, DoS, and other types of security threats [221, 222].

Firewalls and IDS/IPS are essential to create a stable and secure network environment for secure V2X communication. Firewalls serve as the initial line of defense, vigilantly monitoring and regulating the data flow between V2X entities and the external network. Firewalls prevent unauthorized access and protect against external threats through their ability to enforce access policies and filter network traffic based on predefined rules. In addition, they support application layer filtering and VPNs [223] for encrypted communication, thus increasing the security and privacy of V2X data exchange. In spite of their benefits, firewalls can be complicated to configure and administer and can have limitations, such as not being able to inspect encrypted traffic.

On the other hand, IDS/IPS systems offer continuous network traffic monitoring for suspicious activity or potential security breaches and real-time threat identification. IDS/IPS can identify known threats and anomalies using signature-based detection and behavioral analysis, enabling proactive and automated responses to reduce the impact of attacks [224]. Some sophisticated IDS/IPS solutions can inspect encrypted traffic, thereby improving the detection of hazards hidden within secure communications [225]. However, IDS/IPS systems may generate false positives and false negatives, necessitating careful calibration and maintenance to reduce these inaccuracies. Additionally, their resource requirements can influence network performance, particularly in environments with high V2X traffic. Integrating firewalls and IDS/IPS systems into an all-encompassing security strategy is crucial for establishing a secure V2X communication ecosystem. V2X networks can fight more effectively against a variety of cyber-threats by combining the network-level protection provided by firewalls with the real-time threat detection and response capabilities of IDS/IPS.

4.8 Virtual Private Networks

VPNs are frequently used in V2X security to provide secure communication over public networks, such as the Internet. VPNs allow V2X communication systems to securely exchange data between geographically dispersed or behind different firewall entities [226]. It offers a private network over a public network, such as the Internet, using encryption

and other security measures. VPNs are utilized in V2X security to secure communication between V2X devices, such as vehicles, roadside units, and cellular networks [227].

VPNs have numerous advantages. They provide end-to-end encryption, ensuring that all data exchanged between vehicles and infrastructure components remains confidential and secure against eavesdropping. By configuring VPNs to allow only authorized devices with valid credentials to access the V2X network, access control can be strengthened. Additionally, VPNs facilitate secure remote connectivity, allowing geographically separated V2X components to communicate over the Internet securely. In addition, VPNs are cost-effective because they do not require the deployment of dedicated communication infrastructure such as leased lines or microwave connections.

VPNs have a number of benefits for safe V2X communication, but there are also some drawbacks to take into account. The possible emergence of latency is one of the main worries. Data transmission delays caused by the encryption and decryption procedures used by VPNs may affect real-time V2X applications where low-latency communication is essential for safety-critical activities. Another disadvantage of VPNs is their complexity in management and implementation. Any improper set-ups could unintentionally lead to security breaches because they require expert knowledge to properly configure and compatibility with the current V2X infrastructure. Furthermore, the V2X network may now have a single point of failure due to the deployment of VPN concentrators or servers. The secure communication of the entire V2X ecosystem may be compromised if the VPN server is interrupted or the subject of an attack.

4.9 ACLs

ACLs are used to regulate and control access to V2X communication systems, resources, and data. An ACL is a set of rules that determines whether a specific user, device, or system is authorized to access a particular resource or set of data. It enforces security policies, such as who is allowed access to which resources, what actions are permitted on the resources, and under what conditions. ACLs are designed to prevent unauthorized access and defend against potential security threats such as eavesdropping, tampering, and denial of service assaults [250, 251]. In order to improve overall network security and resource protection, ACLs offer a fine-tuned control that blocks illegal access and reduces potential risks. ACLs also make network segmentation easier by allowing the division of V2X services and applications into separate zones, which minimizing the risk of security breaches. ACLs are appropriate for complete, largescale V2X deployments due to their scalability. However, for large V2X networks with many users, creating and administering ACLs can be challenging, and errors in rule setups or misconfigurations may unintentionally obstruct valid traffic or introduce vulnerabilities. These difficulties can be reduced and the advantages of ACLs in V2X communication can be maximized by ensuring diligent management, careful planning, and organization. Therefore, ACLs make a substantial contribution to the development of a safe and effective V2X ecosystem by balancing access control with usability and promoting trust and confidence in communication between connected vehicles and infrastructure components.

4.10 GPS-based Authentication

The V2X communication authentication process is based on GPS, a satellite-based navigation system used to determine the location of a device. Here, communicating devices first establish a GPS connection in order to determine their locations. The location of the devices is then transmitted with the authentication request, allowing the authentication server to validate the location of the communicating devices [174]. This ensures that communication between devices is protected from potential security threats such as impersonation and man-in-the-middle attacks. GPS-based authentication presents several kinds of advantages. It offers dependable and precise position determination, which is a critical factor in V2X communication where accurate location information is vital. Additionally, GPS-based authentication is characterized by its user-friendly nature, as it does not require the use of specialist hardware or software. Also, it is resistant to tampering and spoofing because the authentication server checks the transferred location information. However, GPS-based authentication also has certain limitations. The coverage of the technology may be limited to specific locations, such as interior environments or regions characterized by towering structures or mountainous terrain. These limitations have the potential to affect the precision of location tracking and the connectivity of devices. Furthermore, the time required for devices to establish a GPS connection and communicate location data can potentially introduce latency into the authentication procedure.

Tables 5 and 7 provides a summary of the various security methods, highlights the techniques used, and reviews the main findings of prior research. Implementing security mechanisms in V2X networks involves a trade-off between financial and computational costs and the security benefits achieved. Lightweight symmetric encryption schemes incur low computational overhead and are relatively inexpensive to deploy, providing high benefits in maintaining data confidentiality with minimal impact on latency. PKI and digital signature schemes offer strong authentication and integrity guarantees but involve moderate to high costs due to certificate management, key distribution, and computational requirements.

AI-based intrusion detection systems can provide adaptive threat detection and resilience but require substantial processing resources and deployment investment.

To assist stakeholders in prioritizing security investments, a feasibility framework can be applied by evaluating each mechanism against three criteria: cost, performance impact, and security benefit. Security mechanisms with low cost and high benefit are generally recommended for early deployment, whereas higher-cost solutions (e.g., blockchain or federated learning-based IDS) may be considered for phased integration in critical V2X applications such as autonomous vehicle coordination or safety-critical intersections. Section 7.2.1 analyzes key QoS parameters, illustrating how latency, bandwidth, and reliability are influenced by security considerations.

5 Security Testing Methods

To validate and refine the security measures implemented to counter V2X security challenges, rigorous testing methods are essential. Various testing methods are available, such as conformance testing [259, 260], function testing [252, 261] and [262], performance testing [253, 263], field and gateway testing [264, 265] and [266]. A comparison and review

Table 5 Review of security mechanisms

Reference Number	Aim and scope of the study	Methodology/ Technique used	Contribution	Benefits	Limitations
[198]	To suggest an edge computing system and an intelligent PKI for a safe V2X communication network.	Intelligent PKI, edge computing	Introduction of edge computing and intelligent PKI in V2X networks Improved performance and security of V2X communication	Enhanced security measures Reduced computational overhead	Specific details on implementation challenges Potential limitations in real-world deployment
[228]	To determine whether using blockchain and decentralized public-key infrastructure (DPKI) for V2X communications is practical and efficient.	Review of the literature, critical analysis, case study	Analysis of the DPKI and blockchain research in the V2X space, with particular emphasis on communications Identification of key concerns and challenges	Enhanced security and privacy in V2X communications through decentralized architecture Potential for improved scalability and resilience through blockchain technology	Potential issues with scalability and performance of blockchain technology Lack of empirical evidence or practical implementation
[229]	In order to examine and contrast the security features of PKI and DAA-based architectures for V2X communication	Comparative analysis, simulation, performance evaluation	PKI and DAA-based architectures compared for V2X communication security Key security strengths and weaknesses are identified.	PKI: Well-established infrastructure, supports scalable key management, enables certificate-based authentication DAA: Preserves user privacy, reduces reliance on centralized infrastructure, enables anonymous authentication	PKI: Scalability challenges, certificate revocation complexity, potential vulnerability to certificate-related attacks DAA: Computational overhead, potential anonymity misuse, limited industry adoption
[230]	To propose and evaluate certificate assignment strategies for a PKI-based security architecture in a vehicular network	Comparative analysis, simulation, performance evaluation	Identification of certificate assignment strategies suitable for vehicular networks Assessment of the performance implications of different strategies	Enhanced security through the use of PKI-based security architecture Different strategies provide options to balance security and performance	Evaluation based on simulations, may not capture real-world deployment challenges Specific to PKI-based security architecture, may not consider alternative security frameworks

Table 5 (continued)

Reference Number	Aim and scope of the study	Methodology/Technique used	Contribution	Benefits	Limitations
[231]	To explore a secure V2I authentication scheme that combines PKI (Public Key Infrastructure) and accumulator techniques	Design and analysis, security, comparative evaluation	Introduction of a new V2I authentication scheme that incorporates PKI and accumulator techniques Demonstration of the scheme's effectiveness through security analysis and comparative evaluation	Strong security guarantees against common V2I authentication attacks Efficient verification process with reduced computational overhead	Evaluation limited to theoretical analysis and simulation Possible reliance on specific assumptions or conditions
[232]	To develop an efficient key distribution system for data fusion in V2X heterogeneous networks	Design and analysis, performance evaluation, comparative study	Introduction of a novel key distribution system tailored for V2X heterogeneous networks Demonstration of the system's efficiency through performance evaluation and comparative study Improved security in key management	Efficient key distribution supporting secure data fusion in V2X heterogeneous networks Reduced communication overhead and latency Potential scalability challenges in large-scale deployments	Evaluation limited to simulation-based performance analysis Consideration of specific network scenarios and assumptions

of traditional testing methods are provided in Tables 8 and 9 respectively. However, these traditional testing methods are not sufficient to overcome the security challenges of V2X. To provide efficient security testing methods for secure V2X communication, we use the open systems interconnect (OSI) model [267–269] and [270]. The OSI model is widely used in network design and comprise various security testing approaches used in V2X communication systems, as shown in Fig. 11.

5.1 Security Testing Based on the OSI Approach

The layered nature of the OSI system presents numerous benefits. The components of the communication system are initially presented in a systematic breakdown, aiding in the planning, selection, and utilization of security testing techniques. This improves the accuracy of security measures by facilitating a thorough assessment of potential vulnerabilities and threats. The layered paradigm also advocates for a multilayered protection strategy. Each layer can be independently protected, contributing to a comprehensive protection plan. By understanding the intricate connections between different levels, one can anticipate the potential spread of security threats throughout the system. Potential breaches in a specific layer have the capacity to impact adjacent layers, both higher and lower in the hierarchy.

Table 6 Review of security mechanisms (Continue)

Reference Number	Aim and scope of the study	Methodology/ Technique used	Contribution	Benefits	Limitations
[198]	To suggest an edge computing system and an intelligent PKI for a safe V2X communication network.	Intelligent PKI, edge computing	Introduction of edge computing and intelligent PKI in V2X networks Improved performance and security of V2X communication	Enhanced security measures Reduced computational overhead	Specific details on implementation challenges Potential limitations in real-world deployment
[228]	To determine whether utilizing blockchain and decentralised public-key infrastructure (DPKI) for V2X communications is practical and efficient.	Literature review, critical analysis, case study	Analysis of the DPKI and blockchain research in the V2X space, with particular emphasis on communications Identification of key concerns and challenges	Enhanced security and privacy in V2X communications through decentralized architecture Potential for improved scalability and resilience through blockchain technology	Potential issues with scalability and performance of blockchain technology Lack of empirical evidence or practical implementation
[229]	In order to examine and contrast the security features of PKI and DAA-based architectures for V2X communication	Comparative analysis, simulation, performance evaluation	PKI and DAA-based architectures compared for V2X communication security Key security strengths and weaknesses are identified.	PKI: Well-established infrastructure, supports scalable key management, enables certificate-based authentication DAA: Preserves user privacy, reduces reliance on centralized infrastructure, enables anonymous authentication	PKI: Scalability challenges, certificate revocation complexity, potential vulnerability to certificate-related attacks DAA: Computational overhead, potential anonymity misuse, limited industry adoption
[239]	An intelligent intrusion detection system for sustainable development in autonomous vehicles	Intelligent intrusion detection system, autonomous vehicles, sustainable development	-Development of an intelligent intrusion detection system for autonomous vehicles Integration of real-time detection algorithms and machine learning techniques	Improved security and privacy protection in autonomous vehicles Real-time detection and prevention of intrusions	Limited discussion on the computational and resource requirements Scalability concerns for large-scale autonomous vehicle deployments

Table 6 (continued)

Reference Number	Aim and scope of the study	Methodology/ Technique used	Contribution	Benefits	Limitations
[240]	To explore an IPv6 in IPv4 Virtual Private Network (VPN) based on virtual devices in a vehicle information system	IPv6, IPv4, VPN, virtual devices, vehicle information system	Design and implementation of an IPv6 in IPv4 VPN solution using virtual devices Evaluation of the proposed solution's performance and effectiveness	Seamless integration of IPv6 functionality in existing IPv4 vehicle information systems	Limited discussion on security aspects of the proposed solution Scalability considerations for large-scale vehicle information systems
[223]	A route handoff method based on a virtual private network (VPN) in a vehicle information network	Route hand-off, VPN, virtual private network, vehicle information network	Design and implementation of a route handoff method based on a VPN in a vehicle information network Performance evaluation and analysis of the proposed method	Seamless route handoff during vehicle mobility in a VPN-based information network Efficient handover without service disruption	Limited discussion on potential delays or latency during route handoff Scalability considerations for large-scale vehicle information networks
[241]	A peer-to-peer communication method between autonomous vehicles using a VPN	Peer-to-peer communication, VPN, virtual private network, autonomous vehicles	Design and implementation of a peer-to-peer communication method using a VPN in autonomous vehicle networks Evaluation of the proposed method's performance and security features	Direct communication and information exchange between autonomous vehicles Privacy and security provided by a VPN-based communication approach	Limited discussion on potential communication overhead or latency Scalability considerations for large-scale autonomous vehicle networks
[242]	A secure communication scheme for DSRC based on watermark and VPN	DSRC, watermark, VPN, secure communication	Design and implementation of a secure communication scheme for DSRC using watermark and VPN	Enhanced security and privacy in DSRC-based communication	-Limited discussion on potential computational overhead
[243]	A blockchain-based access control mechanism for the Internet of Things	Blockchain, access control, IoT, security	Design and implementation of a blockchain-based access control mechanism for IoT	Improved security and access control in IoT systems	Limited discussion on potential performance overhead

Table 6 (continued)

Reference Number	Aim and scope of the study	Methodology/ Technique used	Contribution	Benefits	Limitations
[233]	An efficient concurrent and multi-stage firewall for OBD-II network traffic filtering	Firewall, OBD-II networks traffic, concurrent filtering, multi-stage filtering	Development of a concurrent and multi-stage firewall for OBD-II network traffic filtering. Efficient filtering techniques and mechanisms	Improved performance and responsiveness of OBD-II network traffic Filtering. Enhanced security and protection for in-vehicle networks	Only effective defense against unauthorized access and attacks. Supports only filtering of multiple OBD-II network packets. Limited flexible configuration and customization options for filtering rules
[234]	To provide a comprehensive survey on CAN-Bus packet filtering firewalls	Literature review, survey	Comprehensive survey on CAN-bus packet filtering firewalls Identification of existing techniques and approaches Analysis of advantages and limitations	Provides insights into different firewall techniques for CAN-bus networks Helps in selecting appropriate packet filtering mechanisms Identifies areas for further research and improvement	Needed to enhance security and protection for CAN-bus networks Helps in mitigating attacks and unauthorized access
[235]	To develop a modular, protocol agnostic firewall for automotive dongles to enhance privacy and security	Software development, experimental evaluation	Development of a firewall solution tailored for automotive dongles Integration of modular and protocol agnostic features Demonstration of enhanced privacy and security through experimental evaluation	Enhances privacy and security of automotive dongles Protects against unauthorized access and data leakage Supports various communication protocols and modules	Requires integration with existing dongle systems May introduce some performance overhead Limited evaluation in specific scenarios and configurations
[236]	To create a lightweight 5G-V2X intra-slice intrusion detection system using knowledge distillation	Knowledge distillation, 5G-V2X, intrusion detection	Development of a lightweight intrusion detection system for 5G-V2X networks Integration of knowledge distillation for efficient model transfer -Performance evaluation and validation of the proposed system	Enhanced intrusion detection capabilities in 5G-V2X networks Reduced computational overhead due to lightweight design Efficient model transfer using knowledge distillation	Limited discussion on potential false positive or false negative rates Scalability considerations for large-scale deployments not explicitly discussed

Table 6 (continued)

Reference Number	Aim and scope of the study	Methodology/ Technique used	Contribution	Benefits	Limitations
[237]	To explore the use of blockchain technology for intrusion detection in vehicular navigation map systems	Blockchain technology, intrusion detection, vehicular navigation map system	Investigation of blockchain technology for intrusion detection in vehicular navigation map systems Design and implementation of a blockchain-enabled intrusion detection mechanism Evaluation of the proposed approach in detecting map tampering attacks	Enhanced security and integrity of vehicular navigation map systems Effective detection and prevention of map tampering attacks Utilization of blockchain technology for traceability and immutability	Performance overhead related to blockchain usage Scalability concerns in large-scale map systems
[238]	To develop a machine learning framework for intrusion detection in VANET communications	Machine learning, intrusion detection, VANET	Development of a machine learning framework for intrusion detection in VANET communications Feature selection and model training using machine learning algorithms	Enhanced intrusion detection accuracy in VANET communications Flexibility and adaptability to dynamic VANET environments	Dependence on labeled training data for supervised learning Scalability considerations for large-scale VANET deployments
[244]	To propose a distributed enforcement mechanism for access control policies in Intelligent Transportation Systems	Access control, ITS, distributed enforcement, situation awareness	Design and implementation of a distributed access control enforcement mechanism in ITS Evaluation and validation of the proposed mechanism's effectiveness	Enhanced access control and security in ITS environments Distributed enforcement of access control policies	Limited discussion on potential scalability concerns Performance considerations in large-scale ITS deployments
[245]	An access control scheme to mitigate Vehicle-to-network connected vehicle attacks	Access control, V2N, connected vehicle attacks	Design and implementation of an access control scheme against V2N connected vehicle attacks Evaluation and analysis of the scheme's effectiveness and security features	Enhanced security and protection against V2N connected vehicle attacks Access control mechanisms to prevent unauthorized access	Limited discussion on potential performance overhead Scalability considerations for large-scale connected vehicle networks

Table 6 (continued)

Reference Number	Aim and scope of the study	Methodology/ Technique used	Contribution	Benefits	Limitations
[246]	To analyze the evolution of V2X communication from IEEE 802.11p to 5G and explore its future prospects towards 6G	V2X communication, IEEE 802.11p, 5G, 6G	Comprehensive analysis of V2X communication technologies from IEEE 802.11p to 5G Evaluation of key features, challenges, and improvements in each generation	Insights into the evolution of V2X communication technologies Identification of key features, challenges, and improvements in each generation	Limited discussion on specific technical details or implementation aspects Future prospects focused on 6G without detailed analysis or predictions
[247]	To assess the performance of IEEE 802.11p at the application level in field tests for V2X that are safety-related	IEEE 802.11p, V2X, field trials	Executing V2X field tests with a focus on safety using IEEE 802.11p Analysis of essential metrics for performance measurement.	Analyzing IEEE 802.11p performance in V2X settings with a focus on safety Understanding of significant performance metrics and difficulties	There has been little discussion of scalability issues for extensive V2X implementations. Safety-related applications' unique technical issues and details are not fully covered.
[248]	To provide an intelligent vehicle network security mechanism based on geographic location	Geographic location -based security mechanism, vehicular networks	Development of a geographical location -based security system for vehicle networks. Evaluation and analysis of the effectiveness and security attributes of the mechanism	Geographic location -based security for intelligent vehicular networks Protection against unwanted access and location-based attacks	There is little discussion of potential scalability issues Large-scale vehicle network performance considerations
[249]	To solve the shortcomings of straightforward mix zones in maintaining location privacy in vehicular networks	Mathematical analysis, simulation-based evaluation	Proposes the concept of cooperative location privacy in vehicular networks Introduces a novel Mix-CoLoP protocol that combines mix zones with collaborative techniques	Improves the location privacy in vehicle networks Takes into account the strategies and movement patterns of attackers	Assumes that vehicle will work together. Requires cooperation and communication between vehicles

5.1.1 Physical Layer

The Physical Layer is primarily concerned with the physical transfer of data, encompassing both wired and wireless communication channels. While digital safeguards are crucial, physical security serves as the primary defense against unauthorized access, manipulation,

Table 7 Testing methods and their comparisons

Sl. No.	Testing methods	Latency	Reliability	Security	Interoperability	Coverage	Advantages	Disadvantages
1	Function testing [252]		Yes				Detects defects early Improves reliability Facilitates market entry	Resource-intensive Time-consuming Limited coverage Potential for false negatives
2	Performance testing [253]	Yes	Yes		Yes		Improved reliability Better Compliance Increased efficiency Enhanced safety	Complexity Cost Expensive Time-consuming Limited Scope
3	Conformance testing [254]		Yes	Yes	Yes		Ensures Interoperability Increase reliability Facilitates market entry	Time Consuming Cost Expensive Limited Coverage
4	Gateway testing [255]		Yes	Yes			Improved Security Early Detection Better Understanding Compliance	Cost and Time Complexity Simulation Challenges Disruptive Limited Real-world Validation

Table 7 (continued)

Sl. No.	Testing methods	Latency	Reliability	Security	Interoperability	Coverage	Advantages	Disadvantages
5	Penetration testing [256]			Yes			Improved Security Compliance Early Detection Improved Resilience Better Understanding	Cost Expensive Time Consuming False Negatives Disruptive
6	Accelerated testing [257]		Yes				Faster Results Early Detection of Issues Cost Saving Improved Quality	Limited Realism Limited Access Complexity
7	Field testing [258]		Yes		Yes	Yes	Real World Conditions User Feedback Solving Identifying Issues	Cost Expensive Logistics Time Consuming Conditional Limitations

and theft [281]. Maintaining robust physical security measures is critical due to the potential vulnerabilities in this domain that compromise the integrity of the entire communication ecosystem [282].

Physical Security Assessment: Physical security evaluations identify potential vulnerabilities within a given system, including, but not limited to, unprotected access points, exposed hardware, and vulnerable cabling [283]. Identifying these vulnerabilities helps to proactively implement safeguards such as access controls, surveillance systems, and tamper-evident techniques [284].

5.1.2 Data Link Layer

The Data Link Layer ensures accurate data transfer through error detection and correction mechanisms. However, this layer can be vulnerable to various threats, including MAC address spoofing, illegal network access, and inadequate encryption protocols.

Security Code Review: A thorough evaluation of the security code is necessary to guarantee the integrity of the firmware and software in the components of the data link. By

Table 8 Review of testing methods

Reference Number	Objective	Methodology/ Technique used	Key findings	Contribution	Advantages	Limitations
[271]	To suggest a conformance testing system for the C-V2X standard protocol	Creating a comprehensive system with modules for test case creation, execution, and result analysis	Effectiveness of the suggested conformance testing system in ensuring C-V2X devices	To provide insights on the creation of a C-V2X compliance testing system	Enables interoperable and reliable C-V2X communication	Limited to the design aspect of the conformance testing system
[272]	To examine how formal conformance testing can be used to create scenarios for autonomous vehicles	Utilizing formal approaches for conformance testing	Successfully generate a variety of realistic autonomous vehicle scenarios using formal conformance testing	Enhancing the knowledge of and use of formal conformance testing to create scenarios for autonomous vehicles	Makes sure the generated scenarios are legitimate and reliable.	Focuses on the use of formal conformance testing, possibly excluding other methods to create scenarios for autonomous vehicles.
[273]	To develop a framework to create a library of functional testing scenarios for connected and automated vehicles.	Development of a scenario generation framework using several methodologies	Successful creation of a framework for creating a functional testing scenario library for connected and automated vehicles	A thorough framework for creating functional testing scenarios for connected and automated vehicles.	Makes it easier to test and validate connected and automated vehicle systems	Primarily focuses on creating functional testing scenarios, and may not cover additional testing and validation issues for connected and automated vehicles.
[263]	To research the LTE-V2X communication performance testing method in an outfield highway	Setting up and conducting an experiment to evaluate LTE-V2X performance on an outfield highway	Performance assessment of LTE-V2X in an outfield highway scenario	Improvement in understanding LTE-V2X communication performance in outfield highways	LTE-V2X performance insights in an outfield highway environment	Only includes performance testing of LTE-V2X connectivity in an outfield highway scenario.

Table 8 (continued)

Reference Number	Objective	Methodology/ Technique used	Key findings	Contribution	Advantages	Limitations
[253]	To evaluate and test the performance of the LTE-V2X network in an industrial park scenario	Methodology and experimental setup for testing network performance	Evaluation of the LTE-V2X network performance in an industrial park scenario, showing the communication technology's benefits and drawbacks	Gives information about how the LTE-V2X network performs in a situation involving an industrial park.	Improves comprehension of LTE-V2X communication technology in an industrial park scenario	Focuses particularly on the LTE-V2X network in an industrial park scenario
[274]	To design a vehicle gateway automatic test system based on CANoe	Designing a test system utilizing the CANoe software	Successful design of a vehicle gateway automatic test system using CANoe, facilitating automated testing of vehicle gateways	Provides a practical solution for automated testing of vehicle gateways using the CANoe software	Automation and efficiency in vehicle gateway testing	Only Focused on the design of a vehicle gateway automatic test system based on CANoe
[275]	In order to investigate the use of artificial intelligence approaches in penetration testing of CAV network	AI algorithms being used to find weaknesses and potential attack vectors	Possibilities for utilizing AI to improve the effectiveness and efficiency of CAV penetration testing	Improved penetration testing techniques for CAV networks	Adaptability and Scalability of AI-based penetration testing	Considered on AI-based penetration testing and might not address other CAV network security issues
[256]	To investigate the application of reinforcement learning methods in penetration testing for VANET	Utilizing reinforcement learning algorithms	Utilizing reinforcement learning techniques to test the vulnerability of the VANET	The use of reinforcement learning techniques to advance the field of VANET penetration testing	Reinforcement learning improved the efficiency and effectiveness of VANET penetration testing.	Focuses particularly on the application of reinforcement learning techniques and may not cover other VANET penetration testing components.

Table 8 (continued)

Reference Number	Objective	Methodology/ Technique used	Key findings	Contribution	Advantages	Limitations
[276]	To promote accelerated testing protocols for the secure and trustworthy deployment of autonomous and connected vehicles	Literature review and analysis of existing protocols	Principal research discoveries that improve accelerated testing procedures for connected and autonomous vehicles	Contribution of the study to the development of testing procedures for the safe and reliable deployment of connected and autonomous vehicles	Enhanced testing procedures for automated and connected vehicles	There are gaps in the specifics of the methodology, approach, and technique adopted, as well as the main conclusions and limits.
[277]	To model the end-to -end delay of a network in accelerated reliability testing	Modelling strategy taking network properties and testing conditions into account	An understanding of how to calculate the end-to-end delay of network systems under accelerated testing conditions	Provides end-to-end delay understanding and prediction for accelerated reliability testing.	Enables the design and optimization of network systems for increased performance and dependability.	
[278]	To evaluate the efficiency of LTE-V2X in facilitating awareness messages between vehicles and vulnerable road users	Field tests using LTE-V2X connectivity in real-world situations	LTE-V2X's promise to improve communication and safety between vehicles and vulnerable road users.	Offers information on how LTE-V2X is used to facilitate awareness messaging in traffic environments.	Application of LTE-V2X in real-world driving scenarios to enhance communication and awareness	Primarily focuses on the field tests and might not include other LTE-V2X communication-related topics
[279]	To provide an overview of V2X communication in Europe	Vehicle communication technology research initiatives, standardization efforts, and field testing analysis	From research to field testing and standardized deployment of V2X communication in Europe	The Perspectives on the development and possibilities of V2X communication in Europe	Knowing how V2X communication is being developed and used in Europe	European context is the primary focus; possible omission of global breakthroughs in V2X communication

Table 8 (continued)

Reference Number	Objective	Methodology/ Technique used	Key findings	Contribution	Advantages	Limitations
[280]	To improve the effectiveness and consistency of data distribution in VANETs	The creation of a protocol taking resource reservation strategies into account and delay limitations	VANETs' proposed protocol will increase the efficiency and dependability of data dissemination.	Enhanced data dissemination in VANETs	Taking delay restrictions and resource reservation methods into account	With a specific focus on data distribution in VANETs, other aspects of VANET communication might not be covered.

meticulously examining the code, potential vulnerabilities can be identified and addressed proactively [285, 286]. This methodology improves the resilience of the software that facilitates the operation of data connection.

5.1.3 Network Layer

The Network layer handles routing and addressing to ensure successful data delivery to its intended destination. Potential vulnerabilities in this context involve IP spoofing, routing attacks, or deliberate network infrastructure failures [287].

- *Penetration testing*: Penetration testing [288] is used as a proactive method to evaluate the vulnerabilities present inside the network layer. This approach involves the simulation of actual attacks to uncover vulnerabilities that may be susceptible to exploitation by hostile individuals. Through the implementation of deliberate and regulated offensive maneuvers, organizations can identify and expose potential vulnerabilities inside their V2X networks, subsequently strengthening their security measures. This process eventually contributes to the increase in the overall resilience of the system.
- *Vulnerability assessment*: A comprehensive vulnerability assessment should be carried out to systematically scan the network layer to discover potential vulnerabilities [289]. By identifying these vulnerabilities, organizations could prioritize their efforts to mitigate them, improving network resilience and reducing the likelihood of security breaches or interruptions [290].
- *Security architecture review*: A thorough examination of the security architecture is to be performed to assess the structure and execution of security protocols at the network layer. This guarantees that the architectural design is in accordance with established industry standards, regulatory obligations, and the particular security objectives of V2X. It also helps to identify existing gaps within the security framework and, subsequently implement measures to address these gaps, thus enhancing the overall cohesiveness and robustness of the security system [291].
- *Network security testing*: The effectiveness of V2X communication depends on the implementation of robust network security protocols that provide security [292]. The examination of these protocols in network security testing aims to detect vulnerabilities and weaknesses that may potentially jeopardize the integrity of the network layer [293, 294].

Table 9 Blockchain potential in V2X security

Reference Number	Aim and scope of the Study	Methodology used	Characteristics and participants	Techniques used to gather and analyze data	Results or outcomes (findings)	Limitations	Future research
[438]	To provide confidentiality, integrity, authentication, and privacy features	Distributed governance method	It does not require trust	D-PKI concepts to vehicular networks	Fully decentralized and anyone can access this technology	It is difficult to optimize the blockchain connection between the vehicle and road users	To develop the Android application to provide interoperability between prototypes
[440]	To improve the security, secure data sharing and communication between the vehicle	Blockchain-enabled V2V communication system	Automatic authentication for users and their vehicles	User authentication and vehicle authentication to provide secure communication	It eliminates the obstacles of key-based and trust-based authentication problems	The unregistered vehicle cannot access this methodology	Autonomous vehicles also achieve more security and effective data-sharing methodology
[441]	To improve privacy and security with a combination of the federated learning and blockchain	Federated edge learning scheme based on the blockchain	Federated learning (FL) enhances privacy protection by allowing fine-tuning of learning	Blockchain (BC) and FL-based edge scheme	Latency and reliability have been increased	The number of vehicles increases offloading and computation may increase	
[442]	To provide privacy and security	Blockchain-based conditional privacy preserving authentication method	Each participant can authenticate anonymously with others	Signature of knowledge scheme for authentication	Distribution of secret key does not require an expensive separate secure channel	The quality of being vague in the real-world applications	To improve the efficiency of signature of knowledge and conditional privacy-preserving authentication
[443]	To ensure the security of privacy and authentication	Blockchain-based energy trading system for secure ITS	Authenticate the messages using decentralizes without using root trusted authority	Consensus algorithm and practical byzantine fault tolerance (PBFT) in the proposed authorization method	It provides security service from internal intruders and usual threats	Need to review and pay more attention to the security, privacy, and exchange of transaction data	Need more Investigation of the effective consensus algorithm

Table 9 (continued)

Reference Number	Aim and scope of the Study	Methodology used	Characteristics and participants	Techniques used to gather and analyze data	Results or outcomes (findings)	Limitations	Future research
[444]	To construct a local intrusion detection system	Micro-block-chain based geo-graphical dynamic Intrusion detection (MBID)	An MBID detection system that adjusts vehicle intrusion detection strategies	Gather examples of local security breaches that can be utilized to develop intrusion detection methods that are suitable for the present situation	dynamic intrusion detection methods that adapt to a vehicle's change in location.	If users upload more intrusion detection information, the greater number of cryptocurrencies is in a limited time	
[445]	To facilitate V2X communication and parking payments	Real-time application is used for communication and payment processing	Blockchain-based application is responsible for communication and parking payments	Android application is used for communication and parking payments	The real-time android application using blockchain is faster and more scalable	The application may not be compatible with different platforms	The application should be platform independent and has a driver assistant system
[446]	To minimize the link breakage and reduce latency	Blockchain-based edge-enabled data integrity management scheme	an optimal container-based data processing scheme, and a blockchain-based data integrity management scheme is used for reducing latency and link breakage	Container-level allocation and edge-based computation reduces the migration	Lower the energy requirement for migration and minimize the latency	The developed framework is limited	To implement the professional version of this framework

5.1.4 Transport Layer

The Transport layer oversees end-to-end connectivity and data integrity. Vulnerabilities such as insufficient encryption, session hijacking, and protocol weaknesses are of significant concern in this layer [295].

- *Cryptographic testing*: Cryptographic testing is the evaluation of cryptographic algorithms and implementations to ensure their security and functionality, which is essential

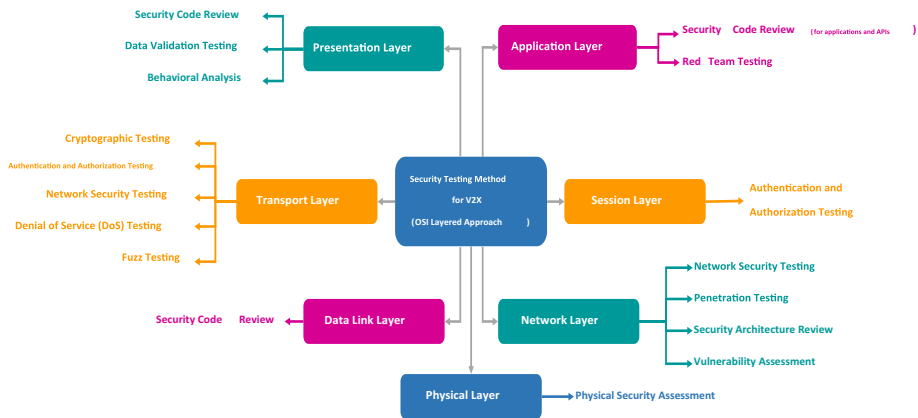


Fig. 11 Testing methods based on the layered approach

to protect the integrity and confidentiality of data [296]. Cryptography serves as the foundation for ensuring security in V2X systems, and cryptographic testing [296] is a crucial process to evaluate the efficacy of encryption and decryption processes used within the transport layer. Therefore, it is possible to ensure that data confidentiality and integrity are maintained throughout transmission by evaluating cryptographic techniques and protocols, increasing the security of V2X communication [297].

- *Authentication and authorization testing*: Authentication and authorization testing examines the methods that grant access to the transport layer. Implementing this approach ensures that only authorized entities can participate in V2X communication, thus mitigating the risk of unauthorized access and potential breaches [298].
- *Network security testing*: Network security testing within the transport layer serves the purpose of verifying the integrity of communication protocols and data transfer. This methodology ensures the integrity of data transmission in V2X communication, thus maintaining its reliability and trustworthiness. This network security test must also be conducted at the network layer to identify vulnerabilities in network protocols [294].
- *Denial of service testing*: Resilience of the transport layer is paramount in mitigating the impact of DoS attacks, which are designed to disrupt service provision. DoS testing involves subjecting the layer to simulated attacks to assess its ability to withstand such attacks. Implementing this approach is crucial to ensuring uninterrupted functionality of V2X systems, even when deliberate efforts are made to overwhelm the transport layer [299, 300].
- *Fuzz testing*: Fuzz testing is a technique that involves subjecting the transport layer to unforeseen and incorrect data input to detect vulnerabilities or crashes that may be exploitable [301]. This technique is used in V2X security to evaluate the ability of the transport layer to handle abnormal data, thus enhancing its resilience and mitigating potential vulnerabilities [302, 303].

5.1.5 Session Layer

The Session layer establishes and terminates communication sessions. This layer is susceptible to vulnerabilities such as session hijacking, session fixation, and unauthorized access [304].

Authentication and authorization testing: The evaluation of authentication and authorization techniques within the session layer of V2X security primarily revolves around assessing their effectiveness in confirming entity identities and granting appropriate access privileges. This approach ensures that only individuals and devices with proper authorization can establish and maintain communication sessions. By conducting comprehensive testing of these elements, it becomes possible to mitigate the risk of unauthorized access and ensure the integrity of V2X communication sessions [305, 306].

5.1.6 Presentation Layer

The presentation layer translates and encrypts the data. Vulnerabilities may arise due to inadequate data validation, attacks that exploits data format, or weaknesses in encryption mechanisms [307].

- *Security code review (for applications and data transformations):* By conducting a comprehensive examination of the security code, it is possible to identify and address potential vulnerabilities within the codebase. The implementation of robust security measures ensures the absence of vulnerabilities in applications and data transformations [308, 309].
- *Data validation testing:* By subjecting data inputs to different situations, it is possible to identify vulnerabilities associated with inadequate input handling, which can result in data breaches or system vulnerabilities [310, 311]. This methodology guarantees the preservation and security of data during the V2X communication procedure. *Behavioral analysis:* Behavioral analysis is a dynamic methodology used for V2X security testing within the presentation layer. By closely observing the actions and behavior of apps and components, it is possible to identify irregularities that could potentially indicate unauthorized or malicious intent. This approach presents a proactive strategy for detecting potential risks and implementing preventive measures to mitigate security breaches [312].

5.1.7 Application Layer

The application layer facilitates user-level interactions. Various vulnerabilities can occur in this layer, such as input validation issues, misconfigurations, and application layer attacks [313].

Security code review (for applications and APIs): The application layer should involve a thorough evaluation of the application and API security through meticulous code inspection. By evaluating the codebase, it is possible to identify and address potential vulnerabilities,

thus improving the security of apps and APIs and protecting them against potential exploits [314].

Red-team Testing: Red team testing simulates actual assaults to assess the security posture of the application layer. By employing strategies that mimic hostile methods, it is possible to detect vulnerabilities and areas of weakness [315, 316].

Effective testing methods increase the use of automotive communication systems by ensuring their reliability and minimizing errors. Consider an example, a V2X communication system specifically developed to improve traffic safety through the exchange of alerts between vehicles using V2V communication (Fig. 12). To conduct a thorough security assessment, various testing methodologies are employed across the different layers of the OSI approach. Starting with the application layer, security professionals diligently examine the code of the V2X application and its APIs, evaluating potential vulnerabilities that could reveal entry points in input validation. In the presentation layer, data validation testing is conducted to examine various data inputs, including invalid ones. The purpose of this test is to verify the system's ability to handle unexpected input without system failures or compromising sensitive data security. Furthermore, behavioral analysis is conducted that assesses the system response to modified data, providing valuable information into possible vulnerabilities. The authentication and authorization methods are carried out at the session layer. Testers check by compromising the system using inaccurate credentials. In the transport layer, cryptographic tests are conducted to analyze the efficiency of encryption algorithms and the secure exchange of keys. Additionally, authentication and authorization procedures are evaluated to determine their effectiveness. In the realm of network infrastructure, the penetration testing process replicates network level attacks, thus exposing weaknesses in routing protocols. Furthermore, the practice of threat modeling helps to assess the system's ability to withstand potential threats, such as DDoS attacks. In the data link layer, a compre-

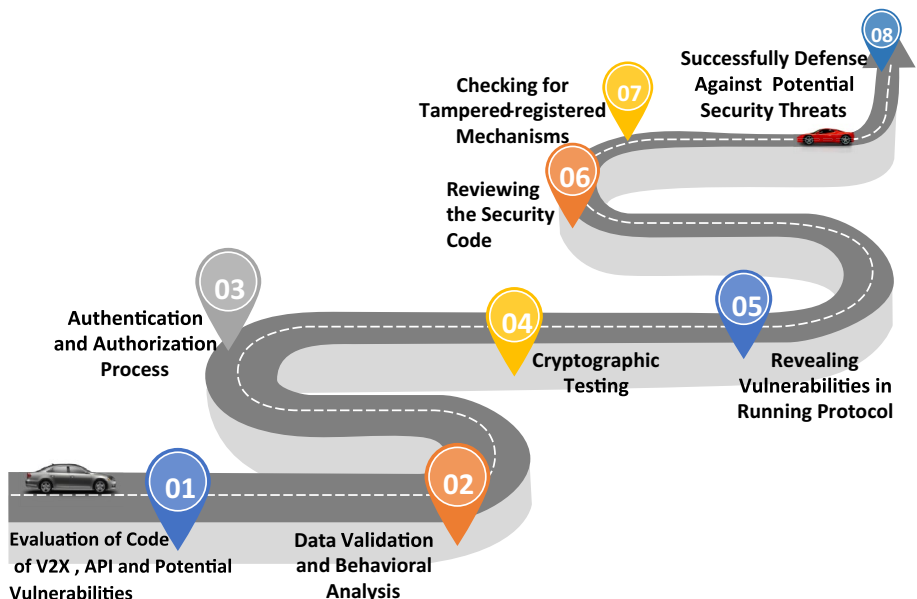


Fig. 12 Testing and exchange of alert messages between V2V

hensive security code review is performed to assess the firmware and software included in the data link components. The ultimate goal of the investigation is to detect potential vulnerabilities, including but not limited to the presence of hard-coded passwords [317]. The primary emphasis of network security testing at this layer pertains to the evaluation of the security methods employed by data connection protocols, such as MAC address filtering. The physical layer is assessed by conducting a physical security assessment, which involves inspecting the equipment responsible for V2X communication. Through the emulation of authentic real-world events and vulnerabilities, OSI testing methodology ensures the resilience of V2X systems, providing a comprehensive safeguard against potential security risks.

5.2 Method Selection and Implementation

Determining suitable testing methodologies for a V2X system involves examining and evaluating the inherent attributes, communication protocols, and potential vulnerabilities of the system. Several important factors must be considered, including aligning methods with established communication protocols, assessing data sensitivity, evaluating potential threats in the environment, complying with relevant regulations and standards, addressing resource limitations, simulating scenarios that mirror real world conditions, and conducting comprehensive testing for the smooth integration of multiple components. Scalability, performance, iterative testing, risk assessment, and keeping up with evolving risks are critical considerations. The proficiency of the testing team is also crucial.

5.3 Challenges and Best Practices

The evaluating security in V2X communication faces complex challenges due to the ever-changing characteristics of vehicular networks, the need for critical system specifications, and the inherent vulnerabilities of wireless technology. The evaluation of testing procedures should encompass real-world scenarios, real-time functionality, and scalability, all while prioritizing vehicular safety. The complexity is further compounded by considerations related to privacy concerns, embedded components, and regulatory compliance. To address these complexities effectively, it is imperative to adopt a comprehensive strategy that integrates conventional methodologies alongside innovative interventions, practical simulations, specialized tools, and active engagement with industry stakeholders and regulatory bodies.

6 QoS Parameters for V2X Security

The development of V2X standards and the use of modern communication techniques are key factors in determining the foundation of V2X networks. However, maintaining standards compliance is not enough to guarantee the security of these systems. It is essential to take into account the QoS elements that govern the reliability and performance of the communication infrastructure in order to fully understand the security of V2X. Hence, we explore the importance of QoS parameters in the section. We gathered significant information on factors impacting a system's ability to withstand security attacks and create a secure, seamless, and reliable V2X environment by examining the impact of QoS metrics on the

overall security of V2X communication. The Fig. 13 shows the QoS parameters and its ensuring techniques for V2X security [318]:

Data confidentiality [174].

Data integrity [319].

Data authenticity [16].

Timing [320].

Reliability [321].

Availability [322].

Bandwidth [323].

Security level [324]. Resilience [325].

All of the aforementioned parameters are essential to ensure the security and dependability of the V2X system. Each parameter plays a distinct role in ensuring that the system runs effectively and securely. Failure in any one parameter can negatively influence the overall performance of the system.

6.1 Data Confidentiality

Ensure that sensitive information exchanged between vehicles is protected from unauthorized access [174]. To maintain QoS to ensure data confidentiality, various techniques can be used [326]. such as,

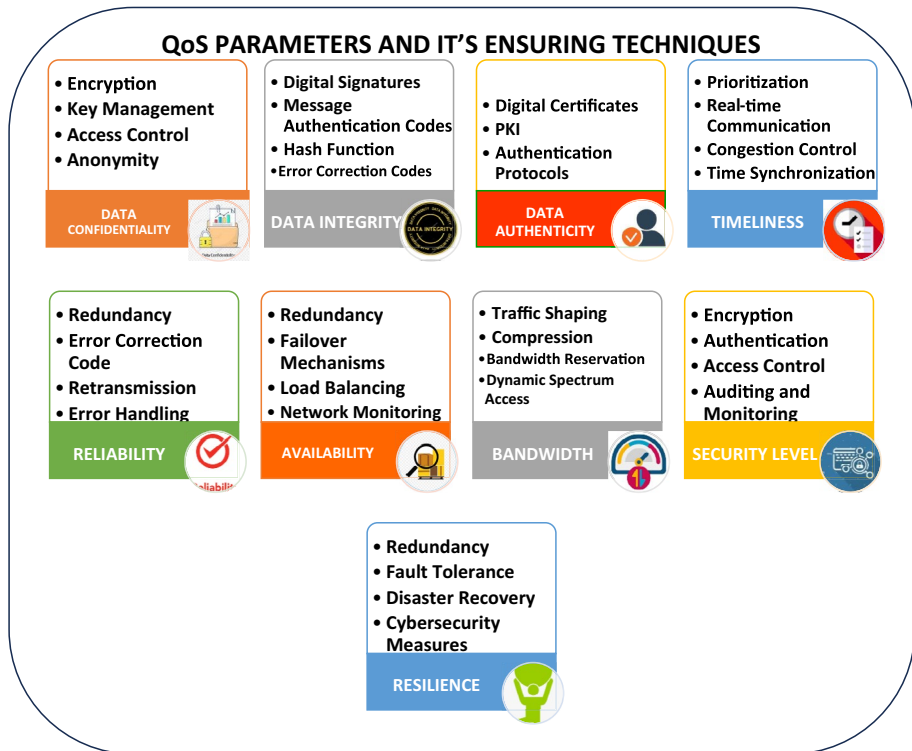


Fig. 13 QoS parameters

- *Encryption*: The use of encryption techniques such as AES can be used to protect the confidentiality of V2X data. This involves encrypting the data before it is transmitted and decrypting it at the receiving end.
- *Key management*: An effective key management scheme is essential to ensure secure transmission of V2X data. This includes the distribution, storage, and revocation of cryptographic keys used for the encryption and decryption of V2X data [327].
- *Access control*: Access control mechanisms can be used to prevent unauthorized access to V2X data. This includes using firewalls, intrusion detection systems, and other access control mechanisms to protect the V2X network from attacks [328, 329].
- *Anonymity*: Anonymity techniques such as pseudonymization can be used to protect the identity of V2X users while still allowing the system to function. This involves replacing the real identity of the users with a temporary identifier or a pseudonym [330].

6.2 Data Integrity

To maintain this parameter, several techniques can be used [319]. such as.

- *Digital signatures*: Digital signatures can be used to ensure data integrity by providing authentication of the source of the message and verifying that the message has not been altered in transit [331].
- *Message authentication codes (MAC)*: Message authentication codes can be used to ensure data integrity by providing a checksum that can be used to verify that the data has not been altered in transit [332].
- *Hash functions*: Hash functions can be used to ensure data integrity by generating a unique fixed-length code from the data, which can be used to verify that the data have not been tampered with [333, 334].
- *Error correction codes (ECCs)*: It can be used to detect and correct errors that may occur during transmission [335, 336].

6.3 Data Authenticity

Data authenticity is one of the V2X security parameters that helps ensure that the data transmitted between vehicles and the infrastructure is from a trusted source. It confirms that the origin of the data is genuine and that the sender is authorized. Digital signatures, certificates, and PKI frameworks are commonly used to validate authenticity. Unlike integrity, authenticity verifies who sent the data, whereas integrity verifies whether the data was altered. To maintain data authenticity in V2X security [16], several techniques can be used such as.

- *Digital certificates*: Digital certificates can be used to ensure the authenticity of the data by providing a means of verifying the identity of the sender and the integrity of the data [337, 338].
- *PKI*: PKI can be used to ensure authenticity of the data by providing a framework for managing digital certificates and associated public and private keys [339].
- *Authentication protocols*: Authentication protocols can be used to ensure the authen-

ticity of the data by verifying the identity of the sender, such as the use of username/password combinations or biometric authentication [332, 340].

6.4 Timeliness

Timeliness is an important parameter of V2X security that helps ensure that messages are transmitted and received promptly [320, 341] and [342]. Several techniques used for achieving this include,

- *prioritization*: This can be achieved through the use of QoS mechanisms that assign different levels of priority to different types of messages. It can be used to ensure that critical messages are given higher priority than noncritical messages [343, 344].
- *Real-time communication*: This can be achieved through the use of protocols that provide low-latency communication, such as the IEEE 802.11p standard. Real-time communication mechanisms can be used to ensure that messages are transmitted and received in a timely manner [345].
- *Congestion control*: This can be achieved through the use of congestion control algorithms that monitor network traffic and adjust the transmission rate of messages to avoid congestion. Congestion control mechanisms can be used to ensure that the V2X network does not become congested and that messages are transmitted promptly [346, 347].
- *Time synchronization*: This can be achieved through the use of protocols that synchronize the clocks of different devices with a common time reference. Time synchronization mechanisms can be used to ensure that messages are transmitted and received synchronously [348, 349].

6.5 Reliability

The reliability of V2X security helps ensure that messages are transmitted and received without errors or loss of data [321, 350]. Several techniques used for to ensure this parameter include,

- *Redundancy*: This can be achieved through the use of redundant systems, such as multiple communication channels or multiple paths for message transmission [351]. Redundancy can be used to ensure that messages are transmitted and received even in the event of failure [349].
- *ECC*: Error correction codes can be used to ensure reliability by detecting and correcting errors that may occur during transmission. It can be used to detect and correct errors in transmitted data, ensuring that received data are the same as the sent data [352].
- *Retransmission*: Retransmission mechanisms can be used to ensure that messages are transmitted and received even if they are lost during transmission. This can be achieved through the use of protocols that require the sender to retransmit the message if it is not acknowledged by the receiver [353, 354].
- *Error handling*: Error handling mechanisms can be used to ensure that errors are detected and handled appropriately. This can be achieved through the use of error detec-

tion mechanisms that detect errors in transmitted data, and error handling mechanisms that handle errors in a way that does not compromise the security or reliability of the V2X system [355].

6.6 Availability

The availability parameter confirms that the communication services are available to authorized users when needed [322, 356]. Several techniques used for ensuring this parameter include:

- *Redundancy*: This can be achieved through the use of redundant systems, such as multiple communication channels or multiple paths for message transmission. Redundancy can be used to ensure that communication services remain available even in the event of failure [351, 357].
- *Failover mechanisms*: This can be achieved through the use of mechanisms that detect and respond to failures, such as automatic failover to backup systems or components. Failover mechanisms can be used to ensure that communication services remain available even if one of the components of the V2X system fails [358].
- *Load balancing*: This can be achieved through the use of load-balancing mechanisms that distribute traffic across multiple systems, ensuring that no single system is overloaded. Load balancing can be used to ensure that communication services remain available even during periods of high traffic [359, 360].
- *Network monitoring*: This can be achieved through the use of monitoring tools that track the performance of the V2X network and alert administrators to potential issues. Network monitoring can be used to ensure that communication services remain available by detecting and responding to problems before they affect the availability of the system [361].

6.7 Bandwidth

Sufficient bandwidth ensures that messages can be transmitted and received without being affected by congestion or insufficient network capacity [362]. Several techniques employed to ensure bandwidth include:

- *Traffic shaping*: Traffic shaping can be used to regulate the flow of traffic on the V2X network and ensure that sufficient bandwidth is available for critical messages. This can be achieved through the use of traffic shaping mechanisms that allocate bandwidth to different types of messages based on their priority [363].
- *Compression*: Compression can be used to reduce the size of messages and conserve bandwidth. This can be achieved through the use of compression algorithms that reduce the size of messages without compromising their security [364].
- *Bandwidth reservation*: Bandwidth reservation can be used to reserve a certain amount

of bandwidth for critical messages, ensuring that they are not affected by congestion or other network issues. This can be achieved through the use of bandwidth reservation mechanisms that allocate a fixed amount of bandwidth to critical messages [365, 366].

- *Dynamic spectrum access*: Dynamic spectrum access can be used to ensure that the V2X network can access available spectrum efficiently and avoid interference. This can be achieved through the use of dynamic spectrum access mechanisms that allow the V2X network to share the available spectrum with other wireless networks in a way that does not compromise the security or reliability of the V2X system [367, 368].

6.8 Security Level

The security level helps to ensure that messages are transmitted and received securely, preventing unauthorized access and manipulation [324]. Several techniques that can be used include,

- *Encryption*: Encryption can be used to protect messages transmitted over the V2X network from unauthorized access or tampering. This can be achieved through the use of encryption algorithms that ensure that messages can only be accessed by authorized parties [369].
- *Authentication*: Authentication can be used to ensure that messages are sent and received by authorized parties. This can be achieved through the use of authentication mechanisms that verify the identity of devices and users before allowing access to the V2X network [370].
- *Access control*: Access control can be used to restrict access to the V2X network to authorized devices and users. This can be achieved through the use of access control mechanisms that ensure that only authorized parties can access the V2X network [371].
- *Auditing and monitoring*: Auditing and monitoring can be used to ensure that the V2X network is secure and that any security incidents are detected and responded to in a timely manner. This can be achieved through the use of auditing and monitoring mechanisms that track the performance of the V2X network and alert administrators to potential security issues [372, 373].

6.9 Resilience

Resilience helps the system continue to operate in the face of various types of disruptions including cyber-attacks, natural disasters, and equipment failures [103, 325]. In order to maintain this parameter, several techniques can be used such as:

- *Redundancy*: Redundancy can be used to ensure that critical components of the V2X system are replicated to provide a backup in the event of a failure. This can be achieved through the use of redundant servers, communication links, and other critical components [351].

- *Fault tolerance*: Fault tolerance can be used to ensure that the V2X system can continue to operate in the face of equipment failures or other types of disruptions. This can be achieved through the use of fault-tolerant systems that are designed to continue to operate even in the face of hardware or software failures [374].
- *Disaster recovery*: Disaster recovery can be used to ensure that the V2X system can recover from natural disasters or other types of disruptions. This can be achieved through the use of disaster recovery mechanisms that provide backup facilities, data, and other critical components in case of a disaster [375, 376].
- *Cybersecurity measures*: Cybersecurity measures can be used to protect the V2X system from cyberattacks and other types of security threats. This can be achieved through the use of firewalls, intrusion detection systems, and other security measures that protect the V2X system from unauthorized access and data breaches [377, 378].

7 Varied Technologies in V2X

A series of ongoing improvements have proven effective in addressing existing obstacles and enhancing the functionalities of this revolutionary technology. The ever-evolving nature of the automobile industry has led to a continuous flow of advancements aimed at improving the efficacy, safety and efficiency of V2X communication. A notable progress can be observed in the continuous development of wireless communication technologies [379]. The combined advancements like AI, 5G, edge Computing, etc. operate in coordination to enhance the overall effectiveness and dependability of V2X communication. The following details some of them:

7.1 Connected and Autonomous Vehicles (CAVs)

CAVs embody a convergence of cutting-edge technologies that are revolutionizing the way vehicles interact with the transportation ecosystem. Due to advanced sensors and processing systems, these entities demonstrate exceptional proficiency in real time data interchange, collaborative decision-making, and proactive driving. CAVs play a crucial role in improving road safety by promptly identifying and responding to potential hazards, thus mitigating accidents resulting from human error. Furthermore, they serve as significant data sources, providing valuable information for traffic management, urban planning, and infrastructure optimization [380, 381].

7.1.1 Security Concerns on CAVs

When conducting an analysis of cyberattacks on CAVs, it is imperative to take into account both the intra vehicle and inter-vehicle perspectives [382]. Intra-vehicle systems focus predominantly on individual vehicles, tightly integrating in-vehicle networks with a variety of components to improve the performance of each vehicle. In contrast, inter-vehicle networks refer to systems that involve several vehicles and prioritize the enhancement of traffic efficiency by facilitating intimate interactions among these vehicles [383]. In relation to

intra-vehicle systems, the increased level of connectivity presents possible vulnerabilities that can be exploited by attackers. For example, malicious actors can remotely inject harmful messages into the in-vehicle network, allowing them to directly affect essential vehicle operations such as braking and steering. This poses a serious threat to the protection of the affected vehicle [384]. Effective communication plays a crucial role in determining the overall performance and dynamics of intervehicle systems. Cyberattacks have the ability to disrupt cooperative interactions among vehicles and impede the seamless flow of traffic through activities like jamming or spoofing inter-vehicle communication. The potential consequences of such interference could have a significant influence on the synchronized functioning of vehicles [385, 386].

Countermeasures are implemented with the objective of preserving the integrity of the interconnections among different vehicles while protecting against any vulnerabilities within these systems. To accomplish this objective, it is essential to provide robust access control protocols that ensure the secure input of information, allowing selective access to designated system elements. Moreover, the inclusion of network monitoring and intrusion detection systems is vital within intra-vehicle systems [387]. Cyberattacks within the inter-vehicle system might arise from either external or internal sources within the platoon of vehicles. Theoretically, the possibility of detecting external attacks is greater due to the implementation of advanced procedures that can efficiently identify and restrict the activities of an external attacker [83].

Communication in CAVs is highly dependent on vehicle sensor data, as sensors provide information about the vehicle internal status and its interaction with the external environment. To address this [388], proposed a taxonomy for detecting anomalies in sensor data for connected vehicles. This taxonomy categorizes anomalies into four components: system component failures, failures resulting from external environmental conditions due to component failures, potential cyber-attacks, and design flaws in sensor manufacturing. This proposed approach investigates the relationship between vehicle safety and sensor data anomalies [389]. proposed a framework to offer valuable recommendations and references for future industrial applications. This framework classified threats based on cybersecurity principles and evaluated CAVs in terms of their cybersecurity environments. In addition, it offered resilient strategies, attack detection, and forensic capabilities.

CAVs do not protect vehicles from external attacks such as spoofing and altering with sensor data. To address this [390], introduced an intrusion detection system based on logical data analysis, which requires fewer computational resources. This technique extracts the normal behavior of CAVs and identifies any deviations from the norm as anomalies. Maintaining the security of confidential information, such as vehicle location and owner details, is critical. However, such security-sensitive information becomes more vulnerable when uploaded to the cloud due to the lack of authentication and encryption measures. To address this problem, the authors [391] developed a fast encryption method based on security attributes that are highly secure against selective plaintext attacks [301]. proposed a fuzz testing method for CAVs to identify vulnerabilities using security metrics. This method assigns weight values to vulnerable components instead of simply expanding code coverage in gray-box testing. Doing so ensured the quality of vehicle software engineering and reduced the risk of missed test cases.

7.2 5G Connectivity

The use of 5G networks to facilitate rapid and low-latency communication between vehicles and infrastructure represents a significant advancement with the potential to fundamentally transform the field of V2X communication. These advances position the 5G as an ideal facilitator for the dynamic and data-intensive interactions that occur between vehicles and their surrounding environments [392, 393]. One of the primary benefits associated with 5G networks is their exceptional speed, surpassing that of previous generation network [394]. With anticipated data rates of up to 20 Gbps, 5G networks provide exceptionally fast connectivity, enabling vehicles to engage in real-time data exchange. Speed is crucial in applications such as autonomous driving, where split-second decisions depend heavily on the ability to process large amounts of data quickly. 5G networks have significantly reduced latency, potentially reaching a minimum of one millisecond, thus reducing the delay between data transmission and reception [395]. When a vehicle detects sudden obstructions on the road, it can quickly relay this information to nearby vehicles and infrastructure, allowing them to take immediate evasive action. The implementation of 5G networks also ensures uninterrupted connectivity in densely populated and congested environments [226]. This is particularly important in situations where multiple vehicles and infrastructure components are in simultaneous communication to effectively regulate traffic patterns and enhance navigation efficiency [396].

7.2.1 Impact of Bandwidth and Latency

The enhanced bandwidth capacity of 5G networks enables efficient transmission of large amount of data at unimaginable speed. This allows for the exchange of a wide range of information, such as high-definition video streams, sensor data, and intricate control signals. Moreover, increased bandwidth supports concurrent communication among multiple vehicles and devices, mitigating the potential for data congestion on heavily trafficked roads and in densely populated urban areas [397]. The ultra-low latency of the 5G network, often measured in milliseconds, guarantees nearly instantaneous V2X communications. This is particularly crucial in safety-critical situations where prompt reactions are required to prevent accidents. For example, a vehicle capable of identifying sudden obstacles or hazardous road conditions can promptly relay this information to neighboring vehicles and infrastructure in near real time allowing instant responses and synchronized reactions. Rapid interactions between vehicles have a significant impact on improving collision avoidance, emergency braking, and overall safety of roadways.

7.2.2 Security in 5G V2X Connectivity

The increased attack surface introduced by 5G connectivity is a significant security consideration. The sheer number of connected devices and the extensive sharing of data increase the potential for unauthorized access, data breaches, and cyberattacks. Therefore, sophisticated encryption mechanisms are essential to protect sensitive data and communications. Even if intercepted by malicious actors, data exchanged between vehicles and infrastructure remains protected by cutting-edge encryption algorithms and protocols [123, 171]. Advancements in secure authentication and identity management are another area of focus.

5G networks enable more robust and multifaceted authentication mechanisms, including multi-factor authentication and biometric identification [398]. These technologies improve vehicle and driver identity verification reducing the risk of unauthorized access and control over V2X communication. Furthermore, the emergence of blockchain technology as a potential solution for to enhance security and trust in V2X communications over 5G is currently underway [399].

The advent of V2X enabled by 5G introduces the potential for secure communication through various security techniques and facilitates the integration of AI-based technologies. A threshold-based cryptographic key exchange methodology [400] is proposed to improve trust and privacy in V2X data exchange. This proposed technique addresses the trust and privacy needs of V2X entities [69]. introduced a technique to enhance the security of 5G-V2X communication by improving key management and authentication. They used a reputation-based mechanism not only to enhance security but also to overcome the real-time limitations inherent in 5G-enabled V2X communication. This approach utilized feedback from authenticated entities within V2X to effectively manage resources.

The authors in [236] explored the potential vulnerabilities associated with the slicing of the 5G network, which allows service providers to create virtual end-to-end networks tailored to specific application requirements. To mitigate these vulnerabilities, the authors proposed a deep learning-based technique to detect intrusions in 5G-V2X network slices. This approach efficiently identifies intrusions and related security threats, offering early detection [401]. introduced an attack detection framework based on machine learning using technologies such as federated learning and smart contracts. To establish trust, the authors used a consensus mechanism based on intelligent incentive techniques. This mechanism helps select trusted nodes within the vehicular environment, ensuring an accurate assessment of trust value.

7.3 Edge Computing

Edge computing plays a pivotal role in efficiently managing the substantial volume of data generated by interconnected vehicles [402]. This is achieved by initially processing data at the edge and then transmitting relevant insights exclusively to the cloud. As a result, the network resources are optimized, leading to improved efficiency and effectiveness. The use of localized data storage keeps sensitive information close to its origin.

7.3.1 Security Challenges

The protection of edge devices and data processing presents unique challenges and requires innovative approaches to ensure the preservation of data integrity, confidentiality, and availability [403]. Edge devices are often placed in remote or publicly accessible locations, making them vulnerable to physical tampering or theft. The challenge lies in finding a balance between security measures and the limitations imposed by available resources. Establishing communication channels between edge devices and central cloud infrastructure introduces inherent vulnerabilities that could be exploited [65, 404]. In addition, edge devices play a critical role in the processing of sensitive data near their source, requiring the implementation of robust mechanisms to protect data privacy and comply with regulatory requirements [405].

Collecting real-time data in dynamic vehicular networks, such as tracking vehicle arrivals and departures, presents a critical challenge. Identifying malicious messages or activities in a limited time is a difficult task. To address this [406], have developed a method that takes advantage of edge computing to enhance storage, computing, and communication processes. Gleichzeitig, they employed blockchain technology to increase the security of the communication network. The authors in [407] have proposed a method that combines edge computing and deep learning for rapid sensor data processing and detection. In this technique, sensor data is partitioned and encrypted into multiple secret shares, each processed on an edge server for detection results. These results are then combined to produce the final outcome from the edge servers.

Traditional security methods often introduce latency. To address this [198], introduced an intelligent PKI that reduces latency through edge computing. They distributed two key pairs, validated on the basis of vehicle and RSU location information. This technique used RNN-based route prediction. The work in [408] introduced a V2X framework that uses distributed edge computing and blockchain technology. In this framework, edge servers are connected to each RSU, reducing computational cost latency through computational offloading [409]. proposed a new lifecycle for vehicular data processing, with the help of UAVs. The author used edge computing for rapid data storage and processing of vehicle data [410]. have developed an offloading technique based on vehicular edge computing, boasting high transmission probability reflected in the sensing characteristics of the vehicles.

7.4 Quantum Computing for V2X Security

The field of V2X communication and its security has witnessed the emergence of quantum computing as a highly promising domain. The need for strong security measures has risen dramatically as V2X systems continue to develop and become essential to the future of transportation. The utilization of quantum mechanics principles in quantum computing has great potential to revolutionize V2X security [176]. Implementing of cryptographic methods that are resistant to quantum attacks guarantees the maintenance of secure V2X communication. The enhanced processing capacities of quantum computers facilitate the advancement of threat modeling, the simulation of intricate cyberattacks, and the formulation of robust security systems. This allows for thorough examination and strengthening of the V2X systems [411].

7.4.1 Challenges in Quantum Computing

Although this technological advancement has the capacity to fundamentally transform encryption and data protection, there exist several challenges that require attention and resolution. Quantum computing is now in its infancy, lacking the widespread availability of practical large-scale quantum computers [412]. The use of quantum-resistant cryptographic algorithms requires meticulous strategic deliberation. quantum key distribution (QKD) [413], necessitates significant infrastructure and is susceptible to environmental factors. Technical issues include error correction and the assurance of interoperability with existing V2X systems. Furthermore, careful management of the transition to post-quantum cryptography is essential. The presence of a shortage of quantum expertise, coupled with the need

for quantum-safe testing, adds more complexity to the current scenario. The complexity is increased by the establishment of regulatory and policy frameworks.

The authors [413] proposed a QKD method incorporated with V2I, providing an efficient end-to-end communication path between QKD and V2I. Additionally, the author integrated QKD with a software-defined network and examined the security performance of QKD in the context of V2I [414]. analyzed two digital signature verification algorithms, FALCON and CRYSTALS-DILITHIUM. After an in-depth analysis, FALCON emerged as the superior algorithm in terms of verification time and signature size [415]. proposed an authentication scheme based on lattice and quantum resistance. The author employed a pseudonym-based authentication scheme due to its ability to provide both security and anonymity.

The authors in [416] identified a challenge in the deployment of post quantum certification (PQC) in V2X communication. The challenge is mainly due to the growing size of the public key and signature, increasing the overall size of the message. This, in turn, reduced the capacity of the communication channel and the transmission range. To address these issues, the authors modified the ETSI and IEEE 1609 certificates to optimize the efficiency of V2X communication [417]. recognized that traditional cryptographic standards used in V2X communication impose high latency constraints. The study highlights the effectiveness of PQC in the context of cooperative ITS.

7.5 Artificial Intelligence (AI) in V2X

The use of machine learning facilitates the implementation of predictive analytics in V2X systems, allowing anticipation of traffic patterns, congestion, and potential road hazards [418, 419]. Furthermore, traffic flow optimization is improved by using AI [420], which has the capability to analyze large amounts of traffic data. This leads to more intelligent route suggestions and improved traffic management tactics [421].

7.5.1 Addressing Security Concerns

AI-powered security mechanisms excel at detecting subtle and complex patterns that conventional methods may overlook [422]. AI quickly identifies anomalous network behaviors, such as unauthorized access attempts, unusual data flows, and aberrant vehicle trajectories. These insights provide real-time alerts, enabling rapid responses and mitigation strategies. Furthermore, AI's ability to learn from historical data enables the development of sophisticated threat models, anticipating emerging threats and vulnerabilities. AI algorithms continuously monitor network traffic and behavior to identify anomalies proactively, preventing the escalation of security breaches [423]. AI-driven automation can initiate predefined actions, such as isolating compromised components or blocking malicious traffic, after threat detection. Integrating AI into security mechanisms also enables adaptive security.

The authors in [424] have proposed an anomaly detection technique by utilizing a multi-agent reinforcement learning technique operating on 6G technology to detect cyber-attacks. Federated Learning (FL) contributes to the privacy and security of vehicles and user data within their network [425]. have developed a training model based on real-time data. FL, as part of an explainable artificial intelligence model, promotes efficient communication intel-

ligence, decentralized availability, and lightweight communication. This model also instills trust among nodes in the network.

The authors of [426] used artificial intelligence to improve trust management. In their proposed work, vehicles send messages to others, and these messages are validated by nearby vehicles. The message exchanged are evaluated using a deep learning algorithm to enable trust management. Finally, the trusted node is updated to the RSU, which verifies the authenticity of the vehicle and its message using blockchain technology [427]. have devised a framework for anomaly detection in the 6G-V2X environment. To improve anomaly prediction, the proposed technique used a deep deterministic policy gradient algorithm that analyzes multiple factors such as driving speed, driving time, etc., thus improving precision [428]. introduced an intelligent physical layer security approach that detects and analyzes the driver's behavior, communication channel characteristics, and information collected from the upper layers [429]. proposed a framework using deep neural networks for secure vehicular communication. In this method, clustering techniques are applied, allowing vehicles in specific clusters to communicate using blockchain techniques. Deep neural networks are used to identify abnormal vehicles by analyzing traffic patterns.

7.6 Reconfigurable Intelligent Surfaces (RIS) for V2X

Incorporation of RIS into automotive communication systems represents a significant breakthrough with wide-ranging consequences. The concept of intelligent reflecting surfaces (IRS), also known as RIS, uses advanced materials and technology to dynamically control electromagnetic waves, thus increasing wireless communication capabilities in diverse environments [430]. It presents a brief overview of the possible collaboration between RIS and vehicular communication, highlighting its ability to improve connectivity and improve security [431].

7.6.1 Security Concerns

Incorporation of RIS within vehicular communication systems exhibits significant potential to improve the efficiency, reliability, and protection of V2X communication. However, this conversion is not without challenges. The logistical and technical challenges presented by the complexity of the deployment, particularly within a dynamic traffic environment, are significant. It is imperative to guarantee seamless integration with preexisting infrastructure and ensure compatibility with outdated systems [432]. Furthermore, it is essential to prioritize mitigation of security risks related to RIS. The adoption of RIS is further complicated by factors such as energy efficiency, scalability, regulatory compliance, and cost considerations [433]. The ongoing research and innovation in the field intends to overcome these obstacles, making RIS a valuable technology for achieving more secure and efficient vehicular communication in the future.

The authors in [434] have identified that RIS plays a crucial role in reducing model aggregation errors in federated learning, particularly in IoT-based environments. RIS is also instrumental in strengthening the propagation of wireless signals [435]. conducted an evaluation of physical layer security in vehicular communication, considering two network types: V2V and VANET. Their evaluation reveals that security parameters depend on factors such as the number of RIS cells and the power source [61]. have determined that RIS has the capa-

bility to enhance QoS in vehicular wireless communication. They achieve this by increasing the count of V2I links, based on QoS requirements, through RIS phase shifts. These phase changes are used to transmit highly critical content and ensure the reliability of V2V communication [436]. conducted an extensive literature review on various use cases of UAV-RIS in different practical scenarios. They developed a new beamforming policy based on machine learning to enhance the performance of communication channels [437]. provided an overview of RIS-assisted backcom. Following a detailed review, they observed that the usage of RIS-assisted backcom resulted in increased channel gain, enhanced energy efficiency, and improved data transmission rates. Fortunately, with the continuous advancement of technology and the resolution of these issues, RIS holds the capacity to assume a crucial position in ensuring the reliability and authenticity of V2X communication making significant contributions towards the creation of safer and more secure vehicular environments.

7.7 Blockchain in V2X

Blockchain is a decentralized, digital ledger that securely and openly logs transactions across numerous computers. It is intended to prevent recorded information from being altered or tampered with. Transactions in a blockchain network are verified and processed by network participants, rather than by a central authority, making it resistant to fraud and censorship [438, 439]. Cryptocurrency, like Bitcoin, is the most well-known application of blockchain technology, but it also has potential applications in industries including finance, supply chain management, and voting systems. V2X communication is the term used to describe wireless communication between vehicles and other entities, including infrastructure, equipment, and other vehicles. By offering a safe and impenetrable transaction ledger, blockchain technology can be leveraged to improve the security of V2X communication [440].

The use of blockchain technology holds considerable promise in increasing the security of V2X systems. A significant application pertains to authentication, in which blockchain technology guarantees the secure identification of vehicles and devices in V2X communication. This serves to prevent illegal access and protect the privacy of data. Moreover, the use of blockchain technology facilitates the establishment of a secure mechanism for data exchange between vehicles, infrastructure, and other interconnected devices, thus increasing both the security and efficiency of vehicular transportation [194]. Furthermore, it has the potential to bring about a significant transformation in payment systems for V2X-related services such as tolls and parking, thereby guaranteeing the security and efficiency of financial operations. Furthermore, within the automotive sector, blockchain technology plays a crucial role in improving supply chain management by facilitating vehicle and component lineage monitoring to ensure authenticity and streamline defects recall management processes [445, 447].

Table 9 outlines the possible benefits of using blockchain technology for V2X security. The table suggests that the blockchain could help mitigate risks associated with cyber-attacks, unauthorized access, and data tampering by providing a decentralized, immutable, and transparent ledger to store V2X data. The integration of blockchain technology with V2X security has many discernible benefits. First, it improves security by establishing a secure and unalterable record of transactions, thereby guaranteeing the secrecy and integrity of V2X communication. Furthermore, the decentralized structure of blockchain technology

obviates the need for a central governing body to supervise and safeguard V2X communication, thus bolstering its ability to withstand any security breaches and operational breakdowns. Furthermore, the implementation of blockchain technology brings about a higher level of openness within the ecosystem. This enhanced transparency facilitates the clear monitoring of transactions and the sharing of data, ultimately promoting accountability and fostering trust between all parties involved. Finally, the implementation of blockchain-based systems has the potential to enhance efficiency through the optimization of processes, the minimization of manual intervention, and the automation of transactions. Consequently, this can lead to increased operational efficiency and reduced expenditures. Although there are many potential advantages to using blockchain in V2X security, there are also some drawbacks that must be taken into account before implementation. To decide whether blockchain is appropriate for particular applications and use cases, it is crucial to carefully weigh its benefits and drawbacks in V2X security.

7.8 Integration of IoT

By connecting vehicles, infrastructure, and devices through the internet of things, V2X communication achieves new efficiency and safety levels. Through a network of sensors, cameras, and communication systems, vehicles become sentient entities capable of exchanging real-time information about their speed, location, and driving behavior. This information exchange supports autonomous driving and allows cooperative collision avoidance [448]. The influence of the IoT extends beyond vehicles to infrastructure elements such as traffic signals and road signs, which integrate into the interconnected ecosystem and communicate valuable information to vehicles. The vast amount of data generated by IoT connected systems fuels advanced analytics and machine learning algorithms [320, 449] and [450]. Additionally, vehicles enabled with IoT can detect environmental conditions such as air quality, weather, and road surface conditions [451].

However, integrating IoT with V2X presents challenges. Ensuring the protection of sensitive data requires a robust approach to data security, privacy concerns, and cybersecurity measures. Managing the influx of vast data requires strong data management strategies while ensuring the compatibility and interoperability of various IoT devices [452] [320]. collected data on driver activity and behavior and, through an evaluation rule, constructed a behavioral model for drivers. This model identifies driver activities and issues warnings when they pose a risk to vehicle and passenger safety. The results of the driver information evaluation are presented to passengers as feedback. This model helps identify and prevent traffic accidents in vehicles enabled with IoT, improving passenger safety [453]. proposed various use cases for integrating blockchain with V2X and IoT. These cases encompass blockchain-based vehicle scoring systems, multitoken-based ownership systems, and the integration of blockchain with IoT to ensure message safety, data integrity, and security. The authors in [454] implemented a multi-protocol gateway that facilitates message transfer between different entities within the V2X environment. This gateway also ensured compatibility between protocols, accommodating variations over time. The proposed architecture promotes efficient communication between nodes, devices, and operating platforms [455]. introduced an IoT-based method to enhance vehicle security, using IoT devices for biometric authentication, such as face recognition. Unauthorized attempts to access the vehicle trigger the biometric device to capture the intruder's image, including location information,

which is then relayed to the vehicle's owner. This innovative approach effectively used IoT devices to enhance vehicle security. The authors in [456] incorporated various sensors, including alcohol detection sensors, eye blink sensors, and pressure sensors, to evaluate data and detect deviations from normal behavior patterns. If anomalies are detected, immediate warnings are issued to passengers and drivers to prevent possible vehicle accidents [457]. proposed a technique to collect real-time location and tolling information, which has a significant impact on identifying current traffic conditions. They used different technologies, including VANET and radio frequency identification (RFID), to gather this information. The goal is to use the data to navigate traffic efficiently and reach destinations without delays.

Emerging technologies such as AI, blockchain, and quantum computing offer new opportunities to strengthen V2X security. Section 8 discusses the open research issues and future directions necessary to integrate these innovations effectively.

8 Open Issues and Research Directions in V2X

It is essential to ensure the safety of V2X communications for its continued success. The primary obstacles encompass authentication and authorization, data integrity and privacy, network resilience, trust and liability allocation among stakeholders, standardization, and secure over-the-air updates. It is essential to address these issues to improve the security and reliability of V2X communication, a technology with the potential to transform the field of transportation.

8.1 Future Research Directions of V2X

Future research in V2X technology should focus on strengthening interoperability, enhancing security, and enabling intelligent, large-scale deployment. A primary area of emphasis is standardization, where unified protocols and interfaces are needed to ensure seamless communication across heterogeneous V2X systems and manufacturers. Establishing such standards will enhance reliability, scalability, and compatibility in future deployments. Security and privacy will remain critical as vehicles exchange increasing volumes of data with infrastructure and cloud services. Future studies should develop lightweight yet robust authentication, authorization, and encryption mechanisms to ensure data integrity and confidentiality without adding significant communication delays. Another key direction involves integration with autonomous driving systems, where V2X can support cooperative perception, collision prediction, and traffic optimization. Research should explore architectures that fuse communication, sensing, and decision-making to enable safer and more efficient vehicle coordination.

Moving beyond simulation studies, large-scale field testing is essential for validating real-world performance under diverse traffic and environmental conditions. Such empirical evaluation will provide insights into system reliability, latency, and security under practical constraints. The application of AI and machine ML offers new opportunities for adaptive and proactive V2X security. AI-driven anomaly detection can recognize abnormal traffic patterns, while federated learning enables distributed model training without sharing sensitive data. Combining these techniques with blockchain-based trust management can support decentralized authentication, secure identity management, and tamper-proof mes-

sage logging. Future work may also explore hybrid AI-blockchain frameworks for autonomous threat detection and distributed trust verification. Finally, to balance performance and security, future work should develop integrated performance-security models that quantify trade-offs between cryptographic strength and Quality QoS parameters such as latency, reliability, and bandwidth under varying traffic and mobility scenarios.

8.2 Future Challenges in V2X Security

The task of ensuring the security of V2X infrastructure poses significant and crucial challenges. The infrastructure in question consists of roadside devices, communication links, and cloud servers that are responsible for the processing and storage of V2X data. The occurrence of any breach within this interconnected sequence has the potential to result in loss of information, disruptions in traffic flow, and possibly even accidents. In order to safeguard V2X infrastructure, it is imperative to implement robust measures including encryption, multifactor authentication, and periodic security audits. The increasing prevalence of V2X communication has led to a corresponding increase in the vulnerability of V2X networks to cyberattacks. Potential adversaries have the ability to carry out several forms of cyberattacks, such as denial-of-service attacks, man-in-the-middle attacks, and data tampering. These malicious activities can have significant repercussions, including disrupting in network traffic and instances of data collisions. In order to mitigate these potential risks, it is imperative that V2X security solutions integrate intrusion detection systems and real-time monitoring capabilities.

Preservation privacy regarding V2X data is a crucial issue, as it involves the transmission of sensitive data, including location, speed, and driving behavior. Achieving a harmonious equilibrium between the importance of data for different entities and the protection of driver privacy requires the implementation of strong encryption, effective anonymization techniques, and the establishment of data exchange protocols based on informed consent. In the realm of privacy protection, novel technologies such as homomorphic encryption and differential privacy have the potential to contribute significantly. With the increasing prevalence of V2X communication, the management of liability concerns becomes of utmost importance. The process of attributing liability in incidents involving vehicles, infrastructure providers, or communication providers can present a multifaceted challenge. To enhance the efficacy of V2X security solutions, it is imperative to implement mechanisms that facilitate the recording and tracking of transactions for the purpose of forensic analysis and responsibility attribution. The adoption of secure measures is a significant challenge in the context of complex V2X systems, which require real-time updates and patch management. Ensuring the safety of updates require the implementation of authentication, authorization, and security procedures. Furthermore, it is imperative to incorporate robust patch management procedures into V2X systems to mitigate the risk of unauthorized manipulation.

9 Inferences on V2X Security

The security of V2X is of utmost importance in facilitating the progress and implementation of V2X technology, which facilitates communication between vehicles and their surrounding environments. Ensuring the security of V2X communications is crucial in this

field to prevent unauthorized access, malicious attacks, and a potential compromise of the safety and privacy of vehicles, passengers, and pedestrians. There are several security risks associated with V2X technology, including illegal access and control of vehicles through hacking attempts. Additionally, there are concerns about events such as jamming, eavesdropping, and spoofing, which involve the sending of deceptive messages. To mitigate these potential risks, it is imperative to implement resilient security measures, such as integrating encryption protocols to protect data during transmission, implementing authentication mechanisms to restrict access to the V2X network exclusively to authorized vehicles and devices, establishing authorization protocols to regulate entry to network components, and deploying intrusion detection systems to ensure constant surveillance and prompt response to suspicious activities. The establishment of uniform standards and certification methods is necessary to guarantee the security of V2X connections.

It is worth mentioning that IEEE has implemented standards such as IEEE 1609.2, which pertains to the management of certificates, and IEEE 1609.3, which focuses on the secure delivery of messages. Certification programs, such as the security credential management system (SCMS) of the US department of transportation, serve the purpose of validating the compliance of V2X devices and systems with defined security requirements, thereby increasing the overall level of security. Additionally, the implementation of V2X communication systems creates privacy problems as a result of the sharing of personal data such as location information and driving behavior. To address these concerns, it is imperative to develop V2X systems that incorporate pseudonyms, which are temporary identifiers that are disconnected from the true identity of the driver. Pseudonyms are utilized in order to safeguard the privacy of drivers while yet enabling effective communication in the context of V2X systems. To achieve an optimal equilibrium between security and performance is of paramount importance. The implementation of robust security protocols is of utmost importance; yet, it is important to acknowledge that these precautions may inadvertently result in the introduction of delays and latency. This can pose significant challenges, particularly in the context of safety-critical applications. Therefore, it is crucial to prioritize the achievement of a balance between security and performance in order to maintain the efficiency and safety of V2X communication. Figure 14 illustrates the inferences from V2X security.

9.1 Inferences on the Communication Technology of V2X

The security of V2X is closely intertwined with the selection of V2X communication technology. The choice of communication technology significantly influences the security framework of V2X communications, as each technology presents distinct security strengths and challenges. DSRC is a widely recognized communication technique that uses a dedicated frequency band specifically for V2X communication. The extensive implementation of this technology has resulted in thorough assessments of its security measures. The DSRC system employs various security measures, such as secure message delivery protocols, encryption, and authentication, to protect V2X communications against unauthorized access and malicious intrusions. On the other hand, C-V2X is a more recent communication system that uses cellular networks for V2X communication. Although it offers benefits such as enhanced coverage and compatibility with complex applications, it also presents inherent security concerns [458].

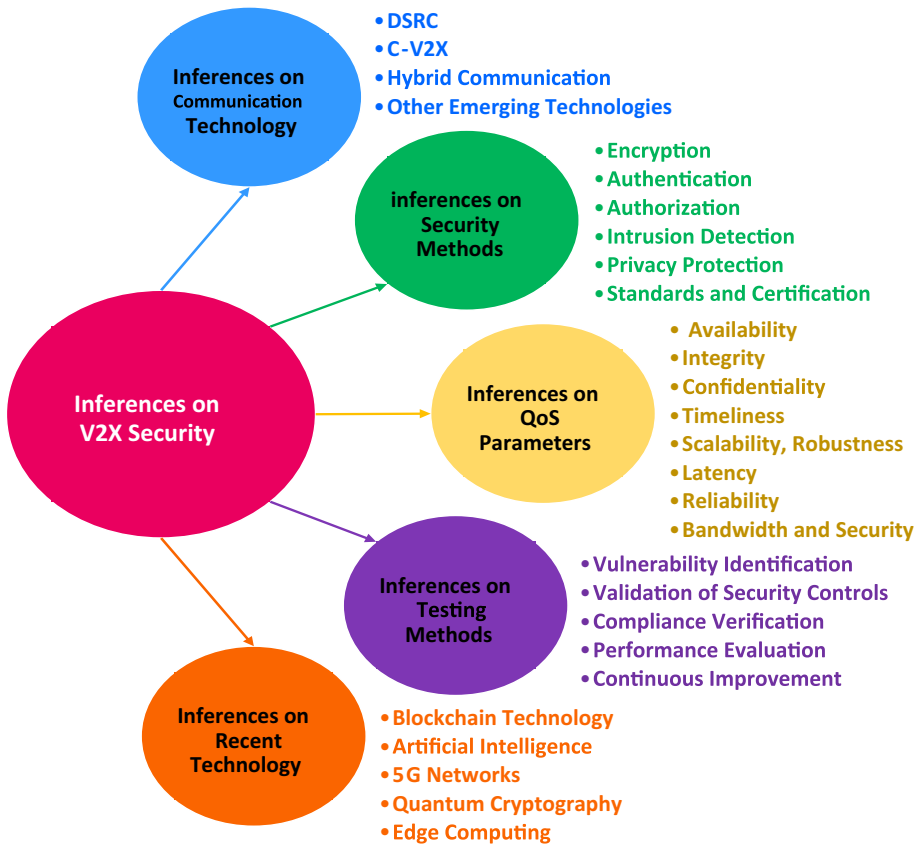


Fig. 14 Inferences on V2X security

The security of C-V2X is depends on cellular networks, which are vulnerable to malware and other attacks. The use of enhanced security measures, such as encryption and authentication, is important to guarantee the integrity and confidentiality of C-V2X communication. Hybrid communication presents a strategic methodology that integrates both the DSRC and C-V2X technologies for V2X communication. This hybridization capitalizes on the advantages of both technologies while minimizing their individual limitations. For example, DSRC has the capability to support applications that are essential for safety, but C-V2X is better suited for handling tasks that are not safety-critical. This distinction leads to the development of a V2X communication system that is more resilient and effective. However, it is imperative to conduct additional study on technologies such as Wi-Fi and Bluetooth to evaluate their suitability for V2X communication and establish the essential security protocols required to guarantee their safety.

9.2 Inferences on Security Mechanisms

Since it is important to protect V2X communications and mitigate the risks of unauthorized access, malicious attacks, and compromise of the safety and privacy of vehicles, passengers,

and pedestrians, it is imperative to implement a variety of security measures. The aforementioned precautions include the implementation of encryption, which serves a crucial function in ensuring the security of V2X communication. Encryption protects transmitted data from being intercepted and accessed by unauthorized people, effectively preventing potential assaults. This is achieved by using techniques such as AES and ECC. Authentication is a crucial security feature that ensures that only authorized vehicles and devices can gain access to the V2X network. This is achieved through the utilization of techniques like digital certificates and PKI, which serve to validate the credibility and reliability of both message senders and recipients. Authorization is a mechanism that regulates access to various components inside the V2X network. Its primary objectives are to thwart privilege-escalation attacks and ascertain vehicles and devices that are authorized to enter certain network segments. Additionally, authorization determines the permissible actions that these authorized entities can undertake within the network.

Furthermore, intrusion detection systems continuously monitor the V2X network for indications of malicious activity, alerting network administrators immediately to take preventive measures against security violations. In order to mitigate privacy concerns arising from the sharing of personal data such as location and transportation patterns, V2X systems have the capability to integrate pseudonyms. These pseudonyms are temporary identifiers that are not linked to the actual identity of the driver. By employing this approach, driver privacy can be protected while still facilitating efficient V2X communication. Furthermore, the establishment of standardized protocols and certification procedures plays a vital role in ensuring the security of V2X communication. IEEE has developed a range of V2X security standards that encompass certificate management and secure message transmission. Certification procedures, such as the SCMS used by the US department of transportation, serve the purpose of validating the adherence to well-defined security standards, hence guaranteeing the security of V2X devices and systems.

9.3 Inferences on QoS Parameters

The establishment of QoS parameters is imperative to delineate the anticipated standard of service. The characteristics in question are of the utmost importance in ensuring the security of V2X systems, as they comprise a wide range of factors that directly impact system performance. The availability metric is a crucial aspect of QoS, as it quantifies the proportion of time during which the V2X security system is in an effective state. The importance of high availability cannot be overstated, particularly in the context of emergency services that rely on V2X technology. Integrity is a critical QoS attribute that plays a crucial role in ensuring the accuracy and completeness of data transfer. Ensuring data integrity is of utmost importance, especially in safety-critical applications such as collision avoidance systems. Confidentiality is the QoS parameter that is responsible for ensuring the protection of data from unwanted access. Security measures for V2X systems should prioritize the protection of sensitive information, such as location data and personal details to maintain privacy. The responsiveness of the V2X security system is evaluated through the timeliness metric. The timely and effective resolution of security concerns is of utmost importance, particularly in the context of emergency services and applications that are vital to ensuring safety. The concept of scalability refers to evaluating the capacity of the system to effectively manage

the increased levels of traffic and demands, which arise due to the increase in the number of interconnected cars and gadgets.

The term robustness refers to the system's ability to withstand and withstand attacks and dangers, therefore guaranteeing its ability to maintain regular functioning even under unfavorable circumstances. Latency is defined as the duration of time necessary for the transmission and reception of a message. Maintenance of low latency is of utmost importance in order to mitigate the risk of security breaches arising from delays in the transmission of security messages. Reliability refers to the degree of consistency and accuracy exhibited by security procedures and processes, which play a crucial role in mitigating security breaches. The significance of bandwidth lies in its role in handling security messages that frequently consist of large amounts of data. Sufficient bandwidth is a prerequisite to mitigate the occurrence of transmission delays or failures. Security measures refer to safeguards implemented through security procedures and systems with the aim of preventing unauthorized access, data tampering, and other forms of security breaches. QoS parameters play a crucial role in ensuring the security and dependability of V2X systems. Performance requirements are explicitly defined criteria that must be fulfilled to meet the expectations and demands of users. The characteristics of availability, integrity, confidentiality, timeliness, scalability, robustness, latency, dependability, bandwidth, and security play a crucial role in ensuring the safety and security of V2X transportation systems, hence ensuring their efficient and secure operation.

9.4 Inference on Security Testing Methods

To ensure the strong security of V2X systems and efficiently mitigate any vulnerabilities, the use of rigorous testing methodologies becomes essential. Several important issues are encompassed in the insights produced by the V2X security testing approaches. Identifying vulnerabilities is of utmost importance. Methods such as penetration testing, vulnerability scan, and threat modeling play a crucial role in identifying potential vulnerabilities within V2X systems. The aforementioned discoveries provide the means to implement preventive steps, enabling the correction of detected vulnerabilities, and thus preventing prospective security concerns. The certification of security mechanisms is of significant importance. Testing methods are utilized as a mechanism for evaluating the effectiveness of security measures that have been established to protect V2X systems.

For example, penetration testing assesses the efficacy of security measures such as encryption and access management in mitigating illegal access and data breaches. Compliance verification plays a critical role. Testing procedures play a crucial role in determining the compliance of V2X systems with relevant security standards and laws, such as ISO 27,001, NIST and SAE J2735. The process of compliance verification guarantees that there is adherence to crucial security measures, thereby mitigating the likelihood of security breaches. Performance assessment is of utmost importance. Various testing methodologies can be employed to assess the efficacy of V2X security solutions. For example, load testing is a valuable method to assess the ability of a system to efficiently manage high volumes of data and traffic while maintaining optimal security measures. Testing methodologies play a significant role in the continuous improvement process. The aforementioned activities involve the identification of areas that require improvement, the facilitation of the adoption of optimal practices, and the optimization of security measures to ensure ongoing advance-

ment of V2X security. The insights obtained from testing approaches play a crucial role in maintaining the security, reliability, and resilience of V2X systems, hence enhancing their capacity to operate securely within a connected environment.

9.5 Inferences on Recent Technology

The incorporation of new technologies into the context of V2X security exhibits considerable potential to increase the safety of interconnected vehicles and infrastructure. These innovations have various implications for the security of V2X. Blockchain technology provides a secure and distributed method for ensuring the integrity and confidentiality of V2X security. The use of this technology can be used to construct a robust system to validate and secure transactions and communications between automobiles, infrastructure, and other entities. By implementing these measures, the preservation of data integrity and the verification of communication authenticity are effectively achieved, hence effectively preventing unauthorized access or tampering. AI plays a pivotal role in the identification and mitigation of security issues within V2X connections. Artificial intelligence systems have the ability to efficiently analyze large amount of data in real time, enabling the detection of trends and abnormalities that could potentially indicate security threats. Artificial intelligence has the capability to function as a proactive alert system, promptly identifying potential security risks and independently implementing appropriate remedial measures.

The implementation of 5G networks yields significant enhancements in V2X communications. 5G networks offer improved data interchange speed and dependability in real-time due to the higher capacity and lower latency they provide. Ensuring safety is of the utmost importance, especially in areas where safety is important. Additionally, the integration of 5G technology facilitates the deployment of sophisticated security measures such as encryption, authentication, and access control. Quantum cryptography improves the security of V2X communications to a significant extent. Through the utilization of quantum key distribution, encryption keys are generated and distributed, making them very difficult to decipher. Quantum cryptography provides a resilient approach to data encryption, effectively thwarting unlawful access and tampering. Edge computing has the ability to provide real-time processing and analysis of V2X data, enhancing the system's ability to promptly detect and respond to security risks. This technological advancement enables the system to rapidly identify and address any security threats, hence increasing the overall level of protection. The integration of these emerging technologies into the security systems of V2X communication holds significant promise to improve the overall safety of interconnected cars and associated infrastructure. Implementing of blockchain technology, AI, 5G networks, IoT and edge computing enables V2X security to effectively defend the integrity, confidentiality, availability, and scalability of the system. These technologies play a crucial role in ensuring the security, dependability, and speed of V2X communications, eventually contributing to the secure and efficient functioning of interconnected vehicles.

10 Conclusion

V2X communication enables seamless data exchange between vehicles and their environment, but it also introduces significant security and privacy challenges such as unauthorized access, malicious attacks, and data breaches. This study systematically analyzed the V2X threats using the STRIDE threat model and mapped them to corresponding mitigation mechanisms, including encryption, authentication, authorization, intrusion detection, and privacy-preserving methods. By examining QoS considerations and relevant security standards, the research emphasizes the importance of balancing safety-critical communication with robust protection. These findings offer practical direction for researchers and practitioners to design scalable, interoperable, and future-ready V2X security architectures.

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References

1. Sobczak, M. (2022). The concept of implementing the vehicle-to-vehicle (v2v) technology in the domestic market. PhD thesis, Zakł ad Inz'ynierii Transportu Lotniczego i Teleinformatyki.
2. Karoui, M., Mannoni, V., Denis, B., & Mayrargue, S. (2022). Performance analysis of v2x-based systems for improved vulnerable road users safety. In: *2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)* (pp. 3368–3373) IEEE.
3. Rakouth, H., Alexander, P., Brown, A. J., Kosiak, W., Fukushima, M., Ghosh, L., Hedges, C., Kong, H., Kopetzki, S., & Siripurapu, R. (2013). V2x communication technology: Field experience and comparative analysis. In: *Proceedings of the FISITA 2012 World Automotive Congress: Volume 12: Intelligent Transport System (ITS) & Internet of Vehicles* (pp. 113–129), Springer.
4. Boran, C., Bostrom, O., Jacobson, B. J., Lie, A., & Sander, U. (2012). Saving lives with v2x versus on-board sensing systems-which will be more effective? Technology leadership brief. Technical report, SAE Technical Paper.
5. Petrov, T., Sevcik, L., Pocta, P., & Dado, M. (2021). A performance benchmark for dedicated short-range communications and lte-based cellular-v2x in the context of vehicle-to-infrastructure communication and urban scenarios. *Sensors*, 21(15), Article 5095.
6. Moradi-Pari, E., Tian, D., Bahramgiri, M., Rajab, S., & Bai, S. (2023). Dsrc versus lte-v2x: Empirical performance analysis of direct vehicular communication technologies. *IEEE Transactions on Intelligent Transportation Systems*. <https://doi.org/10.1109/TITS.2023.3247339>
7. Khezri, R., S. D. (2022). A review on implementation of vehicle to everything (v2x): Benefits, barriers and measures. In: *2022 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe)* (pp. 1–6). IEEE.
8. Farag, M. M., & Rakha, H. A. (2023). Development and evaluation of a cellular vehicle-toeverything enabled energy-efficient dynamic routing application. *Sensors*, 23(4), Article 2314.
9. Affane, M., Satori, A. R., Boutazart, H., Ezzine, Y., & Satori, A. (2024). Machine learning-based attack detection for wireless sensor network security using hidden Markov models. *Wireless Personal Communications*, 135(4), 1965–1992.

10. Hasrouny, H., Samhat, A. E., Bassil, C., & Laouti, A. (2017). Vanet security challenges and solutions: A survey. *Vehicular Communications*, 7, 7–20.
11. Kumar, A., Bansal, M. (2017). A review on vanet security attacks and their countermeasure. In: *2017 4th International Conference on Signal Processing, Computing and Control (ISPCC)* (pp. 580–585). IEEE.
12. Wang, J., Shao, Y., Ge, Y., & Yu, R. (2019). A survey of vehicle to everything (v2x) testing. *Sensors*, 19(2), Article 334.
13. Sheikh, M. S., Liang, J., & Wang, W. (2019). A survey of security services, attacks, and applications for vehicular ad hoc networks (vanets). *Sensors*, 19(16), Article 3589.
14. Lu, Z., Qu, G., & Liu, Z. (2018). A survey on recent advances in vehicular network security, trust, and privacy. *IEEE Transactions on Intelligent Transportation Systems*, 20(2), 760–776.
15. Huang, J., Fang, D., Qian, Y., & Hu, R. Q. (2020). Recent advances and challenges in security and privacy for v2x communications. *IEEE Open Journal of Vehicular Technology*, 1, 244–266.
16. Gyawali, S., Xu, S., Qian, Y., & Hu, R. Q. (2020). Challenges and solutions for cellular based v2x communications. *IEEE Communications Surveys and Tutorials*, 23(1), 222–255.
17. Masood, A., Lakew, D. S., & Cho, S. (2020). Security and privacy challenges in connected vehicular cloud computing. *IEEE Communications Surveys and Tutorials*, 22(4), 2725–2764.
18. Kim, K., Kim, J. S., Jeong, S., Park, J.-H., & Kim, H. K. (2021). Cybersecurity for autonomous vehicles: Review of attacks and defense. *Computers & Security*, 103, Article 102150.
19. Asra, S. A. (2022). Security issues of vehicular ad hoc networks (vanet): A systematic review. *TIERs Information Technology Journal*, 3(1), 17–27.
20. Sedar, R., Kalalas, C., Vázquez-Gallego, F., Alonso, L., & Alonso-Zarate, J. (2023). A comprehensive survey of v2x cybersecurity mechanisms and future research paths. *IEEE Open Journal of the Communications Society*, 4, 325–391. } <https://doi.org/10.1109/OJCOMS.2023.3239115>
21. Yoshizawa, T., Singel'ee, D., Muehlberg, J. T., Delbruel, S., Taherkordi, A., Hughes, D., & Preneel, B. (2023). A survey of security and privacy issues in v2x communication systems. *ACM Computing Surveys*, 55(9), 1–36.
22. Pham, M., & Xiong, K. (2021). A survey on security attacks and defense techniques for connected and autonomous vehicles. *Computers & Security*, 109, Article 102269.
23. Al-Shareeda, M. A., Manickam, S., Mohammed, B. A., Al-Mekhlafi, Z. G., Qtaish, A., Alzahrani, A. J., Alshammari, G., Sallam, A. A., & Almekhlafi, K. (2022). Cm-cppa: Chaotic map-based conditional privacy-preserving authentication scheme in 5genabled vehicular networks. *Sensors*, 22(13), Article 5026.
24. Ali, A. M., Ngadi, M. A., Sham, R., & Al Barazanchi, I. I. (2023). Enhanced qos routing protocol for an unmanned ground vehicle, based on the aco approach. *Sensors*, 23(3), Article 1431.
25. Kabil, A., Rabieh, K., Kaleem, F., & Azer, M. A. (2022). Vehicle to pedestrian systems: Survey, challenges and recent trends. *IEEE Access*, 10, 123981–123994.
26. Barrachina, J., Sanguesa, J. A., Fogue, M., Garrido, P., Martinez, F. J., Cano, J. C., Calafate, C. T., & Manzoni, P. (2013). V2x-d: A vehicular density estimation system that combines v2v and v2i communications. In: *2013 IFIP Wireless Days (WD)* (pp. 1–6) IEEE.
27. Kim, G. H., Pae, D. S., Ahn, W. J., Ko, K. S., Lim, M. T., & Kang, T. K. (2020). Vehicle positioning system using v2x that combines v2v and v2i communications. In: *IOP Conference Series: Materials Science and Engineering* (vol. 922, p. 012009) IOP Publishing.
28. Zhenyu, L., Lin, P., Konglin, Z., & Lin, Z. (2015). Design and evaluation of v2x communication system for vehicle and pedestrian safety. *The Journal of China Universities of Posts and Telecommunications*, 22(6), 18–26.
29. Jang, W. M. (2022). The 5g cellular downlink v2x implementation using v2n with spatial modulation. *IEEE Access*, 10, 129105–129115.
30. Rosenstatter, T., Olovsson, T., & Almgren, M. (2021). V2c: A trust-based vehicle to cloud anomaly detection framework for automotive systems. In: *Proceedings of the 16th International Conference on Availability, Reliability and Security* (pp. 1–10).
31. Rajasekaran, A. S., Azees, M., & Al-Turjman, F. (2023). A comprehensive survey on security issues in vehicle-to-grid networks. *Journal of Control and Decision*, 10(2), 150–159.
32. Kim, S., & Kim, B.-J. (2020). Crash risk-based prioritization of basic safety message in dsrc. *IEEE Access*, 8, 211961–211972. <https://doi.org/10.1109/ACCESS.2020.3039516>
33. Li, H., Liu, F., Zhao, Z., & Karimzadeh, M. (2022). Effective safety message dissemination with vehicle trajectory predictions in v2x networks. *Sensors*, 22(7), Article 2686.
34. Brandl, O. (2016). V2x traffic management. *E & I Elektrotechnik Und Informationstechnik*, 133(7), 353–355.
35. Magdum, S. S., Franklin, A., Tamma, B. R., & Pawar, D. S. (2020). Safenav: a cooperative v2x system using cellular and 802.11 p based radios opportunistically for safe navigation. In: *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)* (pp. 1–6), IEEE.

36. Maglogiannis, V., Naudts, D., Hadiwardoyo, S., Van Den Akker, D., MarquezBarja, J., & Moerman, I. (2021). Experimental v2x evaluation for c-v2x and its-g5 technologies in a real-life highway environment. *IEEE Transactions on Network and Service Management*, 19(2), 1521–1538.
37. Miao, L., Chien, S. C., Chang, F. C., & Hua, K. L. (2022). C-v2x solution for spat application and maintenance. In: *2022 IEEE International Conference on Consumer Electronics-Taiwan* (pp. 405–406), IEEE.
38. Storck, C. R., & Duarte-Figueiredo, F. (2018). 5 g v2x ecosystem providing entertainment on board using mm wave communications. In: *2018 IEEE 10th Latin-American Conference on Communications (LATINCOM)* (pp. 1–6), IEEE.
39. Kiela, K., Barzdenas, V., Jurgo, M., Macaitis, V., Rafanavicius, J., Vasjanov, A., Kladovscikov, L., & Navickas, R. (2020). Review of v2xiot standards and frameworks for its applications. *Applied Sciences*. <https://doi.org/10.3390/app10124314>
40. Chen, F., & Zhao, D. (2022). Computing technology in autonomous vehicle. *Advanced driver assistance systems and autonomous vehicles: From Fundamentals to Applications* (pp. 49–114). Springer.
41. Kwon, Y. (2016). Improving multi-channel wave-based v2x communication to support advanced driver assistance system (adas). *International Journal of Automotive Technology*, 17, 1113–1120.
42. Hecker, T., Zech, J., Sch'aufele, B., Gr'afe, R., & Radusch, I. (2011). Model car testbed for development of v2x applications. *Journal of Communication*, 6(1), 115–124.
43. Kokuti, A., Hussein, A., Mar'in-Plaza, P., La Escalera, A., & Garc'ia, F. (2017). V2x communications architecture for off-road autonomous vehicles. In: *2017 IEEE International Conference on Vehicular Electronics and Safety (icves)* (pp. 69–74), IEEE.
44. Ben Khadra, A. (2023). Road traffic management using vehicle-to-everything (v2x) communication.
45. Zhao, L., Chai, H., Han, Y., Yu, K., & Mumtaz, S. (2022). A collaborative v2x data correction method for road safety. *IEEE Transactions on Reliability*, 71(2), 951–962.
46. Soto, I., Calderon, M., Amador, O., & Uruen'a, M. (2022). A survey on road safety and traffic efficiency vehicular applications based on c-v2x technologies. *Vehicular Communications*, 33, Article 100428.
47. Motallebiaraghi, F., Rabinowitz, A., Rojas, J. F., Kadav, P., Miller, D. A., Bradley, T., Meyer, R., & Asher, Z. (2023). Autonomous eco-driving evaluation of an electric vehicle on a chassis dynamometer. Technical report, SAE Technical Paper.
48. Liu, Q., Song, M., Xv, X., & Qiu, J. (2021). High accuracy positioning for c-v2x. In: *IOP Conference Series: Earth and Environmental Science* (vol. 693, p. 012100), IOP Publishing.
49. Sch'orner, P., Conzelmann, M., Fleck, T., Zofka, M., & Z'ollner, J. M. (2021). Park my car! automated valet parking with different vehicle automation levels by v2x connected smart infrastructure. In: *2021 IEEE International Intelligent Transportation Systems Conference (ITSC)* (pp. 836–843), IEEE.
50. Agbaje, P., Anjum, A., Mitra, A., Oseghale, E., Bloom, G., & Olufowobi, H. (2022). Survey of interoperability challenges in the internet of vehicles. *IEEE Transactions on Intelligent Transportation Systems*, 23(12), 22838–22861.
51. Sindhwan, R., Bhatnagar, A., Soni, A., Sisodia, A., Singh, P. L., Kaushik, V., Sharma, S., & Phanden, R. K. (2021). Design and optimization of suspension for formula society of automotive engineers (fsae) vehicle. *Materials Today: Proceedings*, 38, 229–233.
52. Anwar, W., Franchi, N., & Fettweis, G. (2019). Physical layer evaluation of v2x communications technologies: 5 g nr-v2x, lte-v2x, ieee 802.11 bd, and ieee 802.11 p. In: *2019 IEEE 90th Vehicular Technology Conference (VTC2019-Fall)* (pp. 1–7), IEEE.
53. Uzair, M. (2022). Vehicular wireless communication standards: Challenges and comparison. *International Journal of Electrical and Computer Engineering Systems*, 13(5), 379–397.
54. Chen, S., Hu, J., Zhao, L., Zhao, R., Fang, J., Shi, Y., & Xu, H. (2023). C-v2x industrial developments and applications. *Cellular Vehicle-to-Everything (C-V2X)* (pp. 329–357). Springer.
55. Twardokus, G., & Rahbari, H. (2022). Vehicle-to-nothing? securing c-v2x against protocol-aware dos attacks. In: *IEEE INFOCOM 2022-IEEE Conference on Computer Communications* (pp. 1629–1638). IEEE.
56. McCarthy, C., Harnett, K. (2014). : National institute of standards and technology (nist) cybersecurity risk management framework applied to modern vehicles. Technical report, United States. National Highway Traffic Safety Administration.
57. Khezri, R., Steen, D. (2022). : Vehicle to everything (v2x)-a survey on standards and operational strategies. In: *2022 IEEE International Conference on Environment and Electrical Engineering and 2022 IEEE Industrial and Commercial Power Systems Europe (EEEIC/I&CPS Europe)* (pp. 1–6). IEEE.
58. Ghosh, U., Maziku, H., Gupta, H. P., Sikdar, B., & Rodrigues, J. J. (2022). Security, trust, and privacy solutions for intelligent internet of vehicular things part I. *IEEE Consumer Electronics Magazine*, 11(6), 39–40.

59. Ghosh, U., Maziku, H., Gupta, H. P., Sikdar, B., & Rodrigues, J. J. (2022). Security, trust and privacy solutions for intelligent internet of vehicular things part II. *IEEE Consumer Electronics Magazine*, 12(1), 68–69.
60. Abou-Zeid, H., Pervez, F., Adinoyi, A., Aljlal, M., & Yanikomeroglu, H. (2019). Cellular v2x transmission for connected and autonomous vehicles standardization, applications, and enabling technologies. *IEEE Consumer Electronics Magazine*, 8(6), 91–98.
61. Chen, Y., Wang, Y., Zhang, J., & Di Renzo, M. (2021). Qos-driven spectrum sharing for reconfigurable intelligent surfaces (riss) aided vehicular networks. *IEEE Transactions on Wireless Communications*, 20(9), 5969–5985.
62. Quan, H.-D., Zhao, B., Yin, Z.-Q., & He, F. (2012). Computation and analysis of vehicular frequency-hopping radio ber under cosite interference. *Dianxun Jishu = Telecommunications Engineering*, 52(8), 1232–1238.
63. Ligo, A. K., & Peha, J. M. (2019). Spectrum for v2x: Allocation and sharing. *IEEE Transactions on Cognitive Communications and Networking*, 5(3), 768–779.
64. Eggers, K. J., Oertel, R., & Hesse, M. (2022). Local traffic safety analyzer (ltsa)-improved road safety and optimized signal control for future urban intersections.
65. Abdel Hakeem, S. A., Hady, A. A., & Kim, H. (2020). 5g-v2x: Standardization, architecture, use cases, network-slicing, and edge-computing. *Wireless Networks*, 26, 6015–6041.
66. Ghafoor, K. Z., Guizani, M., Kong, L., Maghdid, H. S., & Jasim, K. F. (2019). Enabling efficient coexistence of dsrc and c-v2x in vehicular networks. *IEEE Wireless Communications*, 27(2), 134–140.
67. Yadav, S., & Rishi, R. Joint mode selection and resource allocation for cellular v2x communication using distributed deep reinforcement learning under 5 g and beyond networks. Available at SSRN 4356711.
68. Ramakrishnan, M. (2023). Signature based v2x communication and authentications using resourceful signcryption and optimised ecc.
69. Hakeem, S. A. A., & Kim, H. (2023). Authentication and encryption protocol with revocation and reputation management for enhancing 5g-v2x security. *Journal of King Saud University - Computer and Information Sciences*, 35(7), Article 101638.
70. Chen, S., Li, Q., Wang, Y., Xu, H., & Jia, X. (2021). C-v2x equipment identification management and authentication mechanism. *China Communications*, 18(8), 297–306.
71. Chen, S., Hu, J., Zhao, L., Zhao, R., Fang, J., Shi, Y., & Xu, H. (2023). V2x network architecture and standards system. *Cellular Vehicle-to-Everything (C-V2X)* (pp. 81–116). Springer.
72. An, N., Yang, F., Cheng, L., Song, J., & Han, Z. (2023). Free space optical communications for intelligent transportation systems: Potentials and challenges. *IEEE Vehicular Technology Magazine*. <https://doi.org/10.1109/MVT.2023.3244032>
73. Yoshizawa, T., & Preneel, B. (2019). Survey of security aspect of v2x standards and related issues. In: *2019 IEEE Conference on Standards for Communications and Networking (CSCN)* (pp. 1–5). IEEE.
74. Noori, H., Shen, R., Khandakar, A., De Geuser, L., & Michelson, D. G. (2023). Detection and reporting of wireless channel congestion and interference in connected vehicle networks. *Transportation Research Record: Journal of the Transportation Research Board*. <https://doi.org/10.1177/03611981221112421>
75. Ivanov, I., Maple, C., Watson, T., & Lee, S. (2018). Cyber security standards and issues in v2x communications for internet of vehicles.
76. Ieee standard for. (2011). Layer 3 transport protocol for time-sensitive applications in local area networks. *IEEE Std, 1733–2011*, 1–21. <https://doi.org/10.1109/IEEESTD.2011.5762653>
77. Ejaz, W., Anpalagan, A., Ejaz, W., & Anpalagan, A. (2019). Communication technologies and protocols for internet of things. Internet of things for smart cities: Technologies. *Big Data and Security*, 17–30.
78. Harounabadi, M., Soleymani, D. M., Bhadauria, S., Leyh, M., & Roth-Mandutz, E. (2021). V2x in 3gpp standardization: Nr sidelink in release-16 and beyond. *IEEE Communications Standards Magazine*, 5(1), 12–21.
79. Chen, S., Hu, J., Shi, Y., Peng, Y., Fang, J., Zhao, R., & Zhao, L. (2017). Vehicle-to-everything (v2x) services supported by lte-based systems and 5g. *IEEE Communications Standards Magazine*, 1(2), 70–76.
80. Chen, W., Lin, X., Lee, J., Toskala, A., Sun, S., Chiasserini, C. F., & Liu, L. (2023). Guest editorial special issue on 3gpp technologies: 5 g-advanced and beyond. *IEEE Journal on Selected Areas in Communications*, 41(6), 1587–1591.
81. Ansari, K. (2021). Joint use of Dsrc and c-v2x for v2x communications in the 5.9 ghz its band. *IET Intelligent Transport Systems*, 15(2), 213–224.
82. Ali, Z., Lag'en, S., Giupponi, L., & Rouil, R. (2021). 3gpp Nr v2x mode 2: Overview, models and system-level evaluation. *Ieee Access : Practical Innovations, Open Solutions*, 9, 89554–89579.
83. Cui, J., Sabaliauskaite, G., Liew, L. S., Zhou, F., & Zhang, B. (2019). Collaborative analysis framework of safety and security for autonomous vehicles. *Ieee Access : Practical Innovations, Open Solutions*, 7, 148672–148683.

84. Liu, Y., Yang, X., Li, M., Wu, M., Sun, C., & Zhou, S. (2022). On the criteria for cybersecurity and risk assessment based on iso/sae 21434 for the application of autonomous vehicle. In: *2022 International Conference on Computer Science, Information Engineering and Digital Economy (CSIEDE 2022)* (pp. 134–143). Atlantis Press.
85. Aliwa, E., Rana, O., Perera, C., & Burnap, P. (2021). Cyberattacks and countermeasures for in-vehicle networks. *ACM Computing Surveys*, 54(1), 1–37.
86. Mavromatis, I., Tassi, A., & Piechocki, R. J. (2019). Operating its-g5 dsrc over unlicensed bands: A city-scale performance evaluation. In: *2019 IEEE 30th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)* (pp. 1–7). IEEE.
87. Mannoni, V., Berg, V., Sesia, S., & Perraud, E. (2019). A comparison of the v2x communication systems: Its-g5 and c-v2x. In: *2019 IEEE 89th Vehicular Technology Conference (VTC2019-Spring)* (pp. 1–5). IEEE.
88. Tassi, A., Mavromatis, I., & Piechocki, R. J. R. (2019). A dataset of full-stack its-g5 Dsrc communications over licensed and unlicensed bands using a large-scale urban testbed. *Data in Brief*, 25, 104368.
89. Bissmeyer, N., Dam, J. F., Zimmermann, C., & Eckert, K. (2018). Security in hybrid vehicular communication based on its-g5, lte-v, and mobile edge computing. In: *AmE 2018-Automotive Meets Electronics; 9th GMM-Symposium* (pp. 1–6). VDE.
90. Sunuwar, D., Kim, S., & Reyes, Z. (2023). Is 30 mhz enough for c-v2x? arXiv preprint arXiv:2302.09536.
91. Anjinappa, C. K., & Guvenc, I. (2018). Millimeter-wave v2x channels: Propagation statistics, beamforming, and blockage. In: *2018 IEEE 88th Vehicular Technology Conference (VTC-Fall)* (pp. 1–6). IEEE.
92. Oh, I., Batzorig, M., Duulga, B., & Yim, K. (2022). Hardware-software interworking real-time v2x dynamic analysis method. In: *Advances on Broad-Band Wireless Computing, Communication and Applications: Proceedings of the 17th International Conference on Broad-Band Wireless Computing, Communication and Applications (BWCCA-2022)*, pp. 215–223 Springer.
93. Al-Shareeda, M. A., Manickam, S., Mohammed, B. A., Al-Mekhlafi, Z. G., Qtaish, A., Alzahrani, A. J., Alshammari, G., Sallam, A. A., & Almekhlafi, K. (2022). Provably secure with efficient data sharing scheme for fifth-generation (5g)-enabled vehicular networks without road-side unit (rsu). *Sustainability*, 14(16), Article 9961.
94. Chen, S., Hu, J., Zhao, L., Zhao, R., Fang, J., Shi, Y., & Xu, H. (2023). Lte-v2x technology. *Cellular Vehicle-to-Everything (C-V2X)* (pp. 117–172). Springer.
95. Xiong, W., & Lagerström, R. (2019). Threat modeling—a systematic literature review. *Computers & Security*, 84, 53–69.
96. Furtado, M. D., Mustrall, R. D., & Liu, H. (2018). Threat analysis of the security credential management system for vehicular communications. In: *2018 IEEE International Symposium on Technologies for Homeland Security (HST)*, pp. 1–5 IEEE.
97. Farooqui, M. N. I., Arshad, J., & Khan, M. M. (2022). A layered approach to threat modeling for 5g-based systems. *Electronics*, 11(12), Article 1819.
98. Kim, S., Shrestha, R., Kim, S., & Shrestha, R. (2020). Security and privacy in intelligent autonomous vehicles. *Automotive Cyber Security: Introduction Challenges and Standardization*, 35–66.
99. Van Landuyt, D., & Joosen, W. (2021). A descriptive study of assumptions in Stride security threat modeling. *Software and Systems Modeling*, 1–18.
100. Krayani, A., Barabino, G., Marcenaro, L., & Regazzoni, C. (2023). Integrated sensing and communication for joint gps spoofing and jamming detection in vehicular v2x networks. *ArXiv Preprint arXiv:230200576*.
101. Awaji, T., Tashima, R., & Kubo, R. (2022). Detection and mitigation of data tampering attacks for cooperative acc systems based on c-v2x. In: *2022 IEEE International Conference on Consumer Electronics-Taiwan* (pp. 149–150). IEEE.
102. Azam, F., Kumar, S., Yadav, K., Priyadarshi, N., & Padmanaban, S. (2020). An outline of the security challenges in vanet. In: *2020 IEEE 7th Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON)* (pp. 1–6). IEEE.
103. Sharma, P. K., Vohra, D., & Rathore, S. (2022). Security and privacy in v2x communications: How can collaborative learning improve cybersecurity? *IEEE Network*, 36(3), 32–39.
104. Li, Y., Hou, R., Lui, K. S., & Li, H. (2018). An mec-based dos attack detection mechanism for c-v2x networks. In: *2018 IEEE Global Communications Conference (GLOBECOM)* (pp. 1–6). IEEE.
105. Kuri, S. K., Islam, T., Jaskolka, J., & Ibnkahla, M. (2022). A threat model and security recommendations for iot sensors in connected vehicle networks. In: *2022 IEEE 95th Vehicular Technology Conference: (VTC2022-Spring)* (pp. 1–5). IEEE.
106. Hasan, R., & Hasan, R. (2021). Towards a threat model and privacy analysis for v2p in 5 g networks. In: *2021 IEEE 4th 5G World Forum (5GWF)* (pp. 383–387). IEEE.

107. Li, Q. (2023). A v2v identity authentication and key agreement scheme based on identity-based cryptograph. *Future Internet*, 15(1), Article 25.
108. Brecht, B., Theriault, D., Weimerskirch, A., Whyte, W., Kumar, V., Hehn, T., & Goudy, R. (2018). A security credential management system for v2x communications. *IEEE Transactions on Intelligent Transportation Systems*, 19(12), 3850–3871.
109. Dey, M. R., & Patra, M. (2023). Hand it over carefully: Security breach during handover in 5 g-v2x. In: *2023 15th International Conference on COMmunication Systems & NETworkS (COMSNETS)* (pp. 360–363). IEEE.
110. Irshad, A., Shafiq, M., Chaudhry, S. A., & Usman, M. (2022). Security Hardened and Privacy Preserved Vehicle-to-Everything (V2X) Communication. Hindawi.
111. Afdhal, A., Ahmadiar, A., & Adriman, R. (2022). Sybil attack detection on its-v2x system using a realistic traffic model-based approach. In: *2022 IEEE International Conference on Communication, Networks and Satellite (COMNETSAT)* (pp. 333–338). IEEE.
112. Khan, A. S., Javed, Y., Saqib, R. M., Ahmad, Z., Abdullah, J., Zen, K., Abbasi, I. A., & Khan, N. A. (2022). Lightweight multifactor authentication scheme for nextgen cellular networks. *IEEE Access*, 10, 31273–31288.
113. Kim, J.-M., & Moon, J.-K. (2014). Secure authentication system for hybrid cloud service in mobile communication environments. *International Journal of Distributed Sensor Networks*, 10(2), Article 828092.
114. Jadav, N. K., Gupta, R., Tanwar, S., & Bhattacharya, P. (2022). Intelligent garlic routing for securing data exchange in v2x communication. In: *2022 IEEE Globecom Workshops (GC Wkshps)* (pp. 286–291). IEEE.
115. Molloy, P., Khodaei, M., Hallgren, P., Thenorio, A., & Papadimitratos, P. (2021). Secprotobuf: Implicit message integrity provision in heterogeneous vehicular systems. In: *2021 IEEE Vehicular Networking Conference (VNC)* (pp. 190–193). IEEE.
116. Mahmood, A., Zhang, W. E., Sheng, Q. Z., Siddiqui, S. A., & Aljubairy, A. (2019). Trust management for software-defined heterogeneous vehicular ad hoc networks. *Security, privacy and trust in the IoT environment*, 203–226.
117. Kumar, A. (2022). Replay attack detection in vanets using machine learning algorithm. PhD thesis, University of Windsor (Canada).
118. Sasikaladevi, N., Geetha, K., Aarthi, S., & Mala, C. (2022). Energy-efficient privacypreserving vehicle registration (entrance) protocol for v2x communication in Vanet. *Cloud IoT* (pp. 185–196). Chapman and Hall/CRC.
119. He, C., Luan, T. H., Lu, R., Su, Z., & Dong, M. (2022). Security and privacy in vehicular digital twin networks: Challenges and solutions. *IEEE Wireless Communications*, 30(4), 154–160. } } <https://doi.org/10.1109/MWC.002.2200015>
120. Miller, C., & Valasek, C. (2015). Remote exploitation of an unaltered passenger vehicle. *Black Hat USA, 2015*(S 91), 1–91.
121. Wang, J., & Liu, J. (2021). Location hijacking attack in software-defined space–air– ground-integrated vehicular network. *IEEE Internet of Things Journal*, 9(8), 5971–5981.
122. Klausling, L. (2022). Impact of motion and visual presentation on the performance of a vehicle roll-tilt task in a virtual reality and motion simulator system.
123. Sharma, V., You, I., & Guizani, N. (2020). Security of 5 g-v2x: Technologies, standardization, and research directions. *IEEE Network*, 34(5), 306–314.
124. Alnasser, A., Sun, H., & Jiang, J. (2019). Recommendation-based trust model for vehicle-to-everything (v2x). *IEEE Internet of Things Journal*, 7(1), 440–450.
125. Taha, M., & Eisenbarth, T. (2015). *Implementation attacks on post-quantum cryptographic schemes*. Cryptology ePrint Archive.
126. Salehi, S., Sheaves, T., Gubbi, K. I., Beheshti, S. A., PD, S. M., Rafatirad, S., Sasan, A., Mohsenin, T., & Homayoun, H. (2022). Neuromorphic-enabled security for iot. In: *2022 20th IEEE Interregional NEWCAS Conference (NEWCAS)* (pp. 153– 157). IEEE.
127. Zhou, M., Qin, Z., Lin, X., Hu, S., Wang, Q., & Ren, K. (2019). Hidden voice commands: Attacks and defenses on the Vcs of autonomous driving cars. *IEEE Wireless Communications*, 26(5), 128–133.
128. Sun, X., Yu, F. R., & Zhang, P. (2021). A survey on cyber-security of connected and autonomous vehicles (cavs). *IEEE Transactions on Intelligent Transportation Systems*, 23(7), 6240–6259.
129. Dibaei, M., Zheng, X., Xia, Y., Xu, X., Jolfaei, A., Bashir, A. K., Tariq, U., Yu, D., & Vasilakos, A. V. (2021). Investigating the prospect of leveraging blockchain and machine learning to secure vehicular networks: A survey. *IEEE Transactions on Intelligent Transportation Systems*, 23(2), 683–700.
130. McGrew, D. A., & Fluhrer, S. R. (2005). *Multiple forgery attacks against message authentication codes*. Cryptology ePrint Archive.

131. Chhatwal, S. S., & Sharma, M. (2015). Detection of impersonation attack in vanets using buck filter and vanet content fragile watermarking (vcfw). In: *2015 International Conference on Computer Communication and Informatics (ICCCI)* (pp. 1–5). IEEE.
132. Zhang, S., Cheng, Q., & Wang, X. (2010). Impersonation attack on two identity-based authenticated key exchange protocols. In: *2010 WASE International Conference on Information Engineering* (Vol. 2, pp. 113–116). IEEE.
133. Bradai, A., & Afifi, H. (2011). A framework using ibc achieving non-repudiation and privacy in vehicular network. In: *2011 Conference on Network and Information Systems Security* (pp. 1–6). IEEE.
134. Song, Y., Venkataramani, A., & Gao, L. (2013). Identifying and addressing protocol manipulation attacks in secure bgp. In: *2013 IEEE 33rd International Conference on Distributed Computing Systems* (pp. 550–559) <https://doi.org/10.1109/ICDCS.2013.32>
135. Karadogan, I., & Das, R. (2015). Analysis of attack types on tcp/ip based networks' via exploiting protocols. In: *2015 23rd Signal Processing and Communications Applications Conference (SIU)* (pp. 1785–1788). IEEE.
136. Hari, A., & Lakshman, T. (2016). The internet blockchain: A distributed, tamper-resistant transaction framework for the internet. In: *Proceedings of the 15th ACM Workshop on Hot Topics in Networks* (pp. 204–210).
137. Singh, D. N., Tejaswi, K., et al. (2013). Real time vehicle theft identity and control system based on arm 9. *International Journal of Latest Trends in Engineering and Technology (IJLTET)*, 2(1), 240–245.
138. Shruthi, K., Ramaprasad, P., Ray, R., Naik, M. A., & Pansari, S. (2015). Design of an anti-theft vehicle tracking system with a smartphone application. In: *2015 International Conference on Information Processing (ICIP)* (pp. 755–760). IEEE.
139. Subke, P., Moshref, M., Vach, A., & Steffelbauer, M. (2017). Measures to prevent unauthorized access to the in-vehicle e/e system, due to the security vulnerability of a remote diagnostic tester. *SAE International Journal of Passenger Cars-Electronic and Electrical Systems*, 10(2017-01-1689), 422–429.
140. Nguyen, V. L., Lin, P. C., & Hwang, R. H. (2019). Physical signal-driven fusion for v2x misbehavior detection. In: *2019 IEEE Vehicular Networking Conference (VNC)* (pp. 1–4). IEEE.
141. Dietzel, S., Petit, J., Heijenk, G., & Kargl, F. (2012). Graph-based metrics for insider attack detection in Vanet multihop data dissemination protocols. *IEEE Transactions on Vehicular Technology*, 62(4), 1505–1518.
142. Noh, J., Kwon, Y., Son, J., & Cho, S. (2022). Blockchain-based one-time authentication for secure v2x communication against insiders and authority compromise attacks. *IEEE Internet of Things Journal*, 10(7), 6235–6248.
143. Ansari, A., Alimohideen, M. A., & Karthik, P. (2021). Deep learning based real time vulnerability fixes verification mechanism for automotive firmware/software. Technical report, SAE Technical Paper.
144. Jesudoss, A., Helen, S., Christy, A., Meena, S., Theresa, M., & Rajesh, S. (2022). Vulnerabilities exploitation in automotive hacking using hardware and software implementation. In: *2022 6th International Conference on Intelligent Computing and Control Systems (ICICCS)* (pp. 50–54). IEEE.
145. Ju'nior, P. R. M., Neves, J. M., Tavares, A. I., & Menotti, D. (2011). Towards an automatic vehicle access control system: License plate location. In: *2011 IEEE International Conference on Systems, Man, and Cybernetics* (pp. 2916–2921). IEEE.
146. Demir, M. O., Kurt, G. K., & Pusane, A. E. (2023). A pseudorange-based GPS spoofing detection using hyperbola equations. *IEEE Transactions on Vehicular Technology*. <https://doi.org/10.1109/TVT.2023.3257228>
147. Mir, Z. H., & Filali, F. (2014). Lte and ieee 802.11 p for vehicular networking: A performance evaluation. *EURASIP Journal on Wireless Communications and Networking*, 2014(1), Article 89.
148. Ahmad, F., Adnane, A., Franqueira, V. N., Kurugollu, F., & Liu, L. (2018). Man-in-the-middle attacks in vehicular ad-hoc networks: Evaluating the impact of attackers strategies. *Sensors (Basel)*, 18(11), Article 4040.
149. El-Rewini, Z., Sadatsharan, K., Selvaraj, D. F., Plathottam, S. J., & Ranganathan, P. (2020). Cybersecurity challenges in vehicular communications. *Vehicular Communications*, 23, Article 100214.
150. Prigoda, Z. C. Z. D. L., & Cekerevac, P. (2017). Techno-economic aspect of the man-in-the-middle attacks. *COMMUNICATIONS* 19.
151. Stepień, K., & Poniszewska-Maranda, A. (2019). Security measures in the vehicular ad-hoc networks—man in the middle attack. In: *Mobile Web and Intelligent Information Systems: 16th International Conference, MobiWIS 2019, Istanbul, Turkey, August 26–28, Proceedings 16*, pp. 136–147 (2019). Springer.
152. Katragadda, S., Darby, P. J., Roche, A., & Gottumukkala, R. (2020). Detecting low-rate replay-based injection attacks on in-vehicle networks. *IEEE Access*, 8, 54979–54993.

153. Xie, M., Ding, D., Ge, X., Han, Q. L., Dong, H., & Song, Y. (2022). Distributed platooning control of automated vehicles subject to replay attacks based on proportional integral observers. *IEEE/CAA Journal of Automatica Sinica*.
154. Kumar, A., Shahid, M. A., Jaekel, A., Zhang, N., & Kneppers, M. (2023). Machine learning based detection of replay attacks in vanet. In: *NOMS 2023–2023 IEEE/IFIP Network Operations and Management Symposium* (pp. 1–6). IEEE.
155. Zhang, Y., Egwede, D., Zhang, G., & Wu, X. (2022). Pseudonym tracking certificateless aggregate signature scheme for preventing replay attacks in a platoon of vehicles. In: *2022 International Conference on Intelligent Data Science Technologies and Applications (IDSTA)* (pp. 109–116). IEEE.
156. Li, Y., Wang, Y., Wu, M., Li, H., et al. (2019). Replay attack and defense of electric vehicle charging on gb/t 27930–2015 communication protocol. *Journal of Computer and Communications*, 7(12), 20.
157. Yao, Y., Zhao, J., Li, Z., Cheng, X., & Wu, L. (2023). Jamming and eavesdropping defense scheme based on deep reinforcement learning in autonomous vehicle networks. *IEEE Transactions on Information Forensics and Security*, 18, 1211–1224.
158. Xing, X., Sun, G., Qian, J., Yu, D., & Cheng, X. (2022). Intermittent jamming for eavesdropping defense in wave based vehicular networks. *Vehicular Communications*, 38, Article 100542.
159. Zhou, F., Li, X., Alazab, M., Jhaveri, R. H., & Guo, K. (2022). Secrecy performance for ris-based integrated satellite vehicle networks with a uav relay and mrc eavesdropping. *IEEE Transactions on Intelligent Vehicles*, 8(2), 1676–1685.
160. Kamal, M., Barua, A., Vitale, C., Laoudias, C., & Ellinas, G. (2021). Gps location spoofing attack detection for enhancing the security of autonomous vehicles. In: *2021 IEEE 94th Vehicular Technology Conference (VTC2021-Fall)* (pp. 1–7). IEEE.
161. Chang, J., Huang, F., Zhang, L., Xu, D., & Hsu, L. T. (2023). Selection of areas for effective Gnss spoofing attacks to a vehicle-mounted Msf system based on scenario classification models. *IEEE Transactions on Vehicular Technology*.
162. Zubaïru, B. (2018). Novel approach of spoofing attack in vanet location verification for non-line-of-sight (nlos). In: *Innovations in Computational Intelligence: Best Selected Papers of the Third International Conference on REDSET 2016* (pp. 45–59). Springer.
163. Sanders, C., & Wang, Y. (2020). Localizing spoofing attacks on vehicular gps using vehicle-to-vehicle communications. *IEEE Transactions on Vehicular Technology*, 69(12), 15656–15667.
164. Kadam, N., & Krovi, R. S. (2021). Machine learning approach of hybrid Ksvn algorithm to detect Ddos attack in Vanet. *International Journal of Advanced Computer Science and Applications* 12(7).
165. Zhao, N., Zhao, X., Chen, M., Zong, G., & Zhang, H. (2023). Resilient distributed event-triggered platooning control of connected vehicles under denial-of-service attacks. *IEEE Transactions on Intelligent Transportation Systems*. <https://doi.org/10.1109/TITS.2023.3250402>
166. Zhang, L., Meng, X., Choo, K.-K.R., Zhang, Y., & Dai, F. (2018). Privacy-preserving cloud establishment and data dissemination scheme for vehicular cloud. *IEEE Transactions on Dependable and Secure Computing*, 17(3), 634–647.
167. Kudva, S., Badsha, S., Sengupta, S., La, H., Khalil, I., & Atiquzzaman, M. (2021). A scalable blockchain based trust management in Vanet routing protocol. *Journal of Parallel and Distributed Computing*, 152, 144–156.
168. Giang, A. T., Tran, H. T., Le, H. T., Doan, N. Q., & Nguyen, M. H. (2022). Jamming attack in vehicular networks: adaptively probabilistic channel surfing scheme. *Wireless Communications and Mobile Computing*.
169. Benslimane, A., & Nguyen-Minh, H. (2016). Jamming attack model and detection method for beacons under multichannel operation in vehicular networks. *IEEE Transactions on Vehicular Technology*, 66(7), 6475–6488.
170. Karagiannis, D., & Argyriou, A. (2018). Jamming attack detection in a pair of rf communicating vehicles using unsupervised machine learning. *Vehicular Communications*, 13, 56–63.
171. Lu, R., Zhang, L., Ni, J., & Fang, Y. (2019). 5G vehicle-to-everything services: Gearing up for security and privacy. *Proceedings of the IEEE*, 108(2), 373–389.
172. Smahi, A., Li, H., Yang, Y., Yang, X., Lu, P., Zhong, Y., & Liu, C. (2023). BVICVS: A privacy-preserving and verifiable federated learning framework for V2X environments using blockchain and ZKSNARKS. *Journal of King Saud University Computer and Information Sciences*. <https://doi.org/10.1016/j.jksuci.2023.03.020>
173. Aliev, H., & Kim, H. W. (2020). Matrix-based dynamic authentication with conditional privacy-preservation for vehicular network security. *Ieee Access : Practical Innovations, Open Solutions*, 8, 200883–200896.
174. Muhammad, M., & Safdar, G. A. (2018). Survey on existing authentication issues for cellular-assisted v2x communication. *Vehicular Communications*, 12, 50–65.

175. Lim, J., Oh, H., Sim, K., Kim, S., & Kim, K. H. (2023). A v2x access authorization mechanism based on decentralized id (did) and verifiable credentials (vc). In: *2023 International Conference on Information Networking (ICOIN)* (pp. 801–805). IEEE.
176. Xu, D., & Ritcey, J. A. (2022). Post-quantum phy-layer authentication for secure initial access in v2x communications. In: *GLOBECOM 2022–2022 IEEE Global Communications Conference* (pp. 1758–1762). IEEE.
177. Ko, T., Ji, C., & Hong, M. (2021). AvoD: Advanced verify-on-demand for efficient authentication against dos attacks in v2x communication. *Security and Communication Networks*, 2021, 1–9.
178. Woodbury, D., & Haltmeyer, N. (2017). Linux-stack based v2x framework: All you need to hack connected vehicles.
179. Di Massa, V., & Foni, S. (2018). Improving its-g5 cybersecurity features starting from hacking ieee 802.11 p v2x communications through low-cost sdr devices. In: *Electronic Components and Systems for Automotive Applications: Proceedings of the 5th CESA Automotive Electronics Congress*, Paris (pp. 275–284). Springer.
180. Casademont, J., Cordero, B., Camps-Mur, D., Conceic,ˆao, L. A. M., Lalos, A., Vitale, C., Laoudias, C., & Khodashenas, P. S. (2020). Multi-radio v2x communications interoperability through a multi-access edge computing (mec). In: *2020 22nd International Conference on Transparent Optical Networks (ICTON)* (pp. 1–4). IEEE.
181. Djaidja, T. E. T., Brik, B., Senouci, S. M., & Ghamri-Doudane, Y. (2022). Adaptive resource reservation to survive against adversarial resource selection jamming attacks in 5 g nr-v2x distributed mode 2. In: *ICC 2022-IEEE International Conference on Communications* (pp. 3406–3411). IEEE.
182. Kihci, B., Wilson, H., & Fall, M. (2021). Experimental results of detecting primitive jamming attacks using machine learning in vehicle-to-everything communication networks. In: *2021 IEEE 7th World Forum on Internet of Things (WF-IoT)* (pp. 530–535). IEEE.
183. Desnitsky, V., & Kotenko, I. (2018). Modeling and analysis of Iot energy resource exhaustion attacks. *Intelligent Distributed Computing XI*, 263–270.
184. Park, S. B., Jo, H. J., & Lee, D. H. (2023). Flooding attack mitigator for in-vehicle can using fault confinement in can protocol. *Computers & Security*, 126, Article 103091.
185. Karthikeyan, H., & Usha, G. (2022). Real-time ddos flooding attack detection in intelligent transportation systems. *Computers and Electrical Engineering*, 101, Article 107995.
186. Kwon, Y.-M., Yu, J., Cho, B.-M., Eun, Y., & Park, K.-J. (2018). Empirical analysis of mavlink protocol vulnerability for attacking unmanned aerial vehicles. *IEEE Access*, 6, 43203–43212.
187. Ioulianou, P. P., Vassilakis, V. G., & Logothetis, M. D. (2019). Battery drain denial-of-service attacks and defenses in the internet of things. *Journal of Telecommunications and Information Technology*, 2, 37–45.
188. Smith, R., Palin, D., Ioulianou, P. P., Vassilakis, V. G., & Shahandashti, S. F. (2020). Battery draining attacks against edge computing nodes in iot networks. *Cyber-Physical Systems*, 6(2), 96–116.
189. Lekidis, A. (2024). Anomaly detection mechanisms for in-vehicle and v2x systems. In: *Proceedings of the 19th International Conference on Availability, Reliability and Security* (pp. 1–8).
190. Mˆoller, D. P. (2020). Cutting-edge digitization challenges in vehicle cyber-physical systems and cyber-security. *Smart Technologies: Scope and Applications* (pp. 17–34).
191. Lyamin, N., Kleyko, D., Delooz, Q., & Vinel, A. (2018). AI-based malicious network traffic detection in Vanets. *IEEE Network*, 32(6), 15–21.
192. Zeydan, E., Turk, Y., Aksoy, B., & Tasbag, Y. Y. (2022). Post-quantum era in v2x security: Convergence of orchestration and parallel computation. *IEEE Communications Standards Magazine*, 6(1), 76–82.
193. Kanchan, S., Singh, G., & Chaudhari, N. S. (2018). Re-encrypting secure and efficient routing in vanet groups using sharable clouds. In: *2018 4th International Conference on Recent Advances in Information Technology (RAIT)* (pp. 1–6). IEEE.
194. Shrestha, R., Nam, S. Y., Bajracharya, R., & Kim, S. (2020). Evolution of v2x communication and integration of blockchain for security enhancements. *Electronics*, 9(9), Article 1338.
195. Khan, R., McLaughlin, K., Laverty, D., & Sezer, S. (2017). Stride-based threat modeling for cyber-physical systems. In: *2017 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe)*, pp. 1–6 IEEE.
196. Al-Shareeda, M. A., Gaber, T., Alqarni, M. A., Alkinani, M. H., Almazroey, A. A., & Almazroi, A. A. (2025). Chebyshev polynomial based emergency conditions with authentication scheme for 5G-assisted vehicular fog computing. *IEEE Transactions on Dependable and Secure Computing*. <https://doi.org/10.1109/TDSC.2025.3553868>
197. Almazroi, A. A., Aldhahri, E. A., Al-Shareeda, M. A., & Manickam, S. (2023). Eca-vfog: An efficient certificateless authentication scheme for 5g-assisted vehicular fog computing. *PLoS One*, 18(6), Article 0287291.

198. Qiu, H., Qiu, M., & Lu, R. (2019). Secure v2x communication network based on intelligent PKI and edge computing. *IEEE Network*, 34(2), 172–178.
199. Gaiduk, P., Ranjan, K. R., Basmer, T., & Tschorsch, F. (2020). Privacy-preserving public key infrastructure for vehicular networks. In: *2020 IEEE 45th Conference on Local Computer Networks (LCN)* (pp. 154–163). IEEE.
200. Pokorny, J., Seda, P., Dvorak, J., Malina, L., & Martinasek, Z. (2021). The security testbed for the purposes of the its-g5 communication attacks prevention. In: *2021 13th International Congress on Ultra Modern Telecommunications and Control Systems and Workshops (ICUMT)* (pp. 167–173). IEEE.
201. Kent, D., Cheng, B. H., & Siegel, J. (2020). Assuring vehicle update integrity using asymmetric public key infrastructure (pki) and public key cryptography (pke). *SAE International Journal of Transportation Cybersecurity and Privacy*, 2(11-02-020013), 141–158.
202. Jivthesh, M., Samuel, R. M., Gaushik, M., Shibu, N. S., Rao, S. N., & Mahesh, A. N. (2023). Smart-verse: Blockchain based crowdsourced v2x message verification and dissemination system. In: *2023 15th International Conference on COMMunication Systems & NETworkS (COMSNETS)* (pp. 84–89). IEEE.
203. Vaidya, B., Makrakis, D., & Mouftah, H. (2014). Effective public key infrastructure for vehicle-to-grid network. In: *Proceedings of the Fourth ACM International Symposium on Development and Analysis of Intelligent Vehicular Networks and Applications* (pp. 95–101).
204. Brecht, B., & Hehn, T. (2019). A security credential management system for v2x communications. *Connected Vehicles: Intelligent Transportation Systems*, 83–115.
205. Yang, Z., Anh Dinh, T. T., Yin, C., Yao, Y., Yang, D., Chang, X., & Zhou, J. (2022). Larp: A lightweight auto-refreshing pseudonym protocol for v2x. In: *Proceedings of the 27th ACM on Symposium on Access Control Models and Technologies* (pp. 49–60).
206. Zhang, L., Shan, L., & Wang, J. (2012). Summary of digital signature. *Advances in control and communication* (pp. 115–120). Springer.
207. Kitsos, P., Sklavos, N., & Koufopavlou, O. (2002). An efficient implementation of the digital signature algorithm. In: *9th International Conference on Electronics, Circuits and Systems* (Vol. 3, pp. 1151–1154). IEEE.
208. Al-shareeda, M. A., Anbar, M., Manickam, S., Hasbullah, I. H., Khalil, A., Alazzawi, M. A., & Al-Hiti, A. S. (2020). Proposed efficient conditional privacy-preserving authentication scheme for v2v and v2i communications based on elliptic curve cryptography in vehicular ad hoc networks. In: *Advances in Cyber Security: Second International Conference, ACeS 2020, Penang, Malaysia, December 8–9, Revised Selected Papers 2* (pp. 588–603) (2021). Springer.
209. Patil, M. J., & Adhiya, K. P. (2022). An enhanced elliptic curve cryptography scheme for secure data transmission to evade entalment of fake vehicles in Vanet. *Cybernetics and Systems*, 1–35.
210. Jain, A., Singh, J., Kumar, S., Florin-Emilian, T., Traian Candin, M., & Chithalur, P. (2022). Improved recurrent neural network schema for validating digital signatures in vanet. *Mathematics*, 10(20), Article 3895.
211. Kumar, P. H., & AnandhaMala, G. (2023). Hmac-r: Hash-based message authentication code and rijndael-based multilevel security model for data storage in cloud environment. *The Journal of Supercomputing*, 79(3), 3181–3209.
212. Wu, H.-T., & Hsieh, W.-S. (2013). Rsu-based message authentication for vehicular ad-hoc networks. *Multimedia Tools and Applications*, 66, 215–227.
213. John Justin, M., & Manimurugan, S. (2012). A survey on various encryption techniques. *International Journal of Soft Computing and Engineering (IJSC)*, 2231, 2307.
214. Sachdeva, A. (2013). A study of encryption algorithms AES, DES and RSA for security. *Global Journal of Computer Science and Technology*, 13(E15), 32–40.
215. David, A. O., & Sulaimon, O. (2023). Text encryption with improved elliptic curve cryptography. *Journal of Advances in Mathematics and Computer Science*, 38(3), 32–41.
216. Manjula, T., & Anand, B. (2021). A secured multiplicative Diffie Hellman key exchange routing approach for mobile ad hoc network. *Journal of Ambient Intelligence and Humanized Computing*, 12, 3621–3631.
217. Mustafa, I., & Huang, C. Y. (2020). Lightweight cryptographic urlle for 5 g-v2x. In: *2020 International Symposium on Networks, Computers and Communications (ISNCC)* (pp. 1–6). IEEE.
218. Weimerskirch, A. (2011). *V2x security & privacy: The current state and its future* (p. 21). In: ITS World Congress.
219. Ma, R., Cao, J., Zhang, Y., Shang, C., Xiong, L., & Li, H. (2022). A group-based multicast service authentication and data transmission scheme for 5 g-v2x. *IEEE Transactions on Intelligent Transportation Systems*, 23(12), 23976–23992.
220. Pes'e, M. D., Schmidt, K., & Zweek, H. (2017). Hardware/software co-design of an automotive embedded firewall. Technical report, SAE Technical Paper.

221. Ondov, A., & Helebrandt, P. (2022). Covert channel detection methods. In: *2022 20th International Conference on Emerging eLearning Technologies and Applications (ICETA)* (pp. 491–496). IEEE.
222. Zelle, D., Rieke, R., & Krauß, C. (2019). Security test platform for autonomous driving. In: *3. ACM COMPUTER SCIENCE IN CARS SYMPOSIUM (CSCS 2019)*, Tech. Rep.
223. Qu, L. D., Cui, X. Y., & Liu, X. N. (2014). A route handoff method based on virtual private network in vehicle information network. *Applied Mechanics and Materials*, 475, 699–702.
224. Shahriar, M. H., Ansari, M. R., Monteuiis, J.-P., Haque, M. S., Chen, C., Petit, J., Hou, Y. T., & Lou, W. (2025). Vehigan: Generative adversarial networks for adversarially robust v2x misbehavior detection systems. *ACM Transactions on Cyber-Physical Systems*, 9(3), 1–26.
225. Kumar, Y., & Kumar, V. (2023). A systematic review on intrusion detection system in wireless networks: Variants, attacks, and applications. *Wireless Personal Communications*, 133(1), 395–452.
226. Abdel Hakeem, S. A., Hady, A. A., & Kim, H. (2020). Current and future developments to improve 5g-newradio performance in vehicle-to-everything communications. *Telecommunication Systems*, 75, 331–353.
227. Hidalgo, C., Vaca, M., Nowak, M. P., Fr'olich, P., Reed, M., Al-Naday, M., Mpatziakas, A., Protogerou, A., Drosou, A., & Tzovaras, D. (2022). Detection, control and mitigation system for secure vehicular communication. *Vehicular Communications*, 34, Article 100425.
228. Kristianto, E., Nguyen, V. L., & Lin, P. C. (2022). Decentralized public-key infrastructure with blockchain in v2x communications: Promising or only euphoria? *IEEE Security & Privacy*, 20(4), 40–50.
229. Angelogianni, A., Krontiris, I., & Giannetsos, T. (2023). Comparative evaluation of pki and daa-based architectures for v2x communication security. In: *2023 IEEE Vehicular Networking Conference (VNC)* (pp. 199–206). IEEE.
230. Bellur, B. (2008). Certificate assignment strategies for a pki-based security architecture in a vehicular network. In: *IEEE GLOBECOM 2008–2008 IEEE Global Telecommunications Conference* (pp. 1–6). IEEE.
231. Heng, X., Qin, S., Xiao, Y., Wang, J., Tao, Y., & Zhang, R. (2022). A strong secure v2i authentication scheme from pki and accumulator. In: *2022 2nd International Conference on Consumer Electronics and Computer Engineering (ICCECE)* (pp. 98–103). IEEE.
232. Han, Q., Meikang, Q., Zhihui, L., & Memmi, G. (2019). An efficient key distribution system for data fusion in v2x heterogeneous networks. *Information Fusion*, 50, 212–220.
233. Klement, F., P'ohls, H. C., & Katzenbeisser, S. (2022). Change your car's filters: Efficient concurrent and multi-stage firewall for obd-ii network traffic. In: *2022 IEEE 27th International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD)* (pp. 19–25). IEEE.
234. Sahana, Y., Gotkhindikar, A., & Tiwari, S. K. (2022). Survey on can-bus packet filtering firewall. In: *2022 International Conference on Edge Computing and Applications (ICECAA)* (pp. 472–478). IEEE.
235. Klement, F., P'ohls, H. C., & Katzenbeisser, S. (2022). Man-in-the-obd: A modular, protocol agnostic firewall for automotive dongles to enhance privacy and security. In: *International Workshop on Attacks and Defenses for Internet-of-Things* (pp. 143–164). Springer.
236. Hossain, S., Boualouache, A., Briki, B., & Senouci, S. M. (2023). A lightweight 5 g-v2x intra-slice intrusion detection system using knowledge distillation. *A Lightweight 5G-V2X Intra-slice Intrusion Detection System Using Knowledge Distillation*.
237. Bodi, B., Chiu, W. Y., & Meng, W. (2022). Towards blockchain-enabled intrusion detection for vehicular navigation map system. In: *International Conference on Information Security Practice and Experience* (pp. 3–20). Springer.
238. Ben Rabah, N., & Idoudi, H. (2022). A machine learning framework for intrusion detection in Vanet communications. *Emerging trends in cybersecurity applications* (pp. 209–227). Springer.
239. Anbalagan, S., Raja, G., Gurumoorthy, S., Suresh, R. D., & Dev, K. (2023). Iids: Intelligent intrusion detection system for sustainable development in autonomous vehicles. *IEEE Transactions on Intelligent Transportation Systems*. <https://doi.org/10.1109/TITS.2023.3271768>
240. Qu, L. D., Cui, X. Y., & Liu, X. N. (2013). IPv6 in IPv4 virtual private network based on virtual devices in vehicle information system. *Applied Mechanics and Materials*, 321, 2910–2913.
241. Guruprasad, K., Guptha, N., & Hema, S. (2021). Peer to peer communication between the autonomous vehicles using virtual private network. In: *Proceedings of the First International Conference on Computing, Communication and Control System, I3CAC 7–8 June 2021, Bharath University, Chennai, India*.
242. Shu, M. L., Liu, Y. X., Yu, W. J., & Fang, H. (2014). The secure communication scheme of dsrc based on watermark and vpn. *Applied Mechanics and Materials*, 475, 824–827.
243. Riabi, I., Ayed, H. K. B., & Saidane, L. A. (2019). A survey on blockchain based access control for internet of things. In: *2019 15th International Wireless Communications & Mobile Computing Conference (IWCMC)* (pp. 502–507). IEEE.

244. Ahmad, T., Morelli, U., & Ranise, S. (2022). Distributed enforcement of access control policies in intelligent transportation system (its) for situation awareness. In: *Proceedings of the 17th International Conference on Availability, Reliability and Security* (pp. 1–10).
245. Lee, S., Chung, B., Choi, J., & Kwon, H. (2019). Design and implementation on the access control scheme against v2n connected car attacks. *Advanced multimedia and ubiquitous engineering: MUE/FutureTech 2018 12* (pp. 705–710). Springer.
246. Munasinghe, G. K., & Murtaza, M. (2020). Analyzing vehicle-to-everything communication for intelligent transportation system: Journey from ieee 802.11 p to 5 g and finally towards 6 g. In: *2020 5th International Conference on Innovative Technologies in Intelligent Systems and Industrial Applications (CITISIA)* (pp. 1–7). IEEE.
247. Klapez, M., Grazia, C. A., & Casoni, M. (2020). Application-level performance of IEEE 802.11 p in safety-related v2x field trials. *IEEE Internet of Things Journal*, 7(5), 3850–3860.
248. Yan, G., Lin, J., Rawat, D. B., & Yang, W. (2011). A geographic location-based security mechanism for intelligent vehicular networks. In: *Intelligent Computing and Information Science: International Conference, ICICIS 2011, Chongqing, China, January 8–9, 2011. Proceedings, Part II* (pp. 693–698). Springer.
249. Khodaei, M., & Papadimitratos, P. (2020). Cooperative location privacy in vehicular networks: Why simple mix zones are not enough. *IEEE Internet of Things Journal*, 8(10), 7985–8004.
250. Xu, C., Liu, H., Li, P., & Wang, P. (2018). A remote attestation security model based on privacy-preserving blockchain for v2x. *IEEE Access*, 6, 67809–67818.
251. Chen, S., Hu, J., Zhao, L., Zhao, R., Fang, J., Shi, Y., & Xu, H. (2023). C-v2x security technology. *Cellular Vehicle-to-Everything (C-V2X)* (pp. 271–308). Springer.
252. Wang, R., Zhang, X., Xu, Z., Zhao, X., & Li, X. (2021). Research on performance and function testing of v2x in a closed test field. *Journal of Advanced Transportation*, 1–18.
253. Wang, Y., Liu, X., Peng, X., & Du, H. (2020). Research on lte-v2x outfield expressway performance testing method in the 5.9 ghz band. In: *2020 IEEE 3rd International Conference on Computer and Communication Engineering Technology (CCET)* (pp. 284–288). IEEE.
254. Zhang, B., Li, J., Zhang, M., Fan, B., Song, K., Jiang, J., & Zhu, C. (2022). Research on conformance test for electric vehicle wireless power transfer communication protocols. In: *2022 IEEE 4th International Conference on Civil Aviation Safety and Information Technology (ICCASIT)* (pp. 802–804). IEEE.
255. Mohammad, H., & Saleem, M. S. (2014). Handling conflicts to test transport protocol's parallel routing on a vehicle gateway system. In: *2014 Federated Conference on Computer Science and Information Systems* (pp. 1559–1568). IEEE.
256. Garrad, P., & Unnikrishnan, S. (2023). Reinforcement learning in vanet penetration testing. *Results in Engineering*, 17, Article 100970.
257. Xu, Y., Zou, Y., & Sun, J. (2018). Accelerated testing for automated vehicles safety evaluation in cut-in scenarios based on importance sampling, genetic algorithm and simulation applications. *Journal of Intelligent and Connected Vehicles*, 1(1), 28–38.
258. Reese, L., Qiu, T., Linzell, D., O'hare, E., & Rado, Z. (2014). Field tests and numerical modeling of vehicle impacts on a boulder embedded in compacted fill. *International Journal of Protective Structures*, 5(4), 435–451.
259. Katranaras, E., Dillinger, M., Abbas, T., Theillaud, R., Calvo, J. L., & Zang, Y. (2021). Standardization and regulation. *Cellular V2X for Connected Automated Driving*, 63–90.
260. Fouchal, H., Bourdy, E., Wilhelm, G., & Ayaida, M. (2017). A validation tool for cooperative intelligent transport systems. *Journal of Computational Science*, 22, 283–288.
261. Kim, H., Kim, T., Kang, S., Yoon, C., & Jung, J. (2014). Design of v2x runtime emulation framework for evaluation of vehicle safety applications. In: *2014 4th IEEE International Conference on Network Infrastructure and Digital Content* (pp. 262–268). IEEE.
262. Schiller, M., & Knoll, A. (2015). Emulating vehicular ad hoc networks for evaluation and testing of automotive embedded systems. *EAI Endorsed Transactions on Smart Cities* 1(2).
263. Fan, Y., Liu, L., Dong, S., Zhuang, L., Qiu, J., Cai, C., & Song, M. (2020). Network performance test and analysis of lte-v2x in industrial park scenario. *Wireless Communications and Mobile Computing*, 1–12.
264. Ghosal, A., & Conti, M. (2020). Security issues and challenges in v2x: A survey. *Computer Networks*, 169, Article 107093.
265. Miao, L., Virtusio, J. J., & Hua, K.-L. (2021). Pc5-based cellular-v2x evolution and deployment. *Sensors*, 21(3), Article 843.
266. Choi, W.-S., & Choi, S.-G. (2022). Thread-based modeling and analysis in multi-corebased v2x communication device. *Sustainability*, 14(14), Article 8277.

267. Li, Y., Li, D., Cui, W., & Zhang, R. (2011). Research based on osi model. In: *2011 IEEE 3rd International Conference on Communication Software and Networks* (pp. 554–557). IEEE.
268. Swetha, M., Thungamani, M., & Krishnan, K. (2018). A survey on different types of Manet attacks in Osi model. *Int J Innovative Res Sci Technol*, 4, 18–23.
269. Kumar, S., Dalal, S., & Dixit, V. (2014). The osi model: Overview on the seven layers of computer networks. *International Journal of Computer Science and Information Technology Research*, 2(3), 461–466.
270. Mughal, A. A. (2020). Cyber attacks on osi layers: Understanding the threat landscape. *Journal of Humanities and Applied Science Research*, 3(1), 1–18.
271. Ge, Y., Xie, Q., Yu, R., Ma, J., Han, Q., & Zhang, Q. (2020). Design of conformance testing system for c-v2x standard protocol. In: *2020 IEEE 91st Vehicular Technology Conference (VTC2020-Spring)* (pp. 1–5). IEEE.
272. Horel, J. B., Laugier, C., Marso, L., Mateescu, R., Muller, L., Paigwar, A., Renzaglia, A., & Serwe, W. (2022). Using formal conformance testing to generate scenarios for autonomous vehicles. In: *2022 Design, Automation & Test in Europe Conference & Exhibition (DATE)* (pp. 532–537). IEEE.
273. Zhu, Y., Wang, J., Guo, X., Meng, F., & Liu, T. (2023). Functional testing scenario library generation framework for connected and automated vehicles. *IEEE Transactions on Intelligent Transportation Systems*. <https://doi.org/10.1109/TITS.2023.3266639>
274. Xu, Y., Ou, P., & Li, Y. (2019). Design of vehicle gateway automatic test system based on canoe. In: *2019 Chinese Automation Congress (CAC)* (pp. 1433–1437). IEEE.
275. Garrad, P., & Unnikrishnan, S. (2022). Artificial intelligence in penetration testing of a connected and autonomous vehicle network. *International Journal of Mechanical and Mechatronics Engineering*, 16(12), 341–346.
276. Chakraborty, S., Khattak, A. J., & Cummings, M. Advancing accelerated testing protocols for safe and reliable deployment of connected and automated vehicles through iterative deployment in physical and digital worlds.
277. Meinan, L., & Ruiying, L. (2013). Modeling end-to-end delay of network in accelerated reliability testing. In: *2013 Third International Conference on Instrumentation, Measurement, Computer, Communication and Control* (pp. 1371–1374). IEEE.
278. Lusvarghi, L., Grazia, C. A., Klapez, M., Casoni, M., & Merani, M. L. (2023). Awareness messages by vulnerable road users and vehicles: Field tests via lte-v2x. *IEEE Transactions on Intelligent Vehicles*. <https://doi.org/10.1109/TIV.2023.3280744>
279. Weiß, C. (2011). V2x communication in europe—from research projects towards standardization and field testing of vehicle communication technology. *Computer Networks*, 55(14), 3103–3119.
280. Jia, K., Hou, Y., Niu, K., Dong, C., & He, Z. (2019). The delay-constraint broadcast combined with resource reservation mechanism and field test in Vanet. *Ieee Access : Practical Innovations, Open Solutions*, 7, 59600–59612.
281. Migliore, M. D. (2021). The world beneath the physical layer: An introduction to the deep physical layer. *IEEE Access*, 9, 77106–77126.
282. Guo, L., Yang, B., Ye, J., Chen, H., Li, F., Song, W., Du, L., & Guan, L. (2020). Systematic assessment of cyber-physical security of energy management system for connected and automated electric vehicles. *IEEE Transactions on Industrial Informatics*, 17(5), 3335–3347.
283. Lautenbach, A., Nowdehi, N., Olovsson, T., & Zaragatzky, R. (2019). A preliminary security assessment of 5 g v2x. In: *2019 IEEE 89th Vehicular Technology Conference (VTC2019-Spring)* (pp. 1–7). IEEE.
284. Ye, J., Guo, L., Yang, B., Li, F., Du, L., Guan, L., & Song, W. (2020). Cyber–physical security of powertrain systems in modern electric vehicles: Vulnerabilities, challenges, and future visions. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 9(4), 4639–4657.
285. McGraw, G. (2008). Automated code review tools for security. *Computer*, 41(12), 108–111.
286. Meneely, A., Tejada, A. C. R., Spates, B., Trudeau, S., Neuberger, D., Whitlock, K., Ketant, C., & Davis, K. (2014). An empirical investigation of socio-technical code review metrics and security vulnerabilities. In: *Proceedings of the 6th International Workshop on Social Software Engineering* (pp. 37–44).
287. Santos, M. G. M., & Marcillo, P. A. A. (2017). Security in the data link layer of the osi model on lans wired cisco. *Journal of Science and Research* 3(CITT 106–112).
288. Utesch, F., Brandies, A., Pekezou Fouopi, P., & Schiefl, C. (2020). Towards behaviour based testing to understand the black box of autonomous cars. *European Transport Research Review*, 12(1), 1–11.
289. Abdullahi, A., Dargahi, T., & Babaie, M. (2020). Vulnerability assessment of vehicle to infrastructure communication: a case study of unmanned ground vehicle. In: *2020 IEEE Globecom Workshops (GC Wkshps)* (pp. 1–6). IEEE.
290. Karoui, M., Berg, V., & Mayrargue, S. (2022). Assessment of v2x communications for enhanced vulnerable road users safety. In: *2022 IEEE 95th Vehicular Technology Conference:(VTC2022-Spring)* (pp. 1–5). IEEE.

291. Wang, L., & Liu, X. (2018). Notsa: Novel Obu with three-level security architecture for internet of vehicles. *IEEE Internet of Things Journal*, 5(5), 3548–3558.
292. So, S., Petit, J., & Starobinski, D. (2019). Physical layer plausibility checks for misbehavior detection in v2x networks. In: *Proceedings of the 12th Conference on Security and Privacy in Wireless and Mobile Networks* (pp. 84–93).
293. Khan, J. (2017). Vehicle network security testing. In: *2017 Third International Conference on Sensing, Signal Processing and Security (ICSSS)* (pp. 119–123). IEEE.
294. Huang, T., Zhou, J., & Bytes, A. (2018). Atg: An attack traffic generation tool for security testing of in-vehicle can bus. In: *Proceedings of the 13th International Conference on Availability, Reliability and Security* (pp. 1–6).
295. Peng, H., Liang, L., Shen, X., & Li, G. Y. (2018). Vehicular communications: A network layer perspective. *IEEE Transactions on Vehicular Technology*, 68(2), 1064–1078.
296. Botella, J., Bouquet, F., Capuron, J. F., Lebeau, F., Legeard, B., & Schadle, F. (2013). Model-based testing of cryptographic components—lessons learned from experience. In: *2013 IEEE Sixth International Conference on Software Testing, Verification and Validation* (pp. 192–201). IEEE.
297. Bhargavan, K., Fournet, C., Corin, R., & Z'ainescu, E. (2012). Verified cryptographic implementations for Tls. *ACM Transactions on Information and System Security (TISSEC)*, 15(1), 1–32.
298. Moosavi, S. R., Gia, T. N., Rahmani, A. M., Nigussie, E., Virtanen, S., Isoaho, J., & Tenhunen, H. (2015). Sea: A secure and efficient authentication and authorization architecture for iot-based healthcare using smart gateways. *Procedia Computer Science*, 52, 452–459.
299. Xuan, Y., Shin, I., Thai, M. T., & Znati, T. (2009). Detecting application denial-of-service attacks: A group-testing-based approach. *IEEE Transactions on Parallel and Distributed Systems*, 21(8), 1203–1216.
300. Verma, K., Hasbullah, H., & Kumar, A. (2013). An efficient defense method against udp spoofed flooding traffic of denial of service (dos) attacks in vanet. In: *2013 3rd IEEE International Advance Computing Conference (IACC)* (pp. 550–555). IEEE.
301. Moukahal, L. J., Zulkernine, M., & Soukup, M. (2021). Vulnerability-oriented fuzz testing for connected autonomous vehicle systems. *IEEE Transactions on Reliability*, 70(4), 1422–1437.
302. Han, J. C., & Zhou, Z. Q. (2020). Metamorphic fuzz testing of autonomous vehicles. In: *Proceedings of the IEEE/ACM 42nd International Conference on Software Engineering Workshops* (pp. 380–385).
303. Nyamdelger, T., Batzorig, M., Albhelil, E. A., Koh, Y., & Yim, K. (2023). Fuzz testing and safe framework development for vehicle security analysis. In *International Conference on Innovative Mobile and Internet Services in Ubiquitous Computing* (pp. 103–111). Springer.
304. Demmer, M., & Fall, K. (2009). The design and implementation of a session layer for delay-tolerant networks. *Computer Communications*, 32(16), 1724–1730.
305. Lu, C. S., Sun, S., Horne, R., Chou, V. Y., Liang, K. H., & Chang, A. T. (1994) Experiences in session layer conformance testing. In: *Proceedings of 1994 International Conference on Parallel and Distributed Systems* (pp. 438–444). IEEE.
306. Bojjagani, S., Reddy, Y. P., Anuradha, T., Rao, P. V., Reddy, B. R., & Khan, M. K. (2022). Secure authentication and key management protocol for deployment of internet of vehicles (IOV) concerning intelligent transport systems. *IEEE Transactions on Intelligent Transportation Systems*, 23(12), 24698–24713.
307. Offutt, J., & Wu, Y. (2010). Modeling presentation layers of web applications for testing. *Software & Systems Modeling*, 9(2), 257–280.
308. Howard, M. A. (2006). A process for performing security code reviews. *IEEE Security & Privacy*, 4(4), 74–79.
309. Ulybyshev, D., Alsalem, A. O., Bhargava, B., Savvides, S., Mani, G., & Othmane, L. B. (2018). Secure data communication in autonomous v2x systems. In: *2018 IEEE International Congress on Internet of Things (ICIoT)* (pp. 156–163). IEEE.
310. Shan, D., Lei, T., Yin, X., Luo, Q., & Gong, L. (2021). Extracting key traffic parameters from UAV video with on-board vehicle data validation. *Sensors*, 21(16), Article 5620.
311. Netter, F., Gauterin, F., & Butterer, B. (2013). Real-data validation of simulation models in a function-based modular framework. In: *2013 IEEE Sixth International Conference on Software Testing, Verification and Validation* (pp. 41–47) IEEE.
312. Weinert, F., & Du'ring, M. (2014). Development and assessment of cooperative v2x applications for emergency vehicles in an urban environment enabled by behavioral models. In: *Modeling Mobility with Open Data: 2nd SUMO Conference 2014 Berlin, Germany*, May 15–16 (pp. 125–153). Springer.
313. Swamy, S. N., Jadhav, D., & Kulkarni, N. (2017). Security threats in the application layer in iot applications. In: *2017 International Conference on i-SMAC (iot in Social, Mobile, Analytics and cloud) (i-SMAC)* (pp. 477–480). IEEE.

314. Eriksson, B. & GROTH, J. (2018). On the road with third-party apps-security, safety and privacy aspects of in-vehicle apps.
315. Mansfield-Devine, S. (2018). The best form of defence—the benefits of red teaming. *Computer Fraud & Security*, 2018(10), 8–12.
316. Abbass, H., Tang, J., Amin, R., Ellejmi, M., & Kirby, S. (2014). The computational air traffic control brain: Computational red teaming and big data for real-time seamless brain-traffic integration. *Journal of Air Traffic Control*, 56(2), 10–17.
317. Al-Shareeda, M. A., Anbar, M., Manickam, S., & Hasbullah, I. H. (2022). Passwordguessing attack-aware authentication scheme based on Chinese remainder theorem for 5g-enabled vehicular networks. *Applied Sciences*, 12(3), Article 1383.
318. Bhadauria, S., Shabbir, Z., Roth-Mandutz, E., & Fischer, G. (2020). Qos based deep reinforcement learning for v2x resource allocation. In: *2020 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)* (pp. 1–6). IEEE.
319. Javed, M. A., Khan, M. Z., Zafar, U., Siddiqui, M. F., Badar, R., Lee, B. M., & Ahmad, F. (2020). Odvp: An efficient protocol to mitigate data integrity attacks in intelligent transport systems. *IEEE Access*, 8, 114733–114740.
320. Dai, S., Zhong, Y., Xu, C., Liu, H., Yuan, J., & Wang, P. (2022). An intelligent security classification model of drivers driving behavior based on V2X in iot networks. *Security and Communication Networks*. <https://doi.org/10.1155/2022/6793365>
321. Ganesan, K., Lohr, J., Mallick, P. B., Kunz, A., & Kuchibhotla, R. (2020). Nr sidelink design overview for advanced v2x service. *IEEE Internet of Things Magazine*, 3(1), 26–30.
322. Alnasser, A., Sun, H., & Jiang, J. (2019). Cyber security challenges and solutions for v2x communications: A survey. *Computer Networks*, 151, 52–67.
323. Yan, X., Ma, M., & Su, R. (2022). Efficient group handover authentication for secure 5 g-based communications in platoons. *IEEE Transactions on Intelligent Transportation Systems*.
324. Lee, D.-H., Kim, C.-M., Song, H.-S., Lee, Y.-H., & Chung, W.-S. (2023). Simulation-based cybersecurity testing and evaluation method for connected car v2x application using virtual machine. *Sensors*, 23(3), Article 1421.
325. Insalaco, C. (2022). Identification of misbehavior detection solutions and risk scenarios in advanced connected and automated driving scenarios. Master's thesis, Universitat Politècnica de Catalunya.
326. Li, G., Lai, C., Lu, R., & Zheng, D. (2021). Seccdv: A security reference architecture for cybertwin-driven 6 g v2x. *IEEE Transactions on Vehicular Technology*, 71(5), 4535–4550.
327. Gallego-Madrid, J., Sanchez-Iborra, R., Santa, J., & Skarmeta, A. (2020). Evaluation of a zone encryption scheme for vehicular networks. *Computer Networks*, 182, Article 107523.
328. Qin, X., Huang, Y., & Li, X. (2020). An ecc-based access control scheme with lightweight decryption and conditional authentication for data sharing in vehicular networks. *Soft Computing*, 24, 18881–18891.
329. Moustafa, H., Bourdon, G., & Gourhant, Y. (2006). Providing authentication and access control in vehicular network environment. In: *Security and Privacy in Dynamic Environments: Proceedings of the IFIP TC-11 21st International Information Security Conference SEC 22–24 May 2006, Karlstad, Sweden 21* (pp. 62–73). Springer.
330. Ying, B., & Nayak, A. (2017). Anonymous and lightweight authentication for secure vehicular networks. *IEEE Transactions on Vehicular Technology*, 66(12), 10626–10636.
331. Yavuz, A. A., Mudgerikar, A., Singla, A., Papapanagiotou, I., & Bertino, E. (2017). Real-time digital signatures for time-critical networks. *IEEE Transactions on Information Forensics and Security*, 12(11), 2627–2639.
332. Liu, D., Ni, J., Lin, X., & Shen, X. (2018). Anonymous group message authentication protocol for lte-based v2x communications. *Internet Technology Letters*, 1(2), 25.
333. Lin, D.-R., Wang, C.-I., & Guan, D. (2011). Efficient vehicle ownership identification scheme based on triple-trapdoor chameleon hash function. *Journal of Network and Computer Applications*, 34(1), 12–19.
334. Cui, J., Tao, X., Zhang, J., Xu, Y., & Zhong, H. (2018). Hcpa-gka: A hash function-based conditional privacy-preserving authentication and group-key agreement scheme for Vanets. *Vehicular Communications*, 14, 15–25.
335. Pankov, K., & Glukhov, M. (2022). Using error-correcting codes to ensure information security of unmanned vehicles and iot systems. In: *2022 Systems of Signal Synchronization, Generating and Processing in Telecommunications (SYNCHROINFO)* (pp. 1–8). IEEE.
336. Chatzoulis, D., Chaikalas, C., Kosmanos, D., Anagnostou, K. E., & Xenakis, A. (2023). 3gpp 5g v2x error correction coding for various propagation environments: A qos approach. *Electronics*, 12(13), Article 2898.
337. Rigazzi, G., Tassi, A., Piechocki, R. J., Tryfonas, T., & Nix, A. (2017). Optimized certificate revocation list distribution for secure v2x communications. In: *2017 IEEE 86th Vehicular Technology Conference (VTC-Fall)* (pp. 1–7). IEEE.

338. Wantoro, J., & Mambo, M. (2022). Efficient and privacy-preserving certificate activation for v2x pseudonym certificate revocation. *Journal of Sensor and Actuator Networks*, 11(3), Article 51.
339. Haidar, F., Kaiser, A., & Lonc, B. (2017). On the performance evaluation of vehicular pki protocol for v2x communications security. In: *2017 IEEE 86th Vehicular Technology Conference (VTC-Fall)* (pp. 1–5). IEEE.
340. Hakeem, S. A. A., Abd El-Gawad, M. A., & Kim, H. (2019). A decentralized lightweight authentication and privacy protocol for vehicular networks. *IEEE Access*, 7, 119689–119705.
341. Howard, D. (2016). Advances in autonomous driving and v2x technologies. In: *SMTA International Conference Proceedings*.
342. Salahdine, F., Han, T., & Zhang, N. (2023). 5G, 6G, and beyond: Recent advances and future challenges. *Annals of Telecommunications*. <https://doi.org/10.1007/s12243-022-00938-3>
343. Porciello, P., & Henry, K. (2019). A secure v2x connected vehicle transponder system for vehicle prioritization. In: *Transportation Association of Canada and ITS Canada 2019 Joint Conference and Exhibition*.
344. Halbach, M., Wesemeyer, D., Merk, L., Lauermann, J., Heß, D., & Kaul, R. (2022). Prioritization of an automated shuttle for v2x public transport at a signalized intersection—a real-life demonstration.
345. Song, C., Zhou, M., Ding, Z., Liu, Z., Cheng, H., Geng, M., & Xu, W. (2022). Lane change maneuver detection considering real-time vehicle dynamic features via V2X communication. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*. <https://doi.org/10.1177/09544070221121835>
346. Yoon, Y., & Kim, H. (2020). Balancing power and rate control for improved congestion control in cellular v2x communication environments. *Ieee Access : Practical Innovations, Open Solutions*, 8, 105071–105081.
347. Toghi, B., Saifuddin, M., Fallah, Y. P., & Mughal, M. O. (2019). Analysis of distributed congestion control in cellular vehicle-to-everything networks. In: *2019 IEEE 90th Vehicular Technology Conference (VTC2019-Fall)* (pp. 1–7). IEEE.
348. Manolakis, K., & Xu, W. (2016). Time synchronization for multi-link d2d/v2x communication. In: *2016 IEEE 84th Vehicular Technology Conference (VTC-Fall)* (pp. 1–6). IEEE.
349. Huang, Y. K., Chennakesavula, P., & Wu, J. M. (2023). Relayed collective perception service with redundancy mitigation and time synchronization for v2x communications networks. In: *2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring)* (pp. 1–5). IEEE.
350. Khan, M. A., Ghosh, S., Busari, S. A., Huq, K. M. S., Dagiuklas, T., Mumtaz, S., Iqbal, M., & Rodriguez, J. (2021). Robust, resilient and reliable architecture for v2x communications. *IEEE Transactions on Intelligent Transportation Systems*, 22(7), 4414–4430.
351. Thandavarayan, G., Sepulcre, M., & Gozalvez, J. (2020). Redundancy mitigation in cooperative perception for connected and automated vehicles. In: *2020 IEEE 91st Vehicular Technology Conference (VTC2020-Spring)* (pp. 1–5). IEEE.
352. Yoshizawa, T., & Preneel, B. (2023). A new approach to pseudonym certificate management in v2x communication. In: *2023 IEEE Vehicular Networking Conference (VNC)* (pp. 25–32). IEEE.
353. Wang, D., Sattiraju, R. R., Qiu, A., Partani, S., & Schotten, H. D. (2020). Effect of retransmissions on the performance of c-v2x communication for 5 g. In: *2020 IEEE 92nd Vehicular Technology Conference (VTC2020-Fall)* (pp. 1–7). IEEE.
354. Kim, J., Jo, O., & Choi, S. W. (2018). Feasibility of index-coded retransmissions for enhancing sidelink channel efficiency of v2x communications. *Ieee Access : Practical Innovations, Open Solutions*, 7, 6545–6552.
355. Serag, K., Bhatia, R., Kumar, V., Celik, Z. B., & Xu, D. (2021). Exposing new vulnerabilities of error handling mechanism in {CAN}. In: *30th USENIX Security Symposium (USENIX Security 21)* (pp. 4241–4258).
356. Pese, M. D. (2022). Bringing practical security to vehicles. PhD thesis.
357. Huang, H., Li, H., Shao, C., Sun, T., Fang, W., & Dang, S. (2020). Data redundancy mitigation in v2x based collective perceptions. *IEEE Access*, 8, 13405–13418.
358. Khvostov, D., Chepinskiy, S., Krasnov, A., Khvostova, K., & Shmigelsky, G. (2016). Design of failover micro aerial vehicle with tilting rotors. In: *2016 8th International Congress on Ultra Modern Telecommunications and Control Systems and Workshops (ICUMT)* (pp. 352–355). IEEE.
359. Bi, J., Qin, X., & Jia, Z. (2023). Energy-efficient resource allocation for d2d-v2v communication with load balancing. *Mathematics*, 11(13), Article 2848.
360. Ortiz, M. T., Salient, O., Camps-Mur, D., Escrig, J., Nasreddine, J., & P'erezRomero, J. (2023). On the application of q-learning for mobility load balancing in realistic vehicular scenarios. In: *2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring)* (pp. 1–7). IEEE.

361. Aissaoui, R., Menouar, H., Dhraief, A., Filali, F., Belghith, A., & Abu-Dayya, A. (2014). Advanced real-time traffic monitoring system based on v2x communications. In: *2014 IEEE International Conference on Communications (ICC)* (pp. 2713–2718). IEEE.
362. Mohan, J. P., Sugunaraj, N., & Ranganathan, P. (2022). Cyber security threats for 5 g networks. In: *2022 IEEE International Conference on Electro Information Technology (eIT)* (pp. 446–454). IEEE.
363. Bellavista, P., Foschini, L., & Zamagni, E. (2014). V2x protocols for low-penetration-rate and cooperative traffic estimations. In: *2014 IEEE 80th Vehicular Technology Conference (VTC2014-Fall)* (pp. 1–6). IEEE.
364. L'ohdefink, J., B'ar, A., Schmidt, N. M., Hu'ger, F., Schlicht, P., & Fingscheidt, T. (2020). Focussing learned image compression to semantic classes for v2x applications. In: *2020 IEEE Intelligent Vehicles Symposium (IV)* (pp. 1641–1648). IEEE.
365. Tatashev, A., Yashina, M., Buslaev, D., & Serebryakov, E. (2022). Evaluation of v2x communication architecture characteristics based on loss queueing system with bandwidth sharing discipline. In: *2022 Systems of Signals Generating and Processing in the Field of on Board Communications* (pp. 1–5). IEEE.
366. Sabeeh, S., Sroka, P., & Wesolowski, K. (2019). Estimation and reservation for autonomous resource selection in c-v2x mode 4. In: *2019 IEEE 30th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)* (pp. 1–6). IEEE.
367. Wang, Y., Li, X., Wan, P., & Shao, R. (2021). Intelligent dynamic spectrum access using deep reinforcement learning for Vanets. *IEEE Sensors Journal*, 21(14), 15554–15563.
368. Gill, K. S., Heath, K. N., Chuke, S., Haider, A., Gegear, R. J., Ryder, E. F., & Wyglinski, A. M. (2020). Bumblebee-inspired c-v2x dynamic spectrum access testbed using openairinterface. In: *2020 IEEE 91st Vehicular Technology Conference (VTC2020-Spring)* (pp. 1–5). IEEE.
369. Aljaeere, H., Gad, A., Elsayed, O., & Hardt, W. (2023). Multi-layer hybrid encryption: A novel approach for enhanced v2x security. *Embedded Selforganising Systems*, 10(7), 28–32.
370. Hakeem, S. A. A., & Kim, H. (2021). Multi-zone authentication and privacy-preserving protocol (mapp) based on the bilinear pairing cryptography for 5g-v2x. *Sensors*, 21(2), Article 665.
371. Gupta, M., Awaysheh, F. M., Benson, J., Alazab, M., Patwa, F., & Sandhu, R. (2020). An attribute-based access control for cloud enabled industrial smart vehicles. *IEEE Transactions on Industrial Informatics*, 17(6), 4288–4297.
372. Rafique, S., Iqbal, M., & Habib, H. A. (2009). Space invariant vehicle recognition for toll plaza monitoring and auditing system. In: *2009 IEEE 13th International Multitopic Conference* (pp. 1–6). IEEE.
373. Zidi, C., Sondi, P., Mitton, N., Wahl, M., & Meddahi, A. (2023). Review and perspectives on the audit of vehicle-to-everything communications. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2023.3301182>
374. Baronti, F., Petri, E., Saponara, S., Fanucci, L., Roncella, R., Saletti, R., D'Abramo, P., & Serventi, R. (2009). Design and verification of hardware Building blocks for high-speed and fault-tolerant in-vehicle networks. *IEEE Transactions on Industrial Electronics*, 58(3), 792–801.
375. Orefice, P., Paura, L., & Scarpiello, A. (2010). Inter-vehicle communication Qos management for disaster recovery. *The internet of things: 20th tyrrhenian workshop on digital communications* (pp. 205–216). Springer.
376. Tuna, G., Nefzi, B., & Conte, G. (2014). Unmanned aerial vehicle-aided communications system for disaster recovery. *Journal of Network and Computer Applications*, 41, 27–36.
377. Kumar, S., Dohare, U., Kumar, K., Dora, D. P., Qureshi, K. N., & Kharel, R. (2018). Cybersecurity measures for geocasting in vehicular cyber physical system environments. *IEEE Internet of Things Journal*, 6(4), 5916–5926.
378. Sersemis, A., Papadopoulos, A., Spanos, G., Lalas, A., Votis, K., & Tzovaras, D. (2023). Cybersecurity oriented architecture to ensure the autonomous vehicles communication. In: *2023 31st Mediterranean Conference on Control and Automation (MED)* (pp. 119–124). IEEE.
379. Hu, J., Chen, S., Zhao, L., Li, Y., Fang, J., Li, B., & Shi, Y. (2017). Link level performance comparison between LTE v2x and Dsrc. *Journal of Communications and Information Networks*, 2(2), 101–112.
380. Kavas-Torris, O., Gelbal, S. Y., Cantas, M. R., Aksun Guvenc, B., & Guvenc, L. (2022). V2x communication between connected and automated vehicles (cavs) and unmanned aerial vehicles (uavs). *Sensors*, 22(22), Article 8941.
381. Correa, A., Avedisov, S. S., Sepulcre, M., Sakr, A. H., Molina-Masegosa, R., Altintas, O., & Gozalvez, J. (2021). On the impact of v2x-based maneuver coordination on the traffic. In: *2021 IEEE 93rd Vehicular Technology Conference (VTC2021-Spring)* (pp. 1–5). IEEE.
382. Hbaieb, A., Rhaïem, O. B., & Chaari, L. (2018). In-car gateway architecture for intra and inter-vehicular networks. In: *2018 14th International Wireless Communications & Mobile Computing Conference (IWCMC)* (pp. 1489–1494). IEEE.
383. Tandon, R., & Gupta, P. (2023). A hybrid security scheme for inter-vehicle communication in content centric vehicular networks. *Wireless Personal Communications*, 129(2), 1083–1096.

384. Chen, H., Liu, J., Wang, J., & Xun, Y. (2022). Towards secure intra-vehicle communications in 5 g advanced and beyond: Vulnerabilities, attacks and countermeasures. *Vehicular Communications*, 100548.
385. Peng, H., Li, D., Ye, Q., Abboud, K., Zhao, H., Zhuang, W., & Shen, X. (2017). Resource allocation for cellular-based inter-vehicle communications in autonomous multiplatoons. *IEEE Transactions on Vehicular Technology*, 66(12), 11249–11263.
386. Nagmode, V. S., & Rajbhoj, S. (2017). An iot platform for vehicle traffic monitoring system and controlling system based on priority. In: *2017 International Conference on Computing, Communication, Control and Automation (ICCCUBEA)* (pp. 1–5). IEEE.
387. Wampler, D., Fu, H., & Zhu, Y. (2009). Security threats and countermeasures for intra-vehicle networks. In: *2009 Fifth International Conference on Information Assurance and Security* (Vol. 2, pp. 153–157). IEEE.
388. Zhao, X., Fang, Y., Min, H., Wu, X., Wang, W., & Teixeira, R. (2023). Potential sources of sensor data anomalies for autonomous vehicles: An overview from road vehicle safety perspective. *Expert Systems with Applications*, 121358.
389. Han, J., Ju, Z., Chen, X., Yang, M., Zhang, H., & Huai, R. (2023). Secure operations of connected and autonomous vehicles. *IEEE Transactions on Intelligent Vehicles*. <https://doi.org/10.1109/TIV.2023.3304762>
390. Kumar, A., & Das, T. K. (2023). Cavid: Real time intrusion detection system for connected autonomous vehicles using logical analysis of data. *Vehicular Communications*, 43, Article 100652.
391. Cui, J., Chen, X., Zhang, J., Zhang, Q., & Zhong, H. (2020). Toward achieving fine-grained access control of data in connected and autonomous vehicles. *IEEE Internet of Things Journal*, 8(10), 7925–7937.
392. Liu, G., Wang, Z., Hu, J., Ding, Z., & Fan, P. (2019). Cooperative Noma broadcasting/multicasting for low-latency and high-reliability 5 g cellular v2x communications. *IEEE Internet of Things Journal*, 6(5), 7828–7838.
393. Zhang, J., Zhong, H., Cui, J., Tian, M., Xu, Y., & Liu, L. (2020). Edge computing-based privacy-preserving authentication framework and protocol for 5 g-enabled vehicular networks. *IEEE Transactions on Vehicular Technology*, 69(7), 7940–7954.
394. Seremet, I. (2019). Benefits of using 5 g network slicing to implement vehicle-to-everything (v2x) technology. In *2019 18th International Symposium INFOTEH-JAHORINA (INFOTEH)* (pp. 1–6). IEEE.
395. Bagheri, H., Noor-A-Rahim, M., Liu, Z., Lee, H., Pesch, D., Moessner, K., & Xiao, P. (2021). 5g nr-v2x: Toward connected and cooperative autonomous driving. *IEEE Communications Standards Magazine*, 5(1), 48–54.
396. Mohammed, B. A., Al-Shareeda, M. A., Manickam, S., Al-Mekhlafi, Z. G., Alayba, A. M., & Sallam, A. A. (2023). Anaa-fog: A novel anonymous authentication scheme for 5g-enabled vehicular fog computing. *Mathematics (Basel)*, 11(6), Article 1446.
397. Lucas-Estanˆ, M. C., Coll-Perales, B., Shimizu, T., Gozalez, J., Higuchi, T., Avedisov, S., Altintas, O., & Sepulcre, M. (2022). An analytical latency model and evaluation of the capacity of 5g nr to support v2x services using v2n2v communications. *IEEE Transactions on Vehicular Technology*, 72(2), 2293–2306.
398. Goswami, B., & Choudhury, H. (2024). A secure and fast handover authentication scheme for 5 g-enabled Iot using blockchain technology. *Wireless Personal Communications*, 138(4), 2155–2181.
399. Chen, H., Zhang, Y., Han, X., Cao, Y., & Zhang, Y. (2024). Blockchain-as-a-service for the security challenges and solutions in the internet of things environment. *Wireless Personal Communications*, 136(4), 2299–2327.
400. Weerasinghe, N., Usman, M. A., Hewage, C., Pfluegel, E., & Politis, C. (2023). Threshold cryptography-based secure vehicle-to-everything (v2x) communication in 5g-enabled intelligent transportation systems. *Future Internet*, 15(5), Article 157.
401. Boualouache, A., Brik, B., Senouci, S.-M., & Engel, T. (2023). On-demand security framework for 5gb vehicular networks. *IEEE Internet of Things Magazine*, 6(2), 26–31.
402. Shaer, I., Haque, A., & Shami, A. (2020). Multi-component v2x applications placement in edge computing environment. In: *ICC 2020–2020 IEEE International Conference on Communications (ICC)* (pp. 1–6). IEEE.
403. Huang, X., Yu, R., Kang, J., & Zhang, Y. (2017). Distributed reputation management for secure and efficient vehicular edge computing and networks. *IEEE Access*, 5, 25408–25420.
404. Srinivasa, R., Naidu, N. K. S., Maheshwari, S., Bharathi, C., & Kumar, A. H. (2019). Minimizing latency for 5 g multimedia and v2x applications using mobile edge computing. In: *2019 2nd International Conference on Intelligent Communication and Computational Techniques (ICCT)* (pp. 213–217). IEEE.

405. Pack, S. (2022). : A lightweight and secure vehicular edge computing framework for v2x services. In: *2022 IEEE 42nd International Conference on Distributed Computing Systems (ICDCS)* (pp. 1316–1317). IEEE.
406. Chavhan, S., Kumar, S., Tiwari, P., Liang, X., Lee, I. H., & Muhammad, K. (2023). Edgeenabled blockchain-based v2x scheme for secure communication within the smart city development. *IEEE Internet of Things Journal*. <https://doi.org/10.1109/JIOT.2023.3281092>
407. Bai, T., Fu, S., & Yang, Q. (2022). Privacy-preserving object detection with secure convolutional neural networks for vehicular edge computing. *Future Internet*, 14(11), Article 316.
408. Vladyko, A., Elagin, V., Spirkina, A., Muthanna, A., & Ateya, A. A. (2022). Distributed edge computing with blockchain technology to enable ultra-reliable low-latency v2x communications. *Electronics*, 11(2), Article 173.
409. Khan, A. A., Laghari, A. A., Shafiq, M., Awan, S. A., & Gu, Z. (2022). Vehicle to everything (v2x) and edge computing: A secure lifecycle for uav-assisted vehicle network and offloading with blockchain. *Drones*, 6(12), Article 377.
410. Feng, W., Lin, S., Zhang, N., Wang, G., Ai, B., & Cai, L. (2022). Joint c-v2x based offloading and resource allocation in multi-tier vehicular edge computing system. *IEEE Journal on Selected Areas in Communications*, 41(2), 432–445.
411. Mangla, C., Rani, S., Qureshi, N. M. F., & Singh, A. (2023). Mitigating 5g security challenges for next-gen industry using quantum computing. *Journal of King Saud University-Computer and Information Sciences*, 35(6), Article 101334.
412. Noor-A-Rahim, M., Liu, Z., Lee, H., Khyam, M. O., He, J., Pesch, D., Moessner, K., Saad, W., & Poor, H. V. (2022). 6g for vehicle-to-everything (v2x) communications: Enabling technologies, challenges, and opportunities. *Proceedings of the IEEE*, 110(6), 712–734.
413. Stavdas, A., Kosmatos, E., Maple, C., Hugues-Salas, E., Epiphaniou, G., Fowler, D. S., Razak, S. A., Matrakidis, C., Yuan, H., & Lord, A. (2023). Quantum key distribution for v2i communications with software-defined network. *IET Quantum Communication*. <https://doi.org/10.1049/qt2.12070>
414. La Manna, M., Perazzo, P., Treccozzi, L., & Dini, G. (2021). Assessing the cost of quantum security for automotive over-the-air updates. In *2021 IEEE Symposium on Computers and Communications (ISCC)* (pp. 1–6). IEEE.
415. Ha, S., Lee, H., Won, D., & Lee, Y. (2020). Quantum-resistant lattice-based authentication for v2x communication in c-its. In: *2020 14th International Conference on Ubiquitous Information Management and Communication (IMCOM)* (pp. 1–8). IEEE.
416. Yoshizawa, T., & Preneel, B. (2023). Post-quantum impacts on v2x certificates—already at the end of the road. In: *2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring)* (pp. 1–6). IEEE.
417. Lonc, B., Aubry, A., Bakhti, H., Christofi, M., & Mehrez, H. A. (2023). Feasibility and benchmarking of post-quantum cryptography in the cooperative its ecosystem. In: *2023 IEEE Vehicular Networking Conference (VNC)* (pp. 215–222). IEEE.
418. Namatherdhal, B., Mazher, N., & Sriram, G. K. (2022). Uses of artificial intelligence in autonomous driving and v2x communication. *International Research Journal of Modernization in Engineering Technology and Science*, 4(7), 1932–1936.
419. Christopoulou, M., Barmounakis, S., Koumaras, H., & Kaloxylou, A. (2022). Artificial intelligence and machine learning as key enablers for v2x communications: A comprehensive survey. *Vehicular Communications*, 100569.
420. Chen, M.-Y., Fan, M.-H., & Huang, L.-X. (2021). Ai-based vehicular network toward 6g and iot: Deep learning approaches. *ACM Transactions on Management Information Systems*, 13(1), 1–12.
421. Miao, L., Chen, S.-F., Hsu, Y.-L., & Hua, K.-L. (2022). How does c-v2x help autonomous driving to avoid accidents? *Sensors*, 22(2), Article 686.
422. Xu, C., Wu, H., Zhang, Y., Dai, S., Liu, H., & Tian, J. (2022). A real-time complex road ai perception based on 5g-v2x for smart city security. *Wireless Communications and Mobile Computing*, 2022, 1–11.
423. Maity, R., Jivthesh, M., Maitra, P. S., Sanjeevi, G., Gaushik, M., NB, S. S., & Menon, K. U. (2023). Maximising highway safety through ai-enabled detection of pedestrians and animals in v2x environments. In: *2023 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET)* (pp. 1–7). IEEE.
424. Kandhro, I. A., Ali, F., Panhwar, A. O., Larik, R. S. A., & Fatima, K. (2023). Artificial intelligence (ai) empowered anomaly detection for autonomous vehicles in 6g-v2x. *Mehran University Research Journal of Engineering & Technology*, 42(3), 79–88.
425. Renda, A., Ducange, P., Marcelloni, F., Sabella, D., Filippou, M. C., Nardini, G., Stea, G., Virdis, A., Micheli, D., Rapone, D., et al. (2022). Federated learning of explainable ai models in 6g systems: Towards secure and automated vehicle networking. *Information*, 13(8), Article 395.
426. Zhang, C., Li, W., Luo, Y., & Hu, Y. (2020). Ait: An ai-enabled trust management system for vehicular networks using blockchain technology. *IEEE Internet of Things Journal*, 8(5), 3157–3169.

427. Raja, G., Begum, M., Gurumoorthy, S., Rajendran, D. S., Srividya, P., Dev, K., & Qureshi, N. M. F. (2022). Ai-empowered trajectory anomaly detection and classification in 6g-v2x. *IEEE Transactions on Intelligent Transportation Systems*, 24(4), 4599–4607.
428. Furqan, H. M., Solaija, M. S. J., Hamamreh, J. M., & Arslan, H. (2019). Intelligent physical layer security approach for v2x communication. arXiv preprint arXiv:1905.05075.
429. Rabcinejad, E., Yazdinejad, A., Dehghantanha, A., Parizi, R. M., & Srivastava, G. (2021). Secure ai and blockchain-enabled framework in smart vehicular networks. In: *2021 IEEE Globecom Workshops (GC Wkshps)* (pp. 1–6). IEEE.
430. Gu, X., Duan, W., Zhang, G., Ji, Y., Wen, M., & Ho, P. H. (2022). Socially aware v2x networks with ris: Joint resource optimization. *IEEE Transactions on Vehicular Technology*, 71(6), 6732–6737.
431. Tang, X., Lan, X., Zhai, D., Zhang, R., & Han, Z. (2021). Securing wireless transmissions with ris-receiver coordination: Passive beamforming and active jamming. *IEEE Transactions on Vehicular Technology*, 70(6), 6260–6265.
432. Chen, Y., Wang, Y., Zhang, J., Zhang, P., & Hanzo, L. (2022). Reconfigurable intelligent surface (ris)-aided vehicular networks: Their protocols, resource allocation, and performance. *IEEE Vehicular Technology Magazine*, 17(2), 26–36.
433. Ai, Y., Felipe, A., Kong, L., Cheffena, M., Chatzinotas, S., & Ottersten, B. (2021). Secure vehicular communications through reconfigurable intelligent surfaces. *IEEE Transactions on Vehicular Technology*, 70(7), 7272–7276.
434. Yang, K., Shi, Y., Zhou, Y., Yang, Z., Fu, L., & Chen, W. (2020). Federated machine learning for intelligent iot via reconfigurable intelligent surface. *IEEE Network*, 34(5), 16–22.
435. Makarfi, A. U., Rabie, K. M., Kaiwartya, O., Li, X., & Kharel, R. (2020). Physical layer security in vehicular networks with reconfigurable intelligent surfaces. In: *2020 IEEE 91st Vehicular Technology Conference (VTC2020-Spring)* (pp. 1–6). IEEE.
436. Ahmad, I., Narmeen, R., Becvar, Z., & Guvenc, I. (2022). Machine learning-based beamforming for unmanned aerial vehicles equipped with reconfigurable intelligent surfaces. *IEEE Wireless Communications*, 29(4), 32–38.
437. Saïam, M., Chowdhury, M. Z., Hasan, S. R., & Jang, Y. M. (2023). Reconfigurable intelligent surface assisted backcom: An overview, analysis, and future research directions. *ICT Express*. <https://doi.org/10.1016/j.icte.2023.07.004>
438. Agudo, I., Montenegro-Gómez, M., & Lopez, J. (2020). A blockchain approach for decentralized v2x (d-v2x). *IEEE Transactions on Vehicular Technology*, 70(5), 4001–4010.
439. Meijers, J., Michalopoulos, P., Motepalli, S., Zhang, G., Zhang, S., Veneris, A., & Jacobsen, H. A. (2022). Blockchain for v2x: Applications and architectures. *IEEE Open Journal of Vehicular Technology*, 3, 193–209.
440. Das, D., Banerjee, S., Chatterjee, P., Ghosh, U., & Biswas, U. (2022). A secure blockchain enabled v2v communication system using smart contracts. *IEEE Transactions on Intelligent Transportation Systems*.
441. Patel, V. A., Bhattacharya, P., Tanwar, S., Jadav, N. K., & Gupta, R. (2022). Bfledge: Blockchain based federated edge learning scheme in v2x underlying 6 g communications. In: *2022 12th International Conference on Cloud Computing, Data Science & Engineering (Confluence)* (pp. 146–152). IEEE.
442. Zhou, X., He, D., Khan, M. K., Wu, W., & Choo, K. K. R. (2022). An efficient blockchainbased conditional privacy-preserving authentication protocol for Vanets. *IEEE Transactions on Vehicular Technology*.
443. Gurung, A. (2021). Data security and privacy in cloud computing focused on transportation sector with the aid of block chain approach. In: *2021 6th International Conference on Innovative Technology in Intelligent System and Industrial Applications (CITISIA)* (pp. 1–9). IEEE.
444. Liang, H., Wu, J., Mumtaz, S., Li, J., Lin, X., & Wen, M. (2019). Mbid: Micro-blockchainbased geographical dynamic intrusion detection for v2x. *IEEE Communications Magazine*, 57(10), 77–83.
445. Jabbar, R., Fetais, N., Kharbeche, M., Krichen, M., Barkaoui, K., & Shinoy, M. (2021). Blockchain for the internet of vehicles: How to use blockchain to secure vehicle-to-everything (v2x) communication and payment? *IEEE Sensors Journal*, 21(14), 15807–15823.
446. Aujla, G. S., Singh, A., Singh, M., Sharma, S., Kumar, N., & Choo, K.-K.R. (2020). Blocked: Blockchain-based secure data processing framework in edge envisioned v2x environment. *IEEE Transactions on Vehicular Technology*, 69(6), 5850–5863.
447. Rao, P. M., Jangirala, S., Pedada, S., Das, A. K., & Park, Y. (2023). Blockchain integration for iot-enabled v2x communications: A comprehensive survey, security issues and challenges. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2023.3281844>
448. Kiela, K., Jurgo, M., & Navickas, R. (2020). Structure of v2x-iot framework for its applications. In: *2020 43rd International Conference on Telecommunications and Signal Processing (TSP)* (pp. 229–234). IEEE.

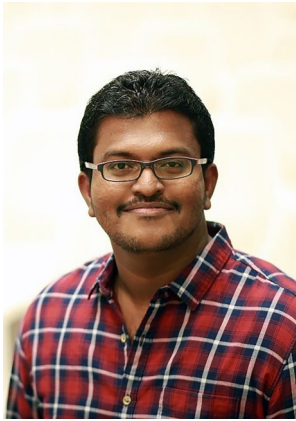
449. Arman, N., Hasan, S. R., & Abedin, M. R. (2023). Vehicle detection using deep learning method and adaptive and dynamic automated traffic system via iot using surveillance camera. In: *Proceedings of International Conference on Information and Communication Technology for Development: ICICTD 2022* (pp. 15–27). Springer.
450. Bai, Y., Yang, K., Chen, X., Guo, J., & Dong, H. (2019). Analysis of the influence of internet of vehicles on driverless technology. In: *Proceedings of 5th International Conference on Vehicle, Mechanical and Electrical EngineeringICVMEE*. Dalian City, China (pp. 28–29).
451. Patidar, S., Kumar, N., & Jindal, R. (2024). Iot data stream handling, analysis, communication and security issues: A systematic survey. *Wireless Personal Communications*, 1–50.
452. Kumar, S., & Singh, J. (2020). Internet of vehicles over vanets: Smart and secure communication using Iot. *Scalable Computing: Practice and Experience*, 21(3), 425–440.
453. Zrikem, M., Hasnaoui, I., & Ellassali, R. (2023). Vehicle-to-blockchain (v2b) communication: Integrating blockchain into v2x and iot for next-generation transportation systems. *Electronics*, 12(16), Article 3377.
454. Ioana, A., Korodi, A., & Silea, I. (2022). Automotive Iot ethernet-based communication technologies applied in a v2x context via a multi-protocol gateway. *Sensors (Basel, Switzerland)*, 22(17), 6382.
455. Elngar, A. A., & Kayed, M. (2020). Vehicle security systems using face recognition based on internet of things. *Open Computer Science*, 10(1), 17–29.
456. Sabri, Y., Siham, A., & Maizate, A. (2021). Internet of things (iot) based smart vehicle security and safety system. *International Journal of Advanced Computer Science and Applications* 12(4).
457. Agarwal, V., Sharma, S., & Agarwal, P. (2021). Iot based smart transport management and vehicle-to-vehicle communication system. In: *Computer Networks, Big Data and IoT: Proceedings of ICCBI 2020* (pp. 709–716). Springer.
458. Limbasiya, T., & Das, D. (2023). Querycom: Secure message communication and data searching protocols for smart transportation. *IEEE Transactions on Intelligent Transportation Systems*, 24(6), 5752–5764. <https://doi.org/10.1109/TITS.2023.3249833>

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